GrassRoute: Online Gradient Routing on the Grassmann Manifold

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Abstract

GrassRoute learns an online Grassmann routing subspace, applies adaptive updates in its coordinates, and preserves the orthogonal gradient residual through a stateless branch. We derive sharp QR pullback and normalized-covariance constants using CS/Poincaré geometry and a Householder identity, while a Ky Fan variational argument controls moving-target tracking. A virtual iterate and an averaged quadratic-root argument analyze the transported Adam recursion; full-state coupling, exact Gaussian orientation costs, Seeger–Maurer inversion and a Haar-mixture prior give conditional stability and PAC-Bayes guarantees. Archived Burgers experiments record best training loss 1.97×10^{-3} versus AdamW's 1.10×10^{-2} and periodic SVD's 3.27×10^{-3}, with 84.1% less optimizer state and a reported $1.78\times$ step cost. One Cora comparison improves on AdamW and corrected periodic routing with 93.0% less state than AdamW; three new initializations do not consistently preserve that accuracy gain. The new SMS comparison retains 87.0% less state at a 0.85 percentage-point macro-F1 cost. The original proposal matrix completed 117 of 162 representative comparisons before a disclosed compute reprioritization. Six additional public-data tasks, controlled tracking and focused seed replications retain complete outcomes and protocol versions alongside selected evidence.

1. Introduction

Adaptive optimizer states can be expensive even when useful gradient directions occupy a much smaller subspace than the parameter matrix. Low-rank optimizers exploit this observation by maintaining adaptive statistics in projected coordinates (Zhao et al., 2024; Hao et al., 2024; Chen et al., 2025; Zhu et al., 2025; Liang et al., 2024; Robert et al., 2025; Rajabi et al., 2025). The scientific question is how to update that subspace as training changes, while accounting for the gradient outside it and the coordinate dependence of the adaptive states.

GrassRoute maintains one right routing subspace per selected matrix. It updates the subspace through a Grassmann tangent step with momentum and QR retraction, applies Adam to projected coordinates, and updates the orthogonal residual with normalized SGD. The parameter remains unconstrained and receives a full-dimensional update. Persistent optimizer state can be much smaller than AdamW's, although routing adds computation.

A rigorous account must distinguish several issues. Normalizing each covariance sample changes the population being estimated. A subspace is invariant to a basis change but diagonal Adam is not. Unbiased gradients do not imply unbiased adaptive directions. A nonzero residual branch also prevents treating its spectral tail as a universally irreducible optimization error. Our theory addresses these issues explicitly. The fixed-covariance landscape has a principal-angle PL bound; stochastic tracking adds noise, drift, and momentum-lag budgets. Parameter convergence retains an adaptive-direction defect. Generalization is analyzed through the full stored state and through an alternative PAC-Bayes perturbation bound. Detailed proofs occupy Appendix A.

The empirical evidence identifies useful operating regimes for selective adaptive state. Archived Burgers/PINN trajectories show a favorable training-loss–state trade-off, while the independent Cora comparison improves on AdamW and corrected periodic routing with substantially less state. These observations concern specific objectives and resource budgets; they do not make GrassRoute the highest-accuracy optimizer in every comparison. Section 5.4 adds six public-data tasks under a frozen protocol and explicitly distinguishes primary-metric gains from Pareto-efficient memory–quality trade-offs. The complete attempted evaluation, including unfavorable and unfinished cases, is retained in Appendices B.7 and B.11.

We supplement that archive with a representative-model matrix designed to cover each task or dataset named in the proposal, using seed 42 and 1000 or 2000 complete epochs. It fixes historical rank overrides and GrassRoute's diagonal covariance transport, uses validation data for selection, and

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includes the official SubTrack++ implementation. After 117 of 162 planned formal runs, we stopped the remaining high-cost queue and documented that post-hoc scope change. An independent follow-up corrects the local periodic comparator's omitted second-moment transport. New and archived algorithm versions remain separate. Tables report only completed formal runs; the coverage audit records pending or failed comparisons instead of filling them with predictions. Additional actual-implementation experiments test tracking against instantaneous and full-covariance EMA spectral estimators, and expose the noise–lag–state trade-off across the entire declared drift grid.

Our contributions are a detailed geometric and conditional convergence analysis, full-state and PAC-Bayesian generalization analyses with explicit scope, an auditable experimental pipeline, and a task-by-task assessment of accuracy, memory, and compute. The angular PL factor is related to existing Rayleigh-quotient theory (Alimisis & Vandereycken, 2024); it is not presented as a uniquely established empirical predictor of optimizer rankings.

2. Related Work

Low-rank optimizer state has become a major strategy for reducing training memory. GaLore projects gradients onto periodically recomputed low-rank subspaces (Zhao et al., 2024), while FLoRA, Fira, APOLLO, OSD, LDAdam, and related methods refine projection, residual, or adaptive-state design (Hao et al., 2024; Chen et al., 2025; Zhu et al., 2025; Liang et al., 2024; Robert et al., 2025; He et al., 2025; Su et al., 2025; Mo et al., 2025). These methods differ in projector updates, error feedback, and adaptive coordinate handling. GrassRoute studies one online Grassmann routing design and explicitly documents both successful and unfavorable regimes.

Grassmann geometry identifies bases related by right orthogonal rotations (Edelman et al., 1998; Absil et al., 2008; Boumal et al., 2014). GROUSE and related streaming algorithms study online subspace estimation (Balzano et al., 2011; 2018). SubTrack++ is the closest optimizer-side work, combining Grassmann geodesic updates, projection-aware Adam, and recovery scaling (Rajabi et al., 2025). GrassRoute instead uses a right-sided QR tracker and a specified normalized residual rule. Online manifold tracking and moment-coordinate handling are established ingredients; our analysis makes their assumptions and approximation errors explicit and the non-LLM experiments examine their empirical trade-offs.

Theoretical ingredients include Rayleigh-quotient geometry (Alimisis & Vandereycken, 2024), spectral perturbation (Yu et al., 2015), adaptive-direction analysis (Reddi et al., 2018; Chen et al., 2019), and stability (Bousquet & Elisseeff, 2002; Hardt et al., 2016). Our analysis composes these tools while distinguishing normalized population covariance, basis-dependent Adam states, and residual-preserving parameter updates. The angular PL dependence is related to existing eigenproblem theory; captured energy is not a substitute for its eigengap and oracle-angle hypotheses. Generalization is treated through full-state stability and a separate PAC-Bayes argument, retaining the Seeger–Maurer binary-KL form (Seeger, 2002; Maurer, 2004). The spectral and statistical bridges use the separated Davis–Kahan theorem and variance-sensitive matrix concentration (Davis & Kahan, 1970; Tropp, 2012); these classical tools are cited at their exact use in the proofs.

3. Problem Setup

For a matrix $W_t \in \mathbb{R}^{m\times d}$ with stochastic gradient G_t, GrassRoute stores $U_t \in \mathbb{R}^{d\times r}$ with $U_t^\top U_t = I_r$. The routing object is $[U_t] \in \mathrm{Gr}(r,d)$, represented by $P_t = U_t U_t^\top$. Gradient routing is the orthogonal decomposition

$$Z_t = G_t U_t, \qquad G_t^{\parallel} = Z_t U_t^\top, \tag{1}$$

$$G_t^{\perp} = G_t - G_t^{\parallel}. \tag{2}$$

The routed moments have $2mr$ entries, while the basis and subspace momentum have $2dr$ entries. Fallback optimizer states are additional. These persistent optimizer-state counts differ from total peak GPU memory.

With $\mathcal{F}_t$ denoting the pre-batch state, the normalized implementation tracks the leading eigenspace of

$$C_t = \mathbb{E}_t\left[\frac{G_t^\top G_t}{\|G_t\|_F^2 + \varepsilon_c}\right], \tag{3}$$

$$\Phi_t(U) = -\tfrac{1}{2}\,\mathrm{tr}(U^\top C_t U). \tag{4}$$

This generally differs from the leading eigenspace of $S_t = \mathbb{E}_t[G_t^\top G_t]$. Let Δ_t be the rank-r eigengap of the covariance under analysis. Relative to its leading subspace, principal angles define $R_t = (\sum_i \theta_i^2)^{1/2}$ and $\theta_{\max,t} = \max_i \theta_i$. The local theory requires spectral separation and angles bounded away from $\pi/2$ on the analyzed interval. It does not infer these hypotheses from small changes between consecutive optimizer bases. Captured sample-gradient energy and compressed-coordinate spectral gaps are useful diagnostics, but neither is the population eigengap.

4. GrassRoute and Theoretical Guarantees

Table 2 specifies the parameter and routing-state updates. The routed branch maintains Adam moments of Z_t, producing $D_t^{\parallel}$. The residual branch uses

$$s_t = \mathrm{clip}\left(\frac{\|G_t\|_F}{\|G_t^{\perp}\|_F + \varepsilon_r}; s_{\min}, s_{\max}\right), \tag{5}$$

Table 1. **Mechanisms in memory-efficient adaptive optimization.** Keywords identify the subspace rule and the use of gradient information outside it. ✓: explicitly present; ✗: absent from the specified design; –: not a coordinate-alignment operation. "No refresh SVD" excludes initialization and refers only to a full-gradient SVD, not a small tangent decomposition. Alignment checks denote projection-aware *approximations*, not exact covariance transport. EF denotes generalized error feedback; it differs from an immediate residual branch.

Method	Subspace rule	Outside subspace	Moment alignment	No refresh SVD
GaLore (Zhao et al., 2024)	Periodic SVD	Discarded	✗	✗
Fira (Chen et al., 2025)	Periodic SVD	Scaled residual ✓	✗	✗
APOLLO (Zhu et al., 2025)	Random / SVD	Full-gradient scaling	–	✓[a]
LDAdam (Robert et al., 2025)	Block power + QR	Generalized EF	✓	✓
SubTrack++ (Rajabi et al., 2025)	Grassmann geodesic	Scaled residual ✓	✓	✓
GrassRoute	**Grassmann QR**	**Norm-matched residual ✓**	✓	✓

[a]Random-projection APOLLO; its SVD variant uses refresh decompositions. GrassRoute's additional analysis covers the normalized-covariance landscape, moving-target tracking, adaptive parameter convergence, and conditional full-state stability/PAC-Bayes generalization. These are separate statements with explicit hypotheses, rather than a universal superiority claim.

Table 2. **One GrassRoute iteration.** The parameter branch preserves the full orthogonal residual; the routing branch changes the basis and aligns stored coordinates. Second-moment transport is a diagonal approximation. Elementwise operations use $\oslash$ and $\odot$.

Stage	Update	Stored state
Project gradient	$Z_t = G_t U_t, \quad G_t^{\perp} = G_t - Z_t U_t^{\top}$	$U_t \in \mathbb{R}^{d\times r}$
Adaptive moments	$M_t = \beta_1 M_{t-1} + (1-\beta_1) Z_t, \quad V_t = \beta_2 V_{t-1} + (1-\beta_2) Z_t^{\odot 2}$	$2mr$ entries
Bias correction	$\widehat{M}_t = M_t/(1-\beta_1^t), \quad \widehat{V}_t = V_t/(1-\beta_2^t)$	Moment ages
Adaptive branch	$D_t^{\parallel} = \widehat{M}_t \oslash (\sqrt{\widehat{V}_t} + \varepsilon)$	No $m\times d$ moments
Residual branch	$s_t = \mathrm{clip}(\|G_t\|_F / (\|G_t^{\perp}\|_F + \varepsilon_r); s_{\min}, s_{\max})$	Stateless
Parameter update	$W_{t+1} = (1-\eta_t\lambda_{\mathrm{wd}}) W_t - \eta_t D_t^{\parallel} U_t^{\top} - \eta_{\perp,t} s_t G_t^{\perp}$	Full W_t
Horizontal direction	$Q_t = (I - U_t U_t^{\top}) G_t^{\top} Z_t / (\|G_t\|_F^2 + \varepsilon_c)$	No full covariance
Transported momentum	$H_t = \mu_H \Pi_{U_t} H_{t-1} + (1-\mu_H) Q_t$	dr entries
Grassmann update	$U_{t+1} = \mathrm{qf}(U_t + \rho_t H_t)$	dr basis entries
Coordinate alignment	$R_t = \mathrm{Polar}(U_{t+1}^{\top} U_t), \quad M_t \leftarrow M_t R_t^{\top}$	$r \times r$ rotation
Diagonal transport	$V_t \leftarrow V_t (R_t^{\top})^{\odot 2}$	Same moment age

$$W_{t+1} = (1-\eta_t\lambda_{\mathrm{wd}}) W_t - \eta_t D_t^{\parallel} U_t^{\top} - \eta_{\perp,t} s_t G_t^{\perp}. \tag{6}$$

For $\widehat{C}_t = G_t^{\top} G_t / (\|G_t\|_F^2 + \varepsilon_c)$, the tracker ascends along $Q_t = (I - P_t)\widehat{C}_t U_t$. Computing $G_t^{\top}(G_t U_t)$ avoids constructing a $d \times d$ covariance.

The positive sign in QR maximizes captured energy. Residual norm matching is relative to the raw gradient, not the Adam update; Proposition 7 shows why the branch step scales can still differ substantially. The second-moment transformation is the diagonal of a rotated diagonal covariance; unavailable cross-correlations are omitted. It is not exact covariance transport, and Adam remains coordinate dependent. Archived amplitude-rotation and hard-reset variants retain separate labels. Lemma 3 quantifies the differences. Lemma 2 proves a cubic quotient-level discrepancy between QR and the exponential map for the same tangent step.

Landscape and tracking. Theorem 1 gives, for $\theta_{\max} \le R_0 < \pi/2$,

$$\|\operatorname{grad} \Phi_t(U)\|_F^2 \ge 2\Delta_t \cos^2 R_0 \{\Phi_t(U) - \Phi_t(U_t^{\star})\}. \tag{7}$$

The proof uses exact principal-coordinate Rayleigh quotients. A CS/Poincaré reduction also gives the sharp QR pullback constant $L_R = \lambda_1(C_t) - \lambda_d(C_t)$, with no ambient-dimension factor. Quadratic growth and PL on this region do not imply geodesic strong convexity throughout it. Theorem 2 proves a stochastic objective-excess recursion with noise, covariance drift, and momentum-bias terms. Its steady-state trade-off contains $\rho\sigma^2/\mu$, B^2/μ, and $v^2/(\rho^2\mu^2\Delta_0)$, where $\mu = \Delta_0 \cos^2 R_0$ and v bounds covariance change.

Convergence. Theorem 4 gives a nonconvex stationarity bound retaining every adaptive-direction defect. Theorem 5 additionally closes that budget for the actual transported moments and current-batch denominator under bounded gradients and conditionally radial unimodal noise. A virtual-iterate identity cancels first-moment history; averaging before Cauchy–Schwarz and Minkowski produces a scalar quadratic-root bound. The denominator defect is bounded by $c_t = (1-\beta_2)/(1-\beta_2^t)$. With $\eta, \rho \asymp T^{-1/2}$ and $\beta_2 = 1 - 1/T$, its bound is $O(T^{-1/2}) + O((\log T)/T) + O(\varepsilon_r^2)$; the last term disappears for a positive residual floor under the same noise assumptions. These additional hypotheses are not asserted for the benchmark runs. Corollary 4 separately treats the deterministic bounded-gain regime. Residual preservation removes a compulsory rank-tail floor (Proposition 6).

Generalization. Theorem 10 couples all stored states on neighboring i.i.d. data sets. Proposition 13 supplies an explicit five-state sensitivity matrix; Proposition 14 derives its spectral and feedback conditions. A separate PAC-Bayes argument keeps the exact Gaussian orientation cost and applies the Seeger–Maurer binary-KL inverse (Theorems 14 and 12). A Davis–Kahan residual bound and McDiarmid–Tropp concentration give its tracking interpretation; the observed-angle certificate itself does not require an eigengap (Theorem 15). For a fixed-feature Gaussian head, one-dimensional conditioning evaluates the empirical Gibbs error exactly (Proposition 17). Validation selection has its own finite-candidate bound in Appendix A.23. These are conditional guarantees, rather than numerical generalization certificates for the present deep-network experiments.

Compatibility and sharper local results. Theorem 6 proves that spherical gradient noise and a fixed positive covariance offset imply $\Delta_{r,t} \le 2\|\nabla f(W_t)\|^2/\varepsilon_c$. A uniform positive-gap tracking theorem therefore cannot simply be extended through the stationarity limit. Theorem 7 supplies a finite-window basin certificate. Theorems 8 and 9 derive, respectively, exact additive-oracle modal error and a nonlinear invariant neighborhood for the actual deterministic rank-one tracker. Theorem 13 allows noncontractive routing states with summable feedback; Theorems 14 and 15 quantify how a valid independent prior turns a tracking-angle bound into Gaussian posterior complexity. A Haar mixture prior and the Gibbs variational principle charge learned orientation through an explicit orbital partition function, whose large-concentration leading term is $\frac{1}{2} r(d-r)\log c$, rather than a loose chart covering. Table 4 identifies the equality cases and the remaining conservative budgets.

5. Experiments

We distinguish archived results from a new validation-selected extension. They use different state-transport versions and, for several tasks, different data/model protocols. Their numerical endpoints are not pooled. All formal runs use seed 42, as specified by the proposal. The extension includes the official SubTrack++ optimizer; the historical B6 was an unofficial geodesic surrogate.

Figure 1 makes the positive loss evidence explicit rather than inferring it from endpoint tables. The stored archived GrassRoute run ends at 2.978×10^{-3}, compared with 1.368×10^{-2} for AdamW and 3.881×10^{-3} for periodic SVD, with every intermediate epoch retained. The independent follow-up panel shows where that advantage transfers and where the strongest competing method remains better. Lower training objective is not equated with lower held-out solution error.

5.1. Scientific-learning and graph evidence

The archived Burgers PINN supplies the clearest joint training-loss–state observation: its tuned best loss is 1.97×10^{-3} versus AdamW's 1.10×10^{-2} and periodic SVD's 3.27×10^{-3}, with 84.1% less persistent optimizer state. Figure 1 separately shows all epochs of the stored trajectories. The archive reports a $1.78\times$ step cost, and its residual-plus-initial-condition objective omits an explicit spatial boundary penalty. This supports a scoped training-loss–memory result, not a claim of superior physical solution accuracy. All recovered endpoints, including the unsuccessful task families, remain in Appendix B.7.

In the independent expanded-search Cora comparison, GrassRoute obtains 78.6% test accuracy versus AdamW's 77.5%, corrected periodic routing's 77.7%, and official SubTrack++'s 77.3%. Its persistent optimizer state is 207,368 bytes versus AdamW's 2,950,224, a 93.0% reduction. SGD reaches 79.7% with 1,475,100 bytes; thus GrassRoute supplies a favorable accuracy–state operating point rather than the highest accuracy across all six methods. The comparison uses the same nine-candidate validation budget and fresh 1000-epoch formal runs (Table 24).

With the seed-42 Cora GraphSAGE hyperparameters frozen, three new initializations give mean final test accuracy 76.90% for GrassRoute, 77.63% for AdamW and 79.23% for SGD-M. GrassRoute exceeds AdamW in 1/3 paired seeds. Corrected periodic SVD records 78.10%, 78.60% on its completed seeds and a retained numerical failure on the other. The favorable seed-42 accuracy–state observation is therefore not consistent across these new initializations; all endpoints and the failure appear in Table 21.

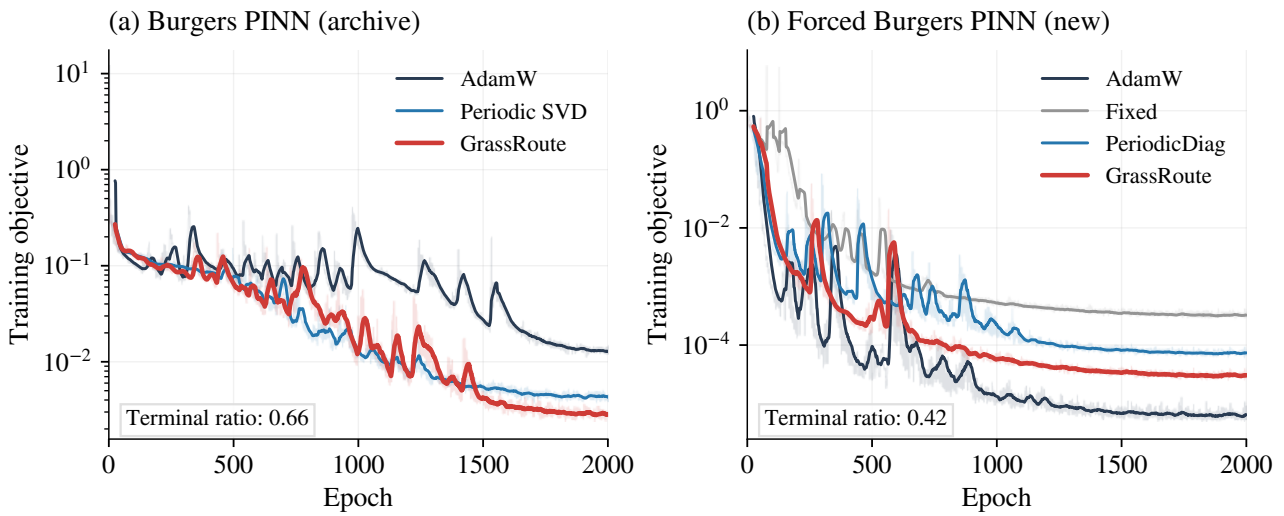


Figure 1. Core loss trajectories supporting the scientific-learning advantage, with the competing outcomes retained. All panels show 2000 stored epochs: faint lines are the raw objectives, solid lines are 25-epoch trailing means. "Terminal ratio" is the ratio of the final-100-epoch mean losses, GrassRoute / periodic comparator; below one favors GrassRoute. (a) Three archived Burgers trajectories: GrassRoute ends below AdamW and periodic SVD. These stored curves are distinct from the tuned scalar summaries in Table 16; the objective is the archived residual-plus-initial-condition loss, with its boundary-condition specification in Appendix B. (b) Independent validation-selected forced Burgers with corrected periodic moment transport. GrassRoute improves on the matched-state periodic run; AdamW remains stronger. The Poisson loss control appears in Figure 3. No best epoch is selected and no cross-protocol loss magnitude is pooled. Seed 42 only.

5.2. Evaluation scope and complete reporting

We distinguish strict primary-metric gains from accuracy–memory trade-offs. The original proposal extension specified 27 representative configurations and six methods, selected by validation before 1000 or 2000 complete formal epochs. After 117 of 162 task–method runs finished, we stopped its remaining high-cost image and continual-learning queue and redirected computation to focused exploratory checks. This scope decision followed inspection of completed results and cannot support a claim of prospective full coverage. The original outcome matrix retains every completed, unfavorable and pending cell in Appendix B.7. A controlled drifting-regression study and a second Cora architecture are reported separately in Appendix B.9. New public-data domains use another frozen protocol below. All experiments use seed 42; finite tuning budgets and one-seed observations do not imply statistical significance or a universal optimizer ranking.

5.3. Testing the mathematical mechanisms

The controlled diagnostics isolate angle dependence at a fixed eigengap, QR/exponential discrepancy, and tracking under known population drift. They use full covariance spectra and actual oracle subspaces, with 600 random landscape checks and a 27-setting tracking bed. A separate neighboring-data experiment replaces one CIFAR-100 image while training a linear probe on frozen random ResNet features for 1000 epochs. It measures parameter and bounded-loss sensitivity, rather than equating a train–test gap with uniform stability. These auxiliary tests are not counted as additional end-to-end task-family wins. Their protocols and diagnostic tables appear in Appendix B.

The complete mechanism grids, resource tables and expanded-search follow-ups are reported in Appendix B.10.

5.4. New public-data task domains

We add six tasks in three domains: raw-text classification (20 Newsgroups and SMS Spam), structured-data classification (Adult and Spambase), and inverse imaging (Optdigits denoising and missing-pixel completion) (scikit-learn developers, 2026; Almeida & Hidalgo, 2011; Becker & Kohavi, 1996; Hopkins et al., 1999; Alpaydin & Kaynak, 1998). The two inverse tasks share the same image corpus and constitute two settings within one domain, not two independent task-family replications. Dataset and split definitions are fixed before formal training.

Every task compares SGD-M, AdamW, fixed routing, thin periodic SVD, GrassRoute, and pinned official SubTrack++. All six receive nine 200-epoch validation candidates and a fresh 1000-epoch selected run. Each epoch passes through the complete training split in one full-data update. The routed grids explicitly vary the residual/parallel learning-rate ratio across $\{1, 10, 100\}$, since an SGD residual and

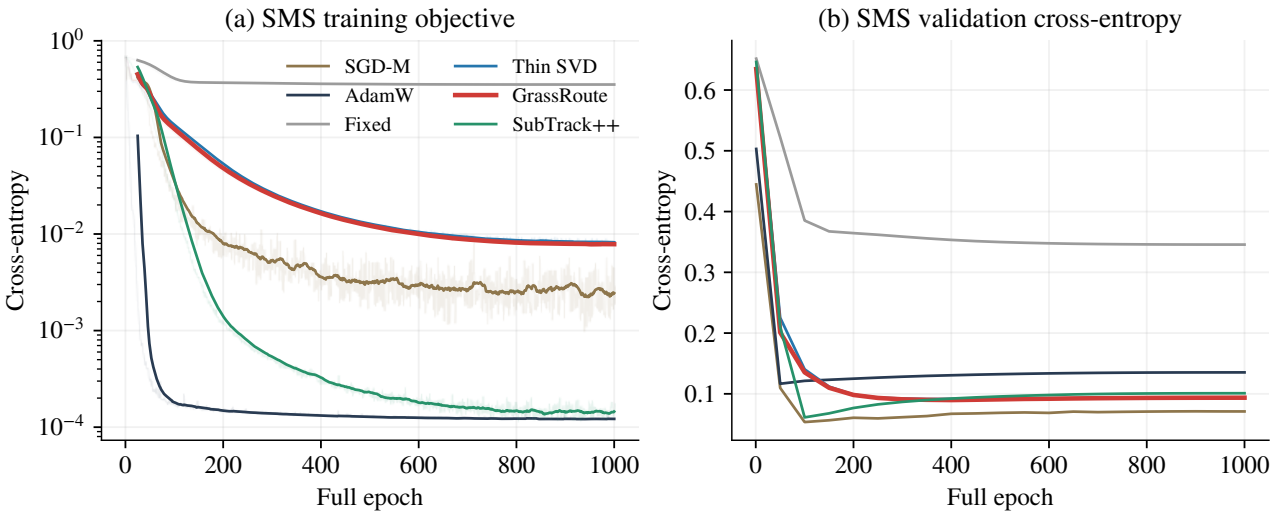


Figure 2. SMS memory–quality operating point, with all six validation-selected methods. Left: the full 1000-epoch training objectives (raw losses faint, 25-epoch trailing means solid). Right: all saved validation cross-entropies. Hyperparameters were selected by final validation macro-F1; secondary cross-entropy did not replace that criterion. The full primary-metric matrix and every classifier endpoint remain in Tables 30 and 31.

an adaptive branch need not have the same useful step-size scale. This changes the search protocol relative to the original extension and is not silently pooled with it.

The periodic comparator directly factorizes the gradient matrix, $G = L\Sigma V^\top$, and uses its top right singular vectors. This avoids forming the full $G^\top G$ and transports both the first moment and the diagonal second moment. Fixed and periodic methods store no unused routing momentum. The complete search, sources and data hashes are frozen in the new-task protocol; Appendix B.11 reports every outcome.

The main-text subset uses an explicit reporting rule. A primary-metric gain requires a better final held-out score than both AdamW and thin periodic SVD after all six methods finish. A separate memory–quality trade-off requires at least 50% less state than AdamW, a primary classification score within 0.01 or an image MSE within 5%, and a non-dominated score–state point among the six methods. The corresponding quality cost is stated explicitly. These are descriptive one-seed criteria; statistical significance is not estimated. The new public-data suite has completed 36 of 36 selected formal runs and 6 of six full comparisons. No completed new task currently establishes the predeclared gain over both AdamW and thin periodic SVD. Separate Pareto-efficient memory–quality trade-offs satisfy the predefined quality band on SMS / TF-IDF MLP. All attempted tasks and outcomes, including numerical failures and unfinished cells, remain in Appendix B.11; none is deleted after observing test results.

On SMS, GrassRoute retains 0.137 MB versus AdamW's 1.050 MB (87.0% less), with macro-F1 94.63 versus 95.48.

Table 3. **SMS memory–quality trade-off.** Validation macro-F1 selects the configurations; test CE is secondary. The complete six-task comparison is in Table 30. Persistent state is decimal MB.

Method	F1 ↑	Test CE ↓	State MB ↓
AdamW	95.48	0.1752	1.050
Thin SVD	94.29	0.0952	0.071
GrassRoute	94.63	0.0931	0.137

Its separately retained test cross-entropy is 0.0931 versus 0.1752, and AUROC is 99.08 versus 98.28. These secondary probabilistic-quality observations did not determine the selected hyperparameters or redefine the primary F1 criterion. Every classifier endpoint is retained in Table 31. With the seed-42 validation choices frozen, the three new SMS initializations yield 94.74% mean macro-F1 for GrassRoute, 95.48% for AdamW and 94.29% for thin periodic SVD. The GrassRoute state is 87.0% smaller than AdamW's; its mean F1 is 0.74 points below AdamW. The one-point quality band holds in 1/3 seeds, and GrassRoute exceeds periodic SVD in 2/3 seeds. This is a descriptive mean memory–quality trade-off, with visible seed variability. All seeds and SGD appear in Table 22.

6. Conclusion

GrassRoute allocates adaptive state to an online Grassmann routing subspace and preserves a full-dimensional residual update. Principal-plane geometry, Householder identities and Ky Fan variational analysis yield sharp local constants; a virtual iterate gives an averaged quadratic-root convergence

bound for the transported adaptive recursion. Full-state coupling and exact Gaussian orientation costs connect the routing dynamics to conditional stability and PAC-Bayes generalization, with Seeger–Maurer inversion and a Haar-mixture prior retaining the geometry of learned directions.

The archived Burgers PINN, independent Cora comparison and new SMS task identify favorable loss–state, accuracy–state and memory–quality operating points under their specified protocols. Additional fixed-hyperparameter seeds show that the Cora accuracy advantage is not consistent across initializations, while the SMS average memory–quality tradeoff varies by seed. The complete drift/noise grid separates temporal averaging from tracking lag and state cost. Completed endpoints, unfinished cells, the subsequent scope change and numerical-recovery policies are recorded with the experimental protocols in the appendix.

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A. Detailed Theoretical Analysis

The proof is organized around three reductions: orthogonal routing to independent principal planes; spectral perturbation to a separated Sylvester equation; and generalization to an exact Gaussian KL followed by binary-KL inversion. Their classical ingredients and their exact roles are explicit. The parameter matrix is unconstrained, and a rank-r optimizer state is not a rank-r parameter constraint. The table distinguishes sharp atomic inequalities from the sufficient adaptive-state budgets used to combine them.

Table 4. **Proof structure and sharpness.** Exactness concerns the stated model or atomic inequality, not universal optimality of the complete optimizer or of every composed finite-sample bound.

Reduction / classical tool	Retained quantity	Equality or scope
CS + Poincaré separation	QR pullback $L_R = \lambda_1 - \lambda_d$	Two extremal eigenvectors; dimension-free and attained.
Ky Fan + square completion	Moving-excess coefficient $(1 + a^{-1})/(4\Delta)$	Optimal to second order in a two-plane perturbation.
Principal-angle Rayleigh quotient	PL factor $2\Delta \cos^2 R$	Rank one at angle R attains it.
Householder involution	$\|D^2\mathcal{C}_\epsilon[E,E]\|_{\mathrm{op}} \le 2\|E\|_F^2/\epsilon$	Attained at zero with rank-one E.
Davis–Kahan / Rosenblum	Off-diagonal residual divided by the mixed gap	Unit residual constant is exact for one Sylvester entry.
Descent + Minkowski	Quadratic-root stationarity	The reference quadratic attains the zero-defect noise coefficient.
Yule–Walker / Polyak	Exact modal stationary covariance	Additive tangent oracle; no multiplicative-noise identification.
Gaussian entropy identity	Exact prior–posterior angle cost	Identity for every positive two-variance covariance.
Seeger–Maurer	Exact Ξ_n and binary-KL inverse	Bernoulli moment is exact; final confidence bound need not be optimal.

The actual Adam defect and the coupled state Lipschitz constants remain sufficient upper bounds; sharpness is not claimed for those worst-case budgets. A positive-gap tracking regime is also kept separate from a stationarity limit in which that gap may vanish.

A.1. Notation, filtration, and the population being tracked

Write $\langle A, B\rangle = \mathrm{tr}(A^\top B)$ and let $\|\cdot\|$ denote the Frobenius norm unless the operator norm is explicitly indicated. The pre-sampling filtration $\mathcal{F}_t$ contains the data set, parameters, bases, and all optimizer states before drawing the current mini-batch. Thus $W_t, U_t, M_{t-1}, V_{t-1}, H_{t-1}$ are measurable with respect to $\mathcal{F}_t$. The sampled gradient G_t generally is not. We use two covariances:

$$S_t = \mathbb{E}[G_t^\top G_t \mid \mathcal{F}_t], \tag{8}$$

$$C_t = \mathbb{E}\left[\frac{G_t^\top G_t}{\|G_t\|^2 + \varepsilon_c}\,\middle|\,\mathcal{F}_t\right]. \tag{9}$$

The implemented normalized tracker is unbiased for the tangent gradient of $\Phi_t(U) = -\mathrm{tr}(U^\top C_t U)/2$. Its eigenspace is not in general the leading eigenspace of S_t. Normalization preserves the eigenvectors of a *single* covariance sample; it need not preserve eigenvectors after expectation. For example, let $G^\top G$ equal $100 e_1 e_1^\top$ with probability 0.1 and $e_2 e_2^\top$ with probability 0.9. With $\varepsilon_c = 0$, the unnormalized population selects e_1, whereas the normalized population selects e_2. All tracking statements refer to C_t; raw residual-energy statements refer to S_t. They coincide up to scale only under an additional structural assumption, such as deterministic gradient norm.

For a fixed covariance $C \succeq 0$, put

$$\lambda_1 \ge \cdots \ge \lambda_d, \quad \Delta = \lambda_r - \lambda_{r+1} > 0, \quad \Lambda = \lambda_1 - \lambda_d, \quad P = UU^\top, \quad P_\star = U_\star U_\star^\top. \tag{10}$$

Here $U_\star$ spans the leading eigenspace. Let $\theta_1, \ldots, \theta_r \in [0, \pi/2]$ be the principal angles and define

$$R(U) = \|\Theta\|_F, \quad \theta_{\max}(U) = \max_i \theta_i, \quad s(U) = \|\sin\Theta\|_F, \quad h_C(U) = \Phi(U) - \Phi(U_\star). \tag{11}$$

The angle $\theta_{\max}$ controls transversality to the optimal subspace; R is geodesic distance. They must not be identified with the distance between two consecutive optimizer bases. Constants in the analysis are specific to a layer, rank, and time interval. A positive average captured energy across layers does not verify a uniform lower bound on their individual eigengaps.

A.2. Quotient geometry and implemented operations

The canonical quotient metric on $\mathrm{Gr}(r, d)$ has horizontal lift

$$\mathcal{H}_U = \{\xi : U^\top \xi = 0\}, \qquad \Pi_U(A) = (I - P)A, \qquad g_U(\xi, \zeta) = \langle \xi, \zeta \rangle. \tag{12}$$

Equivalently, for projector tangent vectors the metric is half the ambient Frobenius inner product. These normalizations agree under $\dot{P} = \xi U^\top + U\xi^\top$ (Bendokat et al., 2024). This metric has nonnegative sectional curvature bounded by 2; the upper bound need not be attained in every low-dimensional case (Zimmermann & Stoye, 2024). The injectivity radius is $\pi/2$. The cut locus relative to $[U_\star]$ consists of subspaces having at least one principal angle $\pi/2$. Consequently $\theta_{\max} < \pi/2$ ensures the usual graph chart exists, while $R < \pi/2$ is a sufficient, stronger condition. Neither condition implies geodesic convexity of Φ throughout that region.

Lemma 1 (Projection and exact energy accounting)**.** *For every G and orthonormal U,*

$$G = GP + G(I - P), \qquad \|G\|^2 = \|GU\|^2 + \|G(I - P)\|^2. \tag{13}$$

For a covariance $S = \mathbb{E}[G^\top G]$, an $\mathcal{F}_t$-measurable projector satisfies

$$\mathbb{E}_t \|G(I - P)\|^2 = \mathrm{tr}(S) - \mathrm{tr}(U^\top S U) = \tau_r(S) + 2h_S(U), \tag{14}$$

where $\tau_r(S) = \sum_{j>r} \lambda_j(S)$.

Proof. The two gradient components are orthogonal because $P(I - P) = 0$. Also $\|GP\|^2 = \mathrm{tr}(U^\top G^\top G U) = \|GU\|^2$. Taking conditional expectation yields the first equality in (14); adding and subtracting the leading r eigenvalues yields the second. If P is chosen from the *current* sampled gradient, the conditional-expectation step requires separate justification. □

Lemma 2 (QR retraction and its quotient error)**.** *Let $\xi \in \mathcal{H}_U$ have compact singular value decomposition $\xi = A\,\mathrm{diag}(\sigma_i)B^\top$. The projector of $\mathrm{qf}(U + \xi)$ is*

$$P_{\mathrm{QR}} = (U + \xi)(I + \xi^\top \xi)^{-1}(U + \xi)^\top. \tag{15}$$

In the principal planes of ξ, QR and the exponential map rotate through $\arctan \sigma_i$ and σ_i, respectively. If $\|\xi\| \le 1/4$,

$$d_{\mathrm{Gr}}([\mathrm{qf}(U + \xi)], \mathrm{Exp}_{[U]}\, \xi) \le \frac{1}{3}\Big(\sum_i \sigma_i^6\Big)^{1/2} \le \frac{1}{3}\|\xi\|^3. \tag{16}$$

Projection transport $\mathcal{T}_{U \to V}\xi = (I - VV^\top)\xi$ is nonexpansive and satisfies

$$\|\mathcal{T}_{U \to V}\xi - \xi\| \le \|VV^\top - UU^\top\|_{\mathrm{op}} \|\xi\|. \tag{17}$$

Proof. Horizontal tangency gives $(U + \xi)^\top (U + \xi) = I + \xi^\top \xi$, which is positive definite for every ξ. The orthogonal projector onto its column span is therefore (15). The polar representative $(U + \xi)(I + \xi^\top \xi)^{-1/2}$ has exactly the same span as its QR representative. In the ith principal plane its coefficients are $(1, \sigma_i)/\sqrt{1 + \sigma_i^2}$, giving angle $\arctan \sigma_i$. The exponential coefficients are $(\cos \sigma_i, \sin \sigma_i)$. The small-step condition prevents angle wrapping. Since $0 \le x - \arctan x = \int_0^x t^2/(1 + t^2)\, dt \le x^3/3$, summing squared angular differences proves (16). Finally, orthogonal projection cannot increase norm and $(I - VV^\top)\xi - \xi = (UU^\top - VV^\top)\xi$. □

This calculation identifies the retraction error on the quotient, where the algorithm actually tracks a subspace. The QR basis itself can also contain a within-subspace rotation. That rotation affects coordinatewise Adam even when it has no effect on the routing objective. Numerical orthogonality error, retraction approximation error, and statistical tracking error are three different quantities; a small value of the first does not establish either of the others.

A.3. The routing landscape: exact principal-angle analysis

Theorem 1 (Landscape, local PL inequality, and its boundary)**.** *For $C \succeq 0$ with gap $\Delta > 0$, the unique minimizing subspace of $\Phi(U) = -\operatorname{tr}(U^\top CU)/2$ is $[U_\star]$. Its horizontal gradient and Hessian quadratic form are*

$$\operatorname{grad}\Phi(U) = -(I-P)CU, \tag{18}$$

$$\langle \xi, \operatorname{Hess}\Phi(U)[\xi]\rangle = \operatorname{tr}(\xi^\top \xi\, U^\top CU) - \operatorname{tr}(\xi^\top C\xi). \tag{19}$$

Globally,

$$\frac{\Delta}{2}s(U)^2 \le h_C(U) \le \frac{\Lambda}{2}s(U)^2, \qquad \frac{2}{\pi}R(U) \le s(U) \le R(U). \tag{20}$$

Whenever $\theta_{\max}(U) \le R_0 < \pi/2$,

$$\|\operatorname{grad}\Phi(U)\|^2 \ge 2\Delta\cos^2 R_0\, h_C(U). \tag{21}$$

At the minimum the Hessian eigenvalues are $\lambda_i - \lambda_j$, $i \le r < j$. Every nonoptimal critical subspace has a tangent direction of curvature at most $-\Delta$. Quadratic growth in (20) and PL in (21) do not *assert geodesic strong convexity on the entire normal neighborhood.*

Proof. The variational characterization of eigenvalues gives $\max_{U^\top U=I}\operatorname{tr}(U^\top CU) = \sum_{i\le r}\lambda_i$. The gap separates this maximizing subspace from its complement, giving uniqueness on the quotient. Differentiating Φ in the ambient space gives $-CU$; horizontal projection proves (18). For a Grassmann geodesic with $\dot U(0) = \xi$, one can choose a horizontal lift satisfying $\ddot U(0) = -U\xi^\top\xi$. Differentiating $-\operatorname{tr}(U(t)^\top CU(t))/2$ twice proves (19).

Choose principal vectors $x_i \in [U_\star]$ and $y_i \in [U_\star]^\perp$ so the columns of a representative of U are

$$u_i = x_i\cos\theta_i + y_i\sin\theta_i. \tag{22}$$

Terms with $\theta_i = 0$ need no complementary vector; when the complement has dimension smaller than r, only the nonzero-angle pairs are used. The x_i form an orthonormal basis for $[U_\star]$ and the relevant y_i are orthonormal. Since this leading subspace is invariant under C, $x_i^\top Cy_j = 0$. Put $a_i = x_i^\top Cx_i$, $b_i = y_i^\top Cy_i$, and $c_i = a_i - b_i$. Then

$$h_C(U) = \frac{1}{2}\sum_i c_i\sin^2\theta_i, \qquad \Delta \le c_i \le \Lambda. \tag{23}$$

This is an exact excess formula in principal coordinates. The coefficients c_i are Rayleigh quotients, not generally individual eigenvalue differences. The bounds on c_i prove the first part of (20); $2x/\pi \le \sin x \le x$ on $[0,\pi/2]$ proves the second.

For every nonzero angle set $e_i = -x_i\sin\theta_i + y_i\cos\theta_i$. The matrix tangent directions which place e_i in column i and zero elsewhere are orthonormal and horizontal. Direct differentiation gives their gradient components $c_i\sin\theta_i\cos\theta_i$. Bessel's inequality therefore gives

$$\|\operatorname{grad}\Phi(U)\|^2 \ge \sum_i c_i^2\sin^2\theta_i\cos^2\theta_i \tag{24}$$

$$\ge \Delta\cos^2 R_0\sum_i c_i\sin^2\theta_i = 2\Delta\cos^2 R_0\, h_C(U). \tag{25}$$

This proves PL without a curvature-comparison shortcut. It is consistent with prior analyses of the block Rayleigh quotient (Alimisis & Vandereycken, 2024); the angular factor is not claimed as a new standalone manifold theorem.

At $U_\star$, diagonalize C in its invariant subspaces. A tangent coordinate moving selected eigenvector i toward unselected eigenvector j has Hessian eigenvalue $\lambda_i - \lambda_j$. For a critical point, $(I-P)CU = 0$, so its selected subspace is invariant. If it is not leading, some selected eigenvector has eigenvalue at most λ_{r+1} and some unselected one has eigenvalue at least λ_r. Rotating the former toward the latter gives Hessian curvature at most $-\Delta$. Finally, for $r = 1, d = 2$,

$$h(\theta) = \frac{\Delta}{2}\sin^2\theta, \qquad h''(\theta) = \Delta\cos(2\theta). \tag{26}$$

Thus the Hessian is negative for $\pi/4 < \theta < \pi/2$ even though quadratic growth and the positive PL bound still hold. □

The distinction in the last sentence is essential for tracking and stability. PL controls objective decrease along one trajectory. Contraction of the distance between *two* trajectories requires additional hypotheses; it does not follow from PL alone. Nor does (21) prove that the scalar $\Delta \cos^2 R_0$ predicts the ranking of complete neural-network optimizers. Gradient noise, covariance normalization, parameter-state conditioning, and computational cost remain independent factors.

A.4. Coordinate transport and the limits of diagonal Adam

Lemma 3 (First moments and diagonal second moments)**.** *Let $U_+ = UQ$ for $Q \in O(r)$, representing a pure basis change. Then $Z_+ = ZQ$ and exact first-moment consistency requires $M_+ = MQ$. For a row-wise full second moment $B_i = \mathbb{E}[z_i^\top z_i]$, the exact transformation is $B_{i,+} = Q^\top B_i Q$. If only its diagonal v_i is stored, the diagonal-covariance approximation is*

$$v_{i,+} = v_i Q^{\odot 2}, \tag{27}$$

and its error is precisely

$$\operatorname{diag}(Q^\top B_i Q) - v_i Q^{\odot 2} = \operatorname{diag}\{Q^\top (B_i - \operatorname{Diag} v_i) Q\}. \tag{28}$$

The error need not be second order in the basis rotation. Furthermore, coordinatewise Adam is not invariant under general Q, even with an exactly transformed diagonal.

Proof. $G(UQ) = (GU)Q$ gives the first claim and $(MQ)(UQ)^\top = MU^\top$ verifies the lifted first moment. For second moments, expand $(z_iQ)^\top (z_iQ)$ and take expectation. Its diagonal under the approximation $B_i = \operatorname{Diag} v_i$ is $\sum_k v_{ik} Q_{kj}^2$, proving (27) and (28). For $Q = I + \epsilon A + O(\epsilon^2)$, the omitted off-diagonal covariance can contribute a linear term through $\operatorname{diag}(B_iA - AB_i)$. Finally, entrywise division by a diagonal square root does not commute with orthogonal mixing. For example, the zero-history, negligible-ε Adam direction for $z = (2, 1)$ is $(1, 1)$. A generic rotation followed by coordinatewise normalization and rotation back does not reproduce that vector. Signed coordinate permutations are a special invariant case. □

For consecutive, genuinely different subspaces we use $R = \operatorname{Polar}(U_+^\top U)$ and align $M_+ = MR^\top$. This minimizes basis mismatch, but cannot restore the part of a historical ambient moment orthogonal to the new span. If an ambient moment is A, then

$$\|AP - AP_+\| \le \|A\| \|P - P_+\|_{\mathrm{op}} = \|A\| \sin \theta_{\max}(U, U_+). \tag{29}$$

This is a first-order perturbation bound, not an exact transport identity. The covariance error from omitted correlations must be added separately.

The archived implementation used $(\sqrt{v}\, Q)^{\odot 2}$ rather than (27). These differ even for diagonal covariance: with $v = (1, 1)$ and a 45-degree rotation the former gives $(2, 0)$ up to column order, while the correct diagonal is $(1, 1)$. New experiments use (27); historical outcomes retain their original algorithm label. Resetting v to zero is another distinct algorithm. A reset is a deterministic state transformation, not a generic additive statistical error, and its effect on descent must be analyzed through the resulting parameter direction.

A.5. Stochastic online tracking with an explicit perturbation budget

The landscape result concerns a fixed covariance. During training both its eigenvectors and eigenvalues change, and the tracked basis is driven by noisy mini-batches. We first establish a finite-horizon bound on objective excess. This avoids silently replacing PL by a two-trajectory contraction property.

Assumption 1 (Bounded, separated tracking interval)**.** *On the interval considered, $C_t \succeq 0$, $\|C_t\|_{\mathrm{op}} \le C_0$, and $\lambda_r(C_t) - \lambda_{r+1}(C_t) \ge \Delta_0 > 0$. Every update remains in the region $\theta_{\max}(U_t, U_t^\star) \le R_0 < \pi/2$. Its descent representation is*

$$U_{t+1} = \operatorname{Retr}_{U_t}\{-\rho_t(g_t + \zeta_t + b_t)\}, \qquad g_t = \operatorname{grad} \Phi_t(U_t), \tag{30}$$

where $\mathbb{E}_t \zeta_t = 0$, $\mathbb{E}_t \|\zeta_t\|^2 \le \sigma_t^2$, and $\mathbb{E}_t \|b_t\|^2 \le B_t^2$. The retraction pullback satisfies

$$\Phi_t(\operatorname{Retr}_U \xi) \le \Phi_t(U) + \langle \operatorname{grad} \Phi_t(U), \xi \rangle + \frac{L_R}{2} \|\xi\|^2 \tag{31}$$

for all actual steps, and $\|C_{t+1} - C_t\| \le v_t$ almost surely, with deterministic v_t.

Proposition 1 verifies (31) with the sharp, dimension-free choice $L_R = \sup_t[\lambda_1(C_t) - \lambda_d(C_t)] \le C_0$. No small-angle assumption is required for this pullback inequality itself; the principal-angle restriction is used only by the PL inequality.

The bias b_t records deviations from a fresh stochastic tangent-gradient step. It includes the Grassmann momentum lag, deterministic approximation errors, and any clipping bias. It is permitted to correlate with ζ_t. This is useful for the implemented momentum, whose error cannot be treated as independent zero-mean noise.

Lemma 4 (Ky Fan completion of squares for moving objective excess)**.** *Let C, C' be symmetric, let C have rank-r gap at least $\Delta_0 > 0$, and put $v = \|C' - C\|_F$. No gap assumption on C' is needed. For every rank-r projector P and every $a > 0$,*

$$h_{C'}(P) \le (1+a)h_C(P) + \frac{1+a^{-1}}{4\Delta_0}v^2. \tag{32}$$

For fixed a, the coefficient $(1+a^{-1})/4$ is optimal to second order.

Proof. Let $P_\star, P'_\star$ maximize the respective Ky Fan traces (Fan, 1951), $E = C' - C$, and $D = P'_\star - P_\star$. Then

$$h_{C'}(P) - h_C(P) = \tfrac{1}{2}\operatorname{tr}\{E(P_\star - P)\} + \tfrac{1}{2}\operatorname{tr}(C'D), \tag{33}$$

$$\tfrac{1}{2}\operatorname{tr}(C'D) = -h_C(P'_\star) + \tfrac{1}{2}\langle E, D\rangle \tag{34}$$

$$\le -\tfrac{1}{4}\Delta_0\|D\|_F^2 + \tfrac{1}{2}v\|D\|_F \le \frac{v^2}{4\Delta_0}. \tag{35}$$

The last line completes a square, retaining the curvature term which a plain perturbation-norm estimate discards. Also,

$$\tfrac{1}{2}|\langle E, P_\star - P\rangle| \le v\sqrt{h_C(P)/\Delta_0} \le ah_C(P) + \frac{v^2}{4a\Delta_0}. \tag{36}$$

For sharpness take $r = 1, d = 2, C = \gamma I + \operatorname{Diag}(\Delta_0, 0)$, $\gamma > 0$, $E_e = \left(\begin{smallmatrix} 0 & e \\ e & 0 \end{smallmatrix}\right)$, and $P_e = u_e u_e^\top$ with $u_e = (\cos\theta_e, \sin\theta_e)$, $\theta_e = -e/(a\Delta_0)$. Direct diagonalization gives

$$\lim_{e\to 0} \frac{\Delta_0\{h_{C+E_e}(P_e) - (1+a)h_C(P_e)\}}{\|E_e\|_F^2} = \frac{1+a^{-1}}{4}. \tag{37}$$

□

Theorem 2 (Tracking with unrounded contraction and perturbation constants)**.** *Under Assumption 1, let $\mu = \Delta_0 \cos^2 R_0$, $\alpha_t = \rho_t\mu$, $L_R\rho_t \le 1/4$, and $\alpha_t \le 1/2$. Define*

$$q_t = (1 + \alpha_t/2)(1 - \alpha_t), \tag{38}$$

$$A_t = 2(1+\alpha_t/2)(L_R\rho_t^2\sigma_t^2 + \rho_t B_t^2) + \left(\frac{1}{2\alpha_t} + \frac{1}{4}\right)\frac{v_t^2}{\Delta_0}. \tag{39}$$

For $h_t = h_{C_t}(U_t)$,

$$\mathbb{E}h_{t+1} \le q_t\mathbb{E}h_t + A_t, \tag{40}$$

$$\mathbb{E}h_T \le h_0\prod_{j<T} q_j + \sum_{t<T} A_t \prod_{j=t+1}^{T-1} q_j. \tag{41}$$

For constant bounds and step ρ, with $\alpha = \rho\mu$ and $\chi_\alpha = (1+\alpha/2)/(1+\alpha)$,

$$\limsup_T \mathbb{E}h_T \le \chi_\alpha\left[\frac{4L_R\rho\sigma^2}{\mu} + \frac{4B^2}{\mu} + \frac{v^2}{\rho^2\mu^2\Delta_0}\right]. \tag{42}$$

Moreover $\mathbb{E}d_{\mathrm{Gr}}(U_T, U_T^\star)^2 \le \pi^2\mathbb{E}h_T/(2\Delta_0)$.

Proof. For $h_t^+ = h_{C_t}(U_{t+1})$, sharp QR pullback smoothness, $\mathbb{E}_t\zeta_t = 0$, and Young's inequality give

$$\mathbb{E}_t h_t^+ \le h_t - \rho_t\|g_t\|^2 - \rho_t\langle g_t, \mathbb{E}_t b_t\rangle + \tfrac{1}{2}L_R\rho_t^2\mathbb{E}_t\|g_t+\zeta_t+b_t\|^2 \tag{43}$$

$$\le h_t - \tfrac{1}{2}\rho_t\|g_t\|^2 + 2L_R\rho_t^2\sigma_t^2 + 2\rho_t B_t^2 \tag{44}$$

$$\le (1-\alpha_t)h_t + 2L_R\rho_t^2\sigma_t^2 + 2\rho_t B_t^2. \tag{45}$$

Here $\rho_t|\langle g_t, \mathbb{E}_t b_t\rangle| \le \rho_t\|g_t\|^2/4 + \rho_t B_t^2$ and $\mathbb{E}_t\|g_t+\zeta_t+b_t\|^2 \le 2\|g_t\|^2 + 4\sigma_t^2 + 4B_t^2$; the step condition gives the second line. Apply Lemma 4 pathwise at U_{t+1} with $a = \alpha_t/2$ and keep its constants to obtain the stated recursion. Unroll it and use $1-q = \alpha(1+\alpha)/2$ for the steady bound. The angle bound follows from quadratic growth and $\sin\theta \ge 2\theta/\pi$. Only the atomic smoothness and excess-drift constants are claimed sharp; the stochastic bias budget is sufficient. □

The noise term increases with ρ, whereas the drift term decreases with ρ^2. For the noise-and-drift terms, dropping $\chi_\alpha \le 1$ gives the envelope $A\rho + D/\rho^2$, whose minimizer is $\rho = (2D/A)^{1/3}$, subject to the admissible-step restrictions. This is an explicit tracking trade-off, not a claim that larger drift always increases GrassRoute's advantage over SVD. If covariance changes are driven by the parameter update and $\|C(W') - C(W)\| \le L_C\|W' - W\|$, then

$$v_t \le L_C\eta_t\|D_t\|. \tag{46}$$

With bounded directions, a sufficient scaling for the drift contribution in (42) to be $O(T^{-1/2})$ is $\rho = O(T^{-1/2})$ and $v = O(T^{-3/4})$. The weaker assertion $v = o(T^{-1/2})$ only implies a vanishing drift floor, without that rate. Different schedules can trade parameter speed for tracking accuracy.

Momentum and retraction have separate costs. For the implemented update, $H_t = \mu_H\Pi_{U_t}H_{t-1} + (1-\mu_H)\widehat{Q}_t$. Writing $\widehat{Q}_t = Q_t + \epsilon_t$ and subtracting Q_t gives

$$H_t - Q_t = \mu_H\Pi_{U_t}(H_{t-1} - Q_{t-1}) + \mu_H(\Pi_{U_t}Q_{t-1} - Q_t) + (1-\mu_H)\epsilon_t. \tag{47}$$

Minkowski's inequality and nonexpansiveness imply an L_2 lag bound $e_t \le \mu_H e_{t-1} + \mu_H D_t + (1-\mu_H)\sigma_t$, where D_t bounds the transported population-direction change. In a stationary region with $D_t \le D$, this gives $\limsup e_t \le \mu_H D/(1-\mu_H) + \sigma$. The familiar variance reduction factor $(1-\mu_H)/(1+\mu_H)$ holds for an EMA of independent noise at a *frozen* basis; it cannot be assumed for moving, data-dependent bases. Equation (47) states exactly which additional terms must be controlled.

If QR is compared with an exponential step of the *same* tangent direction, Lemma 2 gives a local $O(\rho_t^3\|H_t\|^3)$ subspace difference. When QR is analyzed directly through (31), that error is already accounted for in the pullback constant; adding it again as independent noise would double count. Neither the curvature upper bound nor positive curvature alone supplies a universal signed correction to the tracking recursion.

Continuous motion and jumps. Davis–Kahan also gives $s(U_{t+1}^\star, U_t^\star) \le \sqrt{2}v_t/\Delta_0$. A small covariance increment therefore bounds target motion. The converse does not hold, since eigenvalues can change without rotating the target. Theorem 2 uses v_t as a conservative observable control. An abrupt change that violates the angle neighborhood invalidates the local guarantee; it need not cause divergence, and no deterministic ordering between online tracking and periodic recomputation follows. Boundary experiments should test the effect of jump size while holding the spectrum and gradient noise fixed.

A.6. Explicit differential calculus, graph coordinates, and discrete contraction

Throughout this subsection $C = C^\top$, $\Delta > 0$, and $\|\cdot\| = \|\cdot\|_F$.

Proposition 1 (Two-plane QR calculus; sharp pullback constant)**.** *Let $C = C^\top$, $\Lambda_C = \lambda_1(C) - \lambda_d(C)$, and $U^\top\xi = 0$. For $P = UU^\top$,*

$$T_P\mathrm{Gr} = \{J = J^\top : PJ + JP = J\}, \tag{48}$$

$$J = \xi U^\top + U\xi^\top, \qquad \|J\|_F^2 = 2\|\xi\|_F^2, \tag{49}$$

$$\mathcal{P}_P(A) = [P,[P,A]], \qquad \dot{P} = [P,[P,C]]. \tag{50}$$

The QR pullback $F_U(s) = \Phi(\mathrm{qf}(U + s\xi))$ obeys, for all real s,

$$|F_U''(s)| \le \Lambda_C \|\xi\|_F^2, \tag{51}$$

$$\Phi(\mathrm{qf}(U+\xi)) \le \Phi(U) + \langle \mathrm{grad}\,\Phi(U), \xi\rangle + \tfrac{1}{2}\Lambda_C\|\xi\|_F^2. \tag{52}$$

The constant Λ_C is optimal uniformly over U, ξ.

Proof. The projector identities follow from $D(P^2 - P)[J] = PJ + JP - J$. For the quantitative bound use the singular-vector/CS decomposition (Golub & Van Loan, 2013, Chapters 2 and 8). After a right orthogonal change, write $\xi_i = \sigma_i v_i$, with (u_i, v_i) orthonormal for each $\sigma_i > 0$. Its QR span has the orthonormal polar representative

$$q_i(s) = \frac{u_i + s\sigma_i v_i}{\sqrt{1 + s^2\sigma_i^2}}. \tag{53}$$

Zero singular values contribute constant columns. Put $a_i = u_i^\top C u_i$, $b_i = u_i^\top C v_i$, $c_i = v_i^\top C v_i$. Poincaré separation for the 2×2 compression of C gives

$$\sqrt{(a_i - c_i)^2 + 4b_i^2} \le \Lambda_C. \tag{54}$$

The trace is a sum of scalar Rayleigh quotients:

$$F_U(s) = -\frac{1}{2}\sum_i \frac{a_i + 2s\sigma_i b_i + s^2\sigma_i^2 c_i}{1 + s^2\sigma_i^2}, \tag{55}$$

$$F_{U,i}''(s) = \sigma_i^2 \frac{(a_i - c_i)(1 - 3z_i^2) + 2b_i z_i (3 - z_i^2)}{(1 + z_i^2)^3}, \qquad z_i = s\sigma_i. \tag{56}$$

The cancellation

$$(1 - 3z^2)^2 + z^2(3 - z^2)^2 = (1 + z^2)^3 \tag{57}$$

yields $|F_{U,i}''(s)| \le \Lambda_C \sigma_i^2/(1 + z_i^2)^{3/2} \le \Lambda_C \sigma_i^2$. Sum and integrate the second derivative. At $s = 0$, take $r = 1$, u a lowest eigenvector and ξ a unit highest eigenvector; $F_U''(0) = -\Lambda_C$. Interchanging them gives $+\Lambda_C$. Thus both the absolute and upper pullback constants are attained. □

Lemma 5 (Stored QR frames and aligned moment transport)**.** *Let $U^\top \xi = 0$, let $Q(s)R(s) = U + s\xi$ be the thin QR decomposition with positive diagonal in $R(s)$, and put $U_+ = Q(1)$, $O = \mathrm{Polar}(U_+^\top U)$. Then*

$$\sigma_{\min}(U + s\xi) \ge 1, \qquad \|U_+ - U\| \le \sqrt{3}\|\xi\|, \tag{58}$$

$$\|O - I\| \le 2\sqrt{3}\|\xi\|, \qquad \|U_+ O - U\|_{\mathrm{op}} = 2\sin(\theta_{\max}/2) \le d_{\mathrm{Gr}}(U_+, U) \le \|\xi\|. \tag{59}$$

The row-vector transport $\mathcal{T}(XU^\top) = XO^\top U_+^\top$ is an isometry on the old routed space and satisfies

$$\|\mathcal{T}(XU^\top) - XU^\top\| \le \|X\|\,\|\xi\|. \tag{60}$$

On a segment of full-rank matrices with $\sigma_{\min} \ge s_0 > 0$,

$$\|\mathrm{qf}(A) - \mathrm{qf}(B)\| \le \frac{\sqrt{3}}{s_0}\|A - B\|. \tag{61}$$

Proof. Differentiate $A = QR$ and set $K = Q^\top \dot{A} R^{-1}$:

$$\dot{Q} = (I - QQ^\top)\dot{A}R^{-1} + Q\Omega, \tag{62}$$

$$\Omega = Q^\top \dot{Q} = -\Omega^\top, \qquad \mathrm{strictlower}(\Omega) = \mathrm{strictlower}(K), \tag{63}$$

$$\|\Omega\|^2 = 2\|\,\mathrm{strictlower}(K)\|^2 \le 2\|K\|^2, \tag{64}$$

$$\|\dot{Q}\|^2 \le 3\|\dot{A}R^{-1}\|^2 \le \frac{3\|\dot{A}\|^2}{\sigma_{\min}(A)^2}. \tag{65}$$

Integrate along $U + s\xi$ or the stipulated segment. For $B = U_+^\top U$,

$$\| \operatorname{Polar}(B) - I\| \le \frac{2\|B - I\|}{\sigma_{\min}(B) + 1} \le 2\|U_+ - U\|. \tag{66}$$

This polar perturbation inequality follows from the Sylvester equation for the polar differential, or its integrated two-point form. Singular-value decomposition of $U_+^\top U$ gives the aligned-frame singular values $2\sin(\theta_i/2)$; Lemma 2 gives $\theta_i = \arctan \sigma_i(\xi)$. Isometry and (60) follow by right multiplication. □

Proposition 2 (An exact graph flow and contraction of the implemented QR step)**.** *Let $E = (E_1, E_2)$ diagonalize C, with blocks $\Lambda_1 = \operatorname{Diag}(\lambda_1, \dots, \lambda_r)$ and $\Lambda_2 = \operatorname{Diag}(\lambda_{r+1}, \dots, \lambda_d)$. For $K \in \mathbb{R}^{(d-r)\times r}$ define*

$$S_K = I + K^\top K, \qquad B_K = S_K^{-1/2}, \tag{67}$$

$$U(K) = (E_1 + E_2 K)B_K, \qquad P(K) = U(K)U(K)^\top. \tag{68}$$

Then

$$\tan\theta_i = \sigma_i(K), \qquad h_C(K) = \tfrac{1}{2}\operatorname{tr}\left[(\Lambda_1 K^\top K - K^\top \Lambda_2 K)S_K^{-1}\right]. \tag{69}$$

The exact continuous Grassmann ascent flow and its solution are

$$\dot K = \Lambda_2 K - K\Lambda_1 =: \mathcal{L}K, \tag{70}$$

$$K(t) = e^{t\Lambda_2}K(0)e^{-t\Lambda_1}, \qquad \|K(t)\| \le e^{-\Delta t}\|K(0)\|. \tag{71}$$

For the discrete update $U_+ = \operatorname{qf}(U + \rho(I - P)CU)$, set

$$A = U^\top CU, \quad T = B_K A B_K^{-1}, \quad J = I + \rho(\Lambda_1 - T), \quad L_J = \|\Lambda_1 - T\|_{\mathrm{op}}. \tag{72}$$

Whenever $\rho L_J < 1$,

$$K_+ = K + \rho\mathcal{L}KJ^{-1}, \tag{73}$$

$$\|K_+ - (K + \rho\mathcal{L}K)\| \le \frac{\rho^2 \Lambda L_J}{1 - \rho L_J}\|K\|. \tag{74}$$

Consequently, if $\rho\Lambda \le 1$, $\rho L_J \le 1/2$, and $\rho \le \Delta/(4\Lambda L_J)$, then

$$\|K_+\| \le (1 - \rho\Delta/2)\|K\|. \tag{75}$$

For $\|K\|_{\mathrm{op}} \le k_0$ one may take $L_J \le C_0(1 + \sqrt{1 + k_0^2})$ after a fixed scalar shift with $\|C\|_{\mathrm{op}} \le C_0$.

Proof.

$$E_1^\top U(K) = B_K, \qquad E_2^\top U(K) = KB_K, \tag{76}$$

$$\operatorname{tr}(U^\top CU) = \operatorname{tr}[(\Lambda_1 + K^\top \Lambda_2 K)S_K^{-1}]. \tag{77}$$

Subtract from $\operatorname{tr}(\Lambda_1)$ to obtain (69). The solution of $\dot X = CX$ is $X(t) = e^{tC}X(0)$; its span satisfies (50). Taking the lower-to-upper coordinate ratio gives (71). Entrywise,

$$K_{ji}(t) = e^{-(\lambda_i - \lambda_{r+j})t}K_{ji}(0), \qquad \frac{d}{dt}\|K\|^2 = -2\sum_{ji}(\lambda_i - \lambda_{r+j})K_{ji}^2 \le -2\Delta\|K\|^2. \tag{78}$$

For the discrete step, QR does not change the span, and

$$E_1^\top(U + \rho Q) = JB_K, \tag{79}$$

$$E_2^\top(U + \rho Q) = (KJ + \rho\mathcal{L}K)B_K. \tag{80}$$

Division yields (73). The Neumann bound gives $\|J^{-1} - I\|_{\mathrm{op}} \le \rho L_J/(1 - \rho L_J)$. Also

$$\|\mathcal{L}K\| \le \Lambda\|K\|, \tag{81}$$

$$\|K + \rho\mathcal{L}K\|^2 = \sum_{ji}[1 - \rho(\lambda_i - \lambda_{r+j})]^2 K_{ji}^2 \le (1 - \rho\Delta)^2\|K\|^2. \tag{82}$$

Thus the additional remainder is at most $2\rho^2\Lambda L_J\|K\| \le \rho\Delta\|K\|/2$. Finally $\|B_K\|_{\mathrm{op}} \le 1$ and $\|B_K^{-1}\|_{\mathrm{op}} \le \sqrt{1 + k_0^2}$. □

A.7. Covariance normalization, eigenspace derivatives, and probabilistic tracking

Lemma 6 (Normalization mismatch and the Sylvester derivative)**.** *Let $N = \|G\|^2$, $\bar{N} = \mathbb{E}N$, $S = \mathbb{E}[G^\top G]$, $\widetilde{S} = S/(\bar{N} + \varepsilon_c)$, and $C = \mathbb{E}[G^\top G/(N + \varepsilon_c)]$. Then*

$$C - \widetilde{S} = \mathbb{E}\frac{(\bar{N} - N)G^\top G}{(N + \varepsilon_c)(\bar{N} + \varepsilon_c)}, \tag{83}$$

$$\|C - \widetilde{S}\| \le \frac{\mathbb{E}|N - \bar{N}|}{\bar{N} + \varepsilon_c} \le \frac{\sqrt{\operatorname{Var} N}}{\bar{N} + \varepsilon_c}. \tag{84}$$

For a differentiable symmetric path $C(s)$ with gap $\Delta(s) > 0$, a horizontal leading-eigenspace derivative satisfies

$$\dot{U}_\star = U_\perp K, \qquad K\Lambda_1 - \Lambda_2 K = U_\perp^\top \dot{C} U_\star, \tag{85}$$

$$K_{ji} = \frac{u_{r+j}^\top \dot{C} u_i}{\lambda_i - \lambda_{r+j}}, \qquad \|\dot{U}_\star\| \le \frac{\|U_\perp^\top \dot{C} U_\star\|}{\Delta}, \tag{86}$$

$$\|\dot{P}_\star\| \le \frac{\sqrt{2}\|\dot{C}\|}{\Delta}. \tag{87}$$

Proof. Use $\|G^\top G\| \le \|G\|^2 = N$ and $N/(N + \varepsilon_c) \le 1$. Differentiate $CU_\star = U_\star \Lambda_1$ and left multiply by $U_\perp^\top$:

$$U_\perp^\top \dot{C} U_\star + \Lambda_2 K = K\Lambda_1. \tag{88}$$

The entrywise solution and the projector tangent norm give the result. □

Corollary 1 (Finite normalization perturbation with a mixed gap)**.** *If $\widetilde{S}$ has gap $\delta > 0$ and $e = \|C - \widetilde{S}\|_{\mathrm{op}} < \delta$, then*

$$\|\sin\Theta(U_C, U_S)\|_F \le \frac{\min\{\|C - \widetilde{S}\|_F, \sqrt{\min(r, d - r)}e\}}{\delta - e} \tag{89}$$

$$\le \frac{\sqrt{\operatorname{Var} N}}{(\delta - e)(\bar{N} + \varepsilon_c)}. \tag{90}$$

Proof. Combine the normalized-covariance mismatch with the mixed-gap Davis–Kahan bound of Lemma 14. □

Theorem 3 (High-probability frozen-target tracking and almost-sure consistency)**.** *Suppose $C_t = C$, $\theta_{\max}(U_t, U_\star) \le R_0 < \pi/2$, $\mu = \Delta \cos^2 R_0$, $g_t = \operatorname{grad}\Phi(U_t)$, and*

$$U_{t+1} = \mathrm{qf}\{U_t - \rho(g_t + \zeta_t + b_t)\}, \quad \mathbb{E}_t\zeta_t = 0, \quad \|\zeta_t\| \le \sigma, \quad \|b_t\| \le B, \quad \|g_t\| \le \Gamma. \tag{91}$$

All three tangent terms are horizontal at U_t. Let $L_R = \Lambda_C = \lambda_1(C) - \lambda_d(C)$, $L_R\rho \le 1/4$, $\mu\rho \le 1$, and $q = 1 - \mu\rho$. For any fixed T and $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$h_T \le q^T h_0 + \frac{2L_R\rho^2\sigma^2 + 2\rho B^2}{1 - q}(1 - q^T) + \rho\Gamma\sigma\sqrt{\frac{2(1 - q^{2T})\log(1/\delta)}{1 - q^2}}. \tag{92}$$

If instead ρ_t varies, $\sum_t \rho_t = \infty$, $\sum_t \rho_t^2\sigma_t^2 < \infty$, $\sum_t \rho_t B_t^2 < \infty$, and the same neighborhood and step conditions hold almost surely, then

$$h_t \longrightarrow 0 \quad a.s., \qquad d_{\mathrm{Gr}}(U_t, U_\star) \longrightarrow 0 \quad a.s. \tag{93}$$

Proof. The pathwise pullback inequality and Young's inequality give

$$h_{t+1} \le h_t - \rho\|g_t\|^2 - \rho\langle g_t, b_t\rangle - \rho\langle g_t, \zeta_t\rangle + \tfrac{1}{2}L_R\rho^2\|g_t + b_t + \zeta_t\|^2 \tag{94}$$

$$\le h_t - \tfrac{1}{2}\rho\|g_t\|^2 + 2L_R\rho^2\sigma^2 + 2\rho B^2 + \mathfrak{m}_t \tag{95}$$

$$\le qh_t + a + \mathfrak{m}_t, \quad a = 2L_R\rho^2\sigma^2 + 2\rho B^2, \quad \mathfrak{m}_t = -\rho\langle g_t, \zeta_t\rangle. \tag{96}$$

Now $\mathbb{E}_t \mathfrak{m}_t = 0$, $|\mathfrak{m}_t| \le \rho\Gamma\sigma$, and iteration yields

$$h_T \le q^T h_0 + a \sum_{k=0}^{T-1} q^k + \sum_{t=0}^{T-1} q^{T-1-t} \mathfrak{m}_t. \tag{97}$$

Conditional Hoeffding bounds imply, for every $u > 0$,

$$\mathbb{E} \exp\left(u \sum_t q^{T-1-t} \mathfrak{m}_t \right) \le \exp\left(\frac{u^2 \rho^2 \Gamma^2 \sigma^2}{2} \sum_t q^{2(T-1-t)} \right). \tag{98}$$

Chernoff minimization proves (92). For varying steps, the conditional expectation recurrence is

$$\mathbb{E}_t h_{t+1} \le h_t - \mu \rho_t h_t + a_t, \qquad \sum_t a_t < \infty. \tag{99}$$

The Robbins–Siegmund almost-supermartingale theorem (Robbins & Siegmund, 1971) gives convergence of h_t and $\sum_t \rho_t h_t < \infty$ almost surely. A positive limiting h_t would contradict $\sum_t \rho_t = \infty$. Quadratic growth then proves (93). The neighborhood is a stated hypothesis; this argument does not silently assert invariance of that neighborhood. □

A.8. Accumulated diagonal-covariance error and tracking-momentum estimation

The reference covariance below is a full $r \times r$ *transported history*; it is not the covariance of gradients discarded before projection. For a fixed row, let $z_t \in \mathbb{R}^r$ be its routed gradient as a column, let $B_0 = 0$, $v_0 = 0$, and define

$$\widetilde{B}_t = \beta_2 B_{t-1} + (1-\beta_2) z_t z_t^\top, \qquad B_t = R_t \widetilde{B}_t R_t^\top, \tag{100}$$

$$\widetilde{v}_t = \beta_2 v_{t-1} + (1-\beta_2) z_t^{\odot 2}, \qquad v_t = R_t^{\odot 2} \widetilde{v}_t, \tag{101}$$

$$e_t^v = \operatorname{diag}(B_t) - v_t, \qquad O_t = \widetilde{B}_t - \operatorname{Diag}(\operatorname{diag} \widetilde{B}_t). \tag{102}$$

Here R_t is exactly the new-coordinate rotation; the column-vector form $v_t = R_t^{\odot 2} \widetilde{v}_t$ equals the row-wise update in the algorithm.

Proposition 3 (Exact covariance-error recursion and rotation accumulation)**.** *With* $\delta_t = \|R_t - I\|_{\mathrm{op}}$,

$$e_t^v = \beta_2 R_t^{\odot 2} e_{t-1}^v + \operatorname{diag}(R_t O_t R_t^\top), \tag{103}$$

$$\|e_t^v\|_2 \le \beta_2 \|e_{t-1}^v\|_2 + 2\delta_t \|O_t\|_F, \tag{104}$$

$$\|e_T^v\|_2 \le 2 \sum_{t=1}^{T} \beta_2^{T-t} \delta_t \|O_t\|_F. \tag{105}$$

Thus if $\delta_t \le \delta$ *and* $\|O_t\|_F \le B_O$,

$$\frac{\|e_T^v\|_2}{1-\beta_2^T} \le \frac{2\delta B_O}{1-\beta_2}. \tag{106}$$

The trace is nevertheless exact:

$$\mathbf{1}^\top e_t^v = 0, \qquad \mathbf{1}^\top v_t = \operatorname{tr}(B_t). \tag{107}$$

Proof. Separate the diagonal and off-diagonal parts before rotation:

$$\operatorname{diag} \widetilde{B}_t - \widetilde{v}_t = \beta_2 (\operatorname{diag} B_{t-1} - v_{t-1}), \tag{108}$$

$$\operatorname{diag}(R_t \widetilde{B}_t R_t^\top) = R_t^{\odot 2} \operatorname{diag} \widetilde{B}_t + \operatorname{diag}(R_t O_t R_t^\top). \tag{109}$$

Subtract the implemented v_t. Since $R_t^{\odot 2}$ is doubly stochastic, $\|R_t^{\odot 2}\|_2 \le \sqrt{\|R_t^{\odot 2}\|_1 \|R_t^{\odot 2}\|_\infty} = 1$. Moreover $\operatorname{diag} O_t = 0$, and

$$R_t O_t R_t^\top - O_t = (R_t - I) O_t R_t^\top + O_t (R_t^\top - I), \tag{110}$$

$$\|\operatorname{diag}(R_t O_t R_t^\top)\|_2 \le \|R_t O_t R_t^\top - O_t\|_F \le 2\delta_t \|O_t\|_F. \tag{111}$$

Unroll the recursion and sum the geometric series. Orthogonal similarity preserves the full covariance trace, and double stochasticity preserves the sum of the diagonal approximation. Both have identical EMA trace injections $(1-\beta_2)\|z_t\|^2$, proving the trace identities. □

Corollary 2 (Coordinatewise adaptive-direction error)**.** *Let $\widehat{b}_t = \mathrm{diag}(B_t)/(1-\beta_2^t)$ and $\widehat{v}_t = v_t/(1-\beta_2^t)$. For a common first-moment row m,*

$$\left\| m \oslash (\sqrt{\widehat{b}_t} + \varepsilon_a) - m \oslash (\sqrt{\widehat{v}_t} + \varepsilon_a) \right\|_2^2 \le \|m\|_\infty^2 \varepsilon_a^{-4} \|\widehat{b}_t - \widehat{v}_t\|_1 \tag{112}$$

$$\le \|m\|_\infty^2 \varepsilon_a^{-4} \sqrt{r}\, \frac{\|e_t^v\|_2}{1-\beta_2^t}. \tag{113}$$

For QR tracking with $\|H_t\| \le H_0$ one may use $\delta_t \le 2\sqrt{3}\rho_t H_0$ in (105). The resulting comparison budget can scale as $\rho/(1-\beta_2)$; it need not vanish under every simultaneous choice of ρ and β_2. Theorem 5 analyzes the implemented diagonal states directly and does not assume this comparison error vanishes.

Proof. For each coordinate use $|1/(\sqrt{x}+\varepsilon_a) - 1/(\sqrt{y}+\varepsilon_a)|^2 \le \varepsilon_a^{-4}|x-y|$, then sum and apply $\|e\|_1 \le \sqrt{r}\|e\|_2$. □

Proposition 4 (A first-order rotation error with exact coefficient)**.** *For*

$$B = \begin{pmatrix} a & c \\ c & b \end{pmatrix} \succeq 0, \quad R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \tag{114}$$

the diagonal-covariance discrepancy is exactly

$$\mathrm{diag}(RBR^\top) - R^{\odot 2}\,\mathrm{diag}\,B = \begin{pmatrix} -2c\sin\theta\cos\theta \\ 2c\sin\theta\cos\theta \end{pmatrix}. \tag{115}$$

Consequently its norm is $\sqrt{2}|c\sin(2\theta)|$ and its derivative norm at $\theta = 0$ is $2\sqrt{2}|c|$. Small rotations alone do not make omitted-covariance error quadratic.

Proof. Multiply the three 2×2 matrices and subtract their diagonal terms. □

Proposition 5 (Moving-basis EMA error with explicit drift and noise)**.** *Let $q_t = (I - P_t)C_tU_t$ and $\widehat{q}_t = q_t + \zeta_t$, with $\mathbb{E}_t\zeta_t = 0$ and $\mathbb{E}_t\|\zeta_t\|^2 \le \sigma_t^2$. For*

$$H_t = \mu\Pi_{U_t}H_{t-1} + (1-\mu)\widehat{q}_t, \quad a_t = H_t - q_t, \quad d_t = \Pi_{U_t}q_{t-1} - q_t, \quad 0 < \mu < 1, \tag{116}$$

one has the exact recursion and conditional bound

$$a_t = \mu\Pi_{U_t}a_{t-1} + \mu d_t + (1-\mu)\zeta_t, \tag{117}$$

$$\mathbb{E}_t\|a_t\|^2 \le (1+\gamma)\mu^2\|a_{t-1}\|^2 + (1+\gamma^{-1})\mu^2\|d_t\|^2 + (1-\mu)^2\sigma_t^2 \qquad (\gamma > 0). \tag{118}$$

If $\|d_t\| \le D$ and $\sigma_t \le \sigma$, put $q_\mu = (1+\mu^2)/2$. Then

$$\mathbb{E}\|a_T\|^2 \le q_\mu^T\mathbb{E}\|a_0\|^2 + \left\{ \frac{2\mu^2(1+\mu^2)}{(1-\mu^2)^2}D^2 + \frac{2(1-\mu)}{1+\mu}\sigma^2 \right\}(1-q_\mu^T). \tag{119}$$

If $D = 0$, the sharper noise-only steady bound is $(1-\mu)\sigma^2/(1+\mu)$.

Proof. Subtract q_t and insert $\Pi_{U_t}q_{t-1}$. Everything except ζ_t is $\mathcal{F}_t$-measurable, so the cross term with current noise has zero conditional expectation. Apply $\|x+y\|^2 \le (1+\gamma)\|x\|^2 + (1+\gamma^{-1})\|y\|^2$ and $\|\Pi_{U_t}a_{t-1}\| \le \|a_{t-1}\|$. Choose $\gamma = (1-\mu^2)/(2\mu^2)$:

$$(1+\gamma)\mu^2 = q_\mu, \tag{120}$$

$$(1+\gamma^{-1})\mu^2 = \frac{\mu^2(1+\mu^2)}{1-\mu^2}, \tag{121}$$

$$1 - q_\mu = (1-\mu^2)/2. \tag{122}$$

Summation gives (119). If $D = 0$, omit Young's inequality: the contraction coefficient is μ^2 exactly. □

Corollary 3 (A closed drift budget from parameter and basis motion)**.** *Suppose* $C_t = C(W_t)$, $\|C(W)\|_{\mathrm{op}} \le C_0$, *and* $\|C(W) - C(W')\| \le L_C\|W - W'\|$. *Then*

$$\|d_t\| \le \|C_t - C_{t-1}\| + 2C_0\|U_t - U_{t-1}\| \tag{123}$$

$$\le L_C\eta_{t-1}D_0 + 2\sqrt{3}C_0\rho_{t-1}H_0. \tag{124}$$

If each sampled gradient is L_g*-Lipschitz in* W*, then the normalized population covariance admits* $L_C = 2L_g/\sqrt{\varepsilon_c}$. *For* $\epsilon = 1 - \mu \in (0, 1)$, *the steady upper bound in* (119) *is at most*

$$\frac{4D^2}{\epsilon^2} + 2\epsilon\sigma^2. \tag{125}$$

If $(4D^2/\sigma^2)^{1/3} < 1$, *this latter expression is minimized at* $\epsilon = (4D^2/\sigma^2)^{1/3}$, *with value* $3\,4^{1/3}D^{2/3}\sigma^{4/3}$.

Proof. Since $q_{t-1} = C_{t-1}U_{t-1} - U_{t-1}A_{t-1}$ with $\|A_{t-1}\|_{\mathrm{op}} \le C_0$,

$$d_t = (I - P_t)\{C_{t-1}U_{t-1} - C_tU_t - U_{t-1}A_{t-1}\}, \tag{126}$$

$$\|(I - P_t)U_{t-1}A_{t-1}\| \le C_0\|U_{t-1} - U_t\|. \tag{127}$$

The other difference contributes $\|C_t - C_{t-1}\| + C_0\|U_t - U_{t-1}\|$. The normalized-covariance Lipschitz constant follows by taking expectation in (330). Finally,

$$\frac{2\mu^2(1+\mu^2)}{(1-\mu^2)^2} \le \frac{4}{(1-\mu)^2}, \qquad \frac{2(1-\mu)}{1+\mu} \le 2(1-\mu), \tag{128}$$

and elementary differentiation proves the optimization claim. These are estimation-error budgets for the tracker momentum; they are not a replacement for the neighborhood assumption in the tracking theorem. □

A.9. Parameter convergence: retaining the adaptive-direction defect

Unlike the routing variable, the parameter is updated in Euclidean space. The appropriate smoothness inequality is always valid for an Adam direction; what is not automatic is a positive conditional alignment between that direction and the population gradient. We make this distinction explicit. Let f be the objective under analysis, either empirical risk on a fixed training set or population risk. Those objectives must not be interchanged when claiming unbiasedness.

Assumption 2 (Smooth objective and a predictable comparison metric)**.** f *is* L*-smooth and bounded below by* $f_\star$. $g_t = \nabla f(W_t)$ *and* $\mathbb{E}_t G_t = g_t$, $\mathbb{E}_t\|G_t - g_t\|^2 \le \sigma_G^2$. *An* $\mathcal{F}_t$*-measurable self-adjoint linear map* $\mathcal{B}_t$ *on matrix parameters satisfies*

$$a\|X\|^2 \le \langle X, \mathcal{B}_t X\rangle, \qquad \|\mathcal{B}_t X\| \le b\|X\|, \qquad 0 < a \le b < \infty. \tag{129}$$

The implemented parameter direction is decomposed exactly *as*

$$D_t = \mathcal{B}_t G_t + e_t, \qquad \mathbb{E}_t\|e_t\|^2 \le E_t^2. \tag{130}$$

The learning rates are deterministic, positive, and $\eta_t \le a/(4Lb^2)$.

This assumption isolates a measurable error budget rather than presuming that Adam is a fresh gradient step. For example, row i can use the predictable operator

$$\mathcal{B}_{t,i} = U_tA_{t,i}U_t^\top + \gamma_t(I - P_t), \quad aI_r \preceq A_{t,i} \preceq bI_r, \quad a \le \gamma_t \le b. \tag{131}$$

Here $A_{t,i}$ may be a clipped inverse square root of the *previous* second moment. It is not the denominator formed from the current batch. The actual direction minus this comparison direction defines e_t; this difference includes moment lag, same-batch denominator dependence, diagonal-transport approximation, and sampled residual normalization. With weight decay, its direction is included in e_t when f is unregularized; alternatively one can formulate the objective and comparison update for the regularized problem. A decoupled decay term must not simply disappear from the proof.

Theorem 4 (Nonconvex convergence with explicit adaptive defects)**.** *Under Assumption 2, let $S_T = \sum_{t<T} \eta_t$ and draw $\tau \in \{0, \dots, T-1\}$ with $\Pr(\tau = t) = \eta_t / S_T$. For $W_{t+1} = W_t - \eta_t D_t$,*

$$\mathbb{E}\|\nabla f(W_\tau)\|^2 \le \frac{4(f(W_0) - f_\star)}{aS_T} + \frac{2}{a^2 S_T} \sum_{t<T} \eta_t \mathbb{E} E_t^2 + \frac{4L}{aS_T} \sum_{t<T} \eta_t^2 (b^2 \sigma_G^2 + \mathbb{E} E_t^2). \tag{132}$$

For $\eta_t = \eta_0/\sqrt{T}$, the noise and initial-suboptimality terms are $O(T^{-1/2})$. The same stationarity rate follows if $T^{-1} \sum_{t<T} \mathbb{E} E_t^2 = O(T^{-1/2})$. For an arbitrary actual Adam run the displayed defect terms remain in the bound; unbiased raw gradients alone do not make them vanish.

Proof. Predictability is needed at precisely the following step:

$$\mathbb{E}_t \langle g_t, \mathcal{B}_t G_t \rangle = \langle g_t, \mathcal{B}_t g_t \rangle \ge a \|g_t\|^2. \tag{133}$$

Using Cauchy–Schwarz and Young on the remaining term gives

$$\mathbb{E}_t \langle g_t, D_t \rangle \ge a\|g_t\|^2 - \|g_t\| \sqrt{\mathbb{E}_t \|e_t\|^2} \tag{134}$$

$$\ge \tfrac{a}{2} \|g_t\|^2 - \tfrac{E_t^2}{2a}. \tag{135}$$

Furthermore,

$$\mathbb{E}_t \|D_t\|^2 \le 2b^2 (\|g_t\|^2 + \sigma_G^2) + 2E_t^2. \tag{136}$$

Smoothness applied to the actual update yields

$$\mathbb{E}_t f(W_{t+1}) \le f(W_t) - \eta_t \mathbb{E}_t \langle g_t, D_t \rangle + \tfrac{L}{2} \eta_t^2 \mathbb{E}_t \|D_t\|^2 \tag{137}$$

$$\le f(W_t) - \tfrac{a}{4} \eta_t \|g_t\|^2 + \tfrac{\eta_t E_t^2}{2a} + L\eta_t^2 (b^2 \sigma_G^2 + E_t^2). \tag{138}$$

The last line uses $L\eta_t b^2 \le a/4$. Taking expectation, summing, and using $f(W_T) \ge f_\star$ proves (132). Weighted output sampling gives its left side exactly. Substituting the constant-in-horizon step proves the rate statements. □

Corollary 4 (A deterministic guarantee for the implemented Adam branch)**.** *Consider full-gradient GrassRoute without weight decay, with zero initial first moments, Procrustes first-moment alignment, and fixed $\beta_1 \in [0, 1)$. Assume $\|\nabla f(W_t)\| \le G_0$, the bias-corrected routed inverse denominators have eigenvalues in $[a, b]$, and the actual residual gain is in the same interval, $a > 0$. Suppose $\|D_t\| \le D_0$, $\|H_t\| \le H_0$, and the horizon-dependent steps η, ρ are constant within the run and satisfy the small-step conditions above. Then the actual first-moment contribution to the direction defect satisfies*

$$E_t \le \frac{b\beta_1}{1 - \beta_1} (L D_0 \eta + 2 G_0 H_0 \rho). \tag{139}$$

Consequently $\eta, \rho = \Theta(T^{-1/2})$ give

$$\frac{1}{T} \sum_{t<T} \|\nabla f(W_t)\|^2 = O(T^{-1/2}). \tag{140}$$

The second moments may depend on the current full gradient: no predictability assumption is needed in this deterministic specialization. This is a guarantee for the actual Adam first-moment update under bounded gains, not a claim for arbitrary stochastic Adam.

Proof. Let $m_t = M_t U_t^\top$ denote the ambient first moment immediately after its EMA update and before transport. Norm preservation of Procrustes alignment and $\|G_t U_t\| \le G_0$ give $\|M_t\| \le (1 - \beta_1^t) G_0$ by induction. Write $\delta_{t-1} = d_{\mathrm{Gr}}(U_t, U_{t-1})$. The aligned frame mismatch has operator norm $2\sin(\theta_{\max}/2) \le \delta_{t-1}$, so the preceding moment's ambient transport error is at most $(1 - \beta_1^{t-1}) G_0 \delta_{t-1}$. Also

$$\|g_t P_t - g_{t-1} P_{t-1}\| \le L D_0 \eta + G_0 \delta_{t-1}. \tag{141}$$

For $r_t = \|m_t - (1-\beta_1^t) g_t P_t\|$, subtract the target from the EMA recursion and use these two inequalities:

$$r_t \le \beta_1 r_{t-1} + \beta_1(1-\beta_1^{t-1})(LD_0\eta + 2G_0\delta_{t-1}). \tag{142}$$

Lemma 2 gives $\delta_{t-1} \le \rho H_0$. Starting from $r_1 = 0$, summing the geometric recursion and dividing by $1-\beta_1^t$ bounds the bias-corrected ambient moment error by $\beta_1(LD_0\eta + 2G_0H_0\rho)/(1-\beta_1)$. Lifting is isometric in the routed coordinates; multiplication by the current inverse denominator has norm at most b. This proves (139). Choose the comparison metric using that current denominator and the actual residual gain. Pathwise positive definiteness gives the alignment step in the proof of Theorem 4, without a stochastic conditional expectation. The resulting bound with $\sigma_G = 0$ has terms $O((\eta T)^{-1}) + O((\eta+\rho)^2)$, proving the stated rate. □

The gain restrictions have concrete interpretations. A positive Adam denominator offset bounds inverse gains above; bounded corrected second moments bound them below. With normalized covariance samples, tangency projection is nonexpansive and $\|\widehat{Q}_t\| \le 1$, so zero-initialized Grassmann EMA has $H_0 \le 1$. A positive residual clipping lower bound ensures residual coercivity. If the implementation instead permits its gain to approach zero, that uniform lower-bound assumption has not been verified. The result therefore applies to its bounded-gain regime or a specified positive-floor configuration, not automatically to every run.

A rigorous specialization and a counterexample. The theorem applies without an adaptive defect to a predictable, bounded routed preconditioner with a positive residual branch. In deterministic training it also applies directly to a current-gradient positive definite preconditioner, using pathwise alignment instead of predictability and the corresponding elementary descent argument. These are genuine sufficient conditions. Standard Adam with exponential moments and a same-batch denominator is more delicate (Reddi et al., 2018; Chen et al., 2019). For a simple illustration, take a scalar unbiased gradient with $G = 10$ with probability 0.1 and $G = -1$ otherwise. Then $\mathbb{E}G = 0.1 > 0$, but a fresh zero-history Adam step with negligible ε has $\mathbb{E}\,\mathrm{sign}(G) = -0.8$. The expected direction is uphill for this positive population gradient. Unbiasedness of G therefore cannot justify the cancellation used above for a denominator depending on G. Momentum can also temporarily point uphill after a gradient sign change. These examples explain why (130) is retained for the implemented algorithm.

A.10. How tracking, moment transport, and rank tails enter convergence

The theorem can be connected to the routing analysis without conflating their covariances. Let $\mathcal{B}_t^\star$ be an ideal predictable metric constructed in a chosen oracle subspace, with the same positive bounds as (129), and write

$$D_t = \mathcal{B}_t^\star G_t + (\mathcal{B}_t - \mathcal{B}_t^\star)G_t + e_t^{\mathrm{state}}. \tag{143}$$

If, after consistent coordinate alignment,

$$\|\mathcal{B}_t - \mathcal{B}_t^\star\|_{\mathrm{op}} \le K_P \|P_t - P_t^\star\|_{\mathrm{op}} + d_t^{\mathrm{metric}}, \tag{144}$$

and $\mathbb{E}_t\|G_t\|^2 \le G_0^2$, then

$$\mathbb{E}_t\|e_t\|^2 \le 3G_0^2K_P^2 \sin^2\theta_{\max,t} + 3G_0^2(d_t^{\mathrm{metric}})^2 + 3\mathbb{E}_t\|e_t^{\mathrm{state}}\|^2. \tag{145}$$

The projector perturbation term is geometric; the metric term is separate because different coordinatewise diagonal preconditioners can act differently in the *same* subspace. Equation (144) is an extra operator-consistency condition, not a consequence of matching projectors. If its oracle is the leading eigenspace of the normalized covariance, Theorem 2 controls the first term. If the oracle instead uses S_t, normalization mismatch adds another projector perturbation, which can be bounded by Davis–Kahan only when the corresponding raw covariance gap is available.

Proposition 6 (Residual preservation removes a mandatory rank-tail floor)**.** *For the metric* (131) *with* $\gamma_t \ge a > 0$, (132) *contains no obligatory* $\overline{\tau_r}$ *term. Even rank zero reduces to a scaled full-gradient method and can converge. If the residual is deliberately removed, the missing population-gradient energy can be bounded by*

$$\|g_t(I-P_t)\|^2 \le \mathbb{E}_t\|G_t(I-P_t)\|^2 = \tau_r(S_t) + 2h_{S_t}(U_t). \tag{146}$$

Consequently a rank-tail floor is a valid upper-bound contribution for a projection-only *analysis, not an impossibility theorem for full GrassRoute.*

Proof. Orthogonality of the two branches gives

$$\langle X, \mathcal{B}_t X\rangle = \sum_i \langle X_i U_t, A_{t,i} X_i U_t\rangle + \gamma_t \|X(I-P_t)\|^2 \geq a\|X\|^2. \tag{147}$$

The comparison direction is therefore coercive on the full parameter space, regardless of r and its covariance tail. Jensen's inequality for the projected gradient, followed by Lemma 1, proves (146) when the residual is omitted. □

For a scalar routed gain a_t and residual gain γ_t, there is an even simpler identity:

$$D_t = a_t G_t + (\gamma_t - a_t) G_t (I - P_t), \qquad \mathbb{E}_t \|D_t - a_t G_t\|^2 = (\gamma_t - a_t)^2 \{\tau_r(S_t) + 2h_{S_t}(U_t)\}. \tag{148}$$

Here a heavy tail increases the discrepancy from an isotropic reference only when the branch gains differ. Large residual energy can hurt an optimization mechanism empirically, but cannot by itself establish an irreducible stationarity error for a method that retains it.

Bounding moment-induced defects. The scalar map $a \mapsto 1/(a+\varepsilon)$ on $a \geq 0$ is ε^{-2}-Lipschitz. Working with the root-amplitude state $A = \sqrt{V}$ therefore avoids an unjustified derivative bound for $V \mapsto V^{-1/2}$ near zero. For bounded $\|M\| \leq M_0$,

$$\left\| \frac{M}{A+\varepsilon} - \frac{M'}{A'+\varepsilon} \right\| \leq \varepsilon^{-1}\|M - M'\| + M_0 \varepsilon^{-2} \|A - A'\|. \tag{149}$$

Bias correction contributes its explicit finite factors $(1-\beta_1^t)^{-1}$ and $(1-\beta_2^t)^{-1/2}$. For a fixed-coordinate momentum EMA,

$$M_t = (1-\beta_1) \sum_{k=0}^{t-1} \beta_1^k Z_{t-k}, \tag{150}$$

and thus

$$\|M_t - (1-\beta_1^t) Z_t\| \leq (1-\beta_1) \sum_{k<t} \beta_1^k \|Z_{t-k} - Z_t\|. \tag{151}$$

If consecutive aligned coordinates change by at most D_Z, the right side is at most $\beta_1 D_Z/(1-\beta_1)$. Moving coordinates additionally contribute the first-moment projection error and second-moment covariance error in Lemma 3. Combining these bounds with (149) gives explicit state contributions to E_t once their magnitudes are controlled. It does not establish that those magnitudes vanish in arbitrary neural training. Resetting a denominator may increase its reciprocal toward $1/\varepsilon$ and amplify precisely these terms.

Proposition 7 (Gradient norm matching does not balance branch steps)**.** *Consider prescribed nonzero gradient histories $G_1, \ldots, G_t$ and $cG_1, \ldots, cG_t$, $c > 0$, with identical initial basis and zero moments. In the zero-offset idealization of normalized covariance routing, diagonal moment transport, and residual norm matching, the two basis histories agree, their first and second moments scale by c and c^2, respectively, and*

$$D_{t,c}^{\parallel} = D_t^{\parallel}, \qquad D_{t,c}^{\perp} = c D_t^{\perp}. \tag{152}$$

The statement applies to coordinates with nonzero second moments, and a zero residual remains zero. For a fixed branch learning-rate ratio, the routed-to-residual update norm ratio consequently scales as $1/c$. With positive offsets this is an approximate statement in the regime where the corresponding scaled norms dominate the offsets.

Proof. Normalized covariance samples are invariant under positive scaling of G, so the tangent directions, subspace momentum, QR bases, and Procrustes rotations are the same. Induction in the moment updates gives $M_{t,c} = cM_t$ and $V_{t,c} = c^2 V_t$; diagonal covariance transport is homogeneous and preserves these relations. Bias correction has identical factors. Thus $c\widehat{M}_t/\sqrt{c^2 \widehat{V}_t} = \widehat{M}_t/\sqrt{\widehat{V}_t}$ coordinatewise. The norm ratio $\|G\|/\|G^{\perp}\|$, including its fixed clipping limits, is scale invariant, whereas its product with $G^{\perp}$ scales by c. This proves (152). □

For an unclipped, negligible-offset residual rule, $\|D_t^{\perp}\| \simeq \|G_t\|$, while the routed Adam direction can remain approximately scale invariant. A small gradient therefore does not make the two branch *update* norms comparable. This provides an

additional finite-budget failure mechanism beyond subspace tracking: residual directions can be updated too slowly relative to the adaptive branch even while their gradient is preserved. In particular, identical normalized eigengaps, captured-energy fractions, and principal angles do not determine the entire parameter dynamics. This observation does not establish a nonzero asymptotic error floor, nor identify the sole cause of any benchmark outcome; measuring actual branch step norms would be needed for that causal claim. It is consistent with the importance of recovery scaling studied in related optimizers (Chen et al., 2025; Rajabi et al., 2025).

Iteration and time complexity. When the theorem's defect condition holds, a target $\mathbb{E}\|\nabla f\|^2 \le \epsilon$ requires $O(\epsilon^{-2})$ iterations for the stated stochastic schedule, with constants depending on a, b, L, σ_G. This is a stationarity statement, not global optimality or a test-accuracy guarantee. Routing costs $O(mdr)$ for GU, $G^\top(GU)$, and lifting, plus $O(dr^2 + r^3)$ for QR and the small Procrustes polar factor. No $d \times d$ covariance need be formed by the optimizer. Persistent routed tensors M, V, U, H use exactly $2mr + 2dr$ scalar entries, before fallback states. Temporary full gradients, activations, and parameter memory are additional. A step-complexity advantage must therefore be multiplied by actual per-step time before making a wall-clock claim.

A.11. Averaging before relaxation: a quadratic-root stationarity bound

Proposition 8 (Descent lemma with Hilbert-space Minkowski)**.** *Let $f \ge f_\star$ be L-smooth, $\mathbb{E}_t G_t = g_t = \nabla f(W_t)$, and let the predictable self-adjoint metric $\mathcal{B}_t$ satisfy $aI \preceq \mathcal{B}_t \preceq bI$. For $D_t = \mathcal{B}_t G_t + e_t$, $W_{t+1} = W_t - \eta D_t$, define*

$$z_T^2 = T^{-1} \sum_{t<T} \mathbb{E}\|g_t\|^2, \qquad e_T^2 = T^{-1} \sum_{t<T} \mathbb{E}\|e_t\|^2, \tag{153}$$

$$s_T^2 = T^{-1} \sum_{t<T} \mathbb{E}\|\mathcal{B}_t(G_t - g_t)\|^2, \qquad A_\eta = a - L\eta b^2/2. \tag{154}$$

If $0 < \eta < 2a/(Lb^2)$, put

$$C_T = \frac{f(W_0) - f_\star}{\eta T} + \frac{L\eta}{2}(s_T + e_T)^2. \tag{155}$$

Then

$$z_T \le \frac{e_T(1 + L\eta b) + \sqrt{e_T^2(1 + L\eta b)^2 + 4A_\eta C_T}}{2A_\eta}. \tag{156}$$

The variance s_T^2 is the actual metric-noise energy, not necessarily the worst-case replacement $b^2\sigma_G^2$.

Proof. The descent lemma, conditional predictability, and Cauchy–Schwarz on the product space of time and randomness give

$$a z_T^2 \le \frac{f(W_0) - f_\star}{\eta T} + z_T e_T + \frac{L\eta}{2T} \sum_t \mathbb{E}\|D_t\|^2. \tag{157}$$

The cross term between $\mathcal{B}_t g_t$ and $\mathcal{B}_t(G_t - g_t)$ has zero mean. Minkowski therefore yields

$$\left(T^{-1} \sum_t \mathbb{E}\|D_t\|^2 \right)^{1/2} \le \sqrt{b^2 z_T^2 + s_T^2} + e_T, \tag{158}$$

$$T^{-1} \sum_t \mathbb{E}\|D_t\|^2 \le b^2 z_T^2 + 2b e_T z_T + (s_T + e_T)^2. \tag{159}$$

In the second line, only the defect cross term uses $\sqrt{b^2 z_T^2 + s_T^2} \le b z_T + s_T$; no artificial cross term between metric-gradient energy and metric-noise energy is introduced. Rearrange to $A_\eta z_T^2 - e_T(1 + L\eta b) z_T - C_T \le 0$ and take its positive root. □

Corollary 5 (A sharp quadratic reference)**.** *If $e_T = 0$, (156) reduces to*

$$z_T^2 \le \frac{f(W_0) - f_\star}{\eta T(a - L\eta b^2/2)} + \frac{L\eta s_T^2}{2(a - L\eta b^2/2)}. \tag{160}$$

For $f(w) = Lw^2/2$, $\mathcal{B}_t = aI$, and additive martingale noise of constant variance σ^2, the stability threshold is exactly $\eta a L < 2$, and the stationary gradient variance is

$$\lim_t \mathbb{E}|\nabla f(w_t)|^2 = \frac{L\eta a\sigma^2}{2 - \eta a L}. \tag{161}$$

This equals the noise term above, since $s_T^2 = a^2\sigma^2$. The step-size threshold and noise coefficient cannot be uniformly improved for this reference class.

Proof. Here $w_{t+1} = (1 - \eta a L)w_t - \eta a \xi_t$. The AR(1) Yule–Walker equation (Brockwell & Davis, 1991) gives $\mathrm{Var}(w) = \eta^2 a^2 \sigma^2 / [1 - (1 - \eta a L)^2]$. The descent lemma is an equality for this quadratic. □

A.12. A closed stochastic bound for the actual transported Adam recursion

We now analyze the current-batch denominator itself. This result uses the additional noise and schedule hypotheses below; it is not a consequence of unbiasedness alone. Indices in this subsection start at $t = 1$.

Assumption 3 (Actual-recursion specialization)**.** *There is one routed matrix $W \in \mathbb{R}^{m\times d}$, no weight decay, and $f \geq f_\star$ is L-smooth on the ambient parameter space. With $g_t = \nabla f(W_t)$,*

$$G_t = g_t + \xi_t, \qquad \mathbb{E}_t \xi_t = 0, \qquad \|G_t\| \leq G_0 \quad \text{a.s.} \tag{162}$$

Conditional on $\mathcal{F}_t$, either $\xi_t = 0$, or its vectorization has an integrable density $p_t(x) = \varphi_t(\|x\|)$ with φ_t nonincreasing. The stored basis U_t is $\mathcal{F}_t$-measurable. The two branch learning rates equal $\eta > 0$, and $\rho > 0$ is constant within the horizon. Let

$$0 \leq \beta_1 < 1, \quad 0 < \beta_2 < 1, \quad \varepsilon_a, \varepsilon_r, \varepsilon_c > 0, \quad s_{\max} \geq 1. \tag{163}$$

All moment states start at zero, QR uses a fixed positive-diagonal convention, and the Procrustes factor is computed exactly. The residual gain is $s_t = \mathrm{clip}(\|G_t\|/(\|G_t(I - P_t)\| + \varepsilon_r); 0, s_{\max})$. The tracker is precisely the normalized-covariance momentum tracker in Table 2, with $0 \leq \mu_H < 1$.

Write M_t^- and V_t^- for the preceding states after alignment to U_t, and M_t, V_t for the fresh moments before the next alignment:

$$Z_t = G_t U_t, \qquad M_t = \beta_1 M_t^- + (1 - \beta_1) Z_t, \tag{164}$$

$$V_t = \beta_2 V_t^- + (1 - \beta_2) Z_t^{\odot 2}, \qquad \widehat{M}_t = M_t/(1 - \beta_1^t), \quad \widehat{V}_t = V_t/(1 - \beta_2^t), \tag{165}$$

$$R_t = \mathrm{Polar}(U_{t+1}^\top U_t), \qquad M_{t+1}^- = M_t R_t^\top, \quad V_{t+1}^- = V_t (R_t^\top)^{\odot 2}. \tag{166}$$

For row-wise coordinate multiplication, define the lifted operator

$$[\mathcal{A}(U, a)X]_{i,:} = (X_{i,:}U)\,\mathrm{Diag}(a_{i,:})U^\top, \quad a_{t,ij} = (\sqrt{\widehat{V}_{t,ij}} + \varepsilon_a)^{-1}. \tag{167}$$

It has support $P = UU^\top$ and is self-adjoint, although it is not strictly positive on the full ambient space.

Lemma 7 (Uniform state bounds without an eigengap)**.** *Under Assumption 3, for every t,*

$$\|M_t\| \leq (1 - \beta_1^t)G_0, \qquad \|V_t\|_1 \leq (1 - \beta_2^t)G_0^2, \tag{168}$$

$$\|\widehat{M}_t\| \leq G_0, \qquad \|\sqrt{\widehat{V}_t}\| \leq G_0, \tag{169}$$

$$a \leq a_{t,ij} \leq b, \qquad a = (G_0 + \varepsilon_a)^{-1}, \quad b = \varepsilon_a^{-1}, \tag{170}$$

$$\|Q_t\| \leq 1, \qquad \|H_t\| \leq 1, \tag{171}$$

$$d_{\mathrm{Gr}}(U_{t+1}, U_t) \leq \rho, \qquad \|U_{t+1} - U_t\| \leq \sqrt{3}\rho, \quad \|R_t - I\| \leq 2\sqrt{3}\rho. \tag{172}$$

Proof. Orthogonality gives $\|MR^\top\| = \|M\|$. Nonnegativity and double stochasticity of $R^{\odot 2}$ give

$$\|V(R^\top)^{\odot 2}\|_1 = \sum_{ijk} V_{ik} R_{jk}^2 = \sum_{ik} V_{ik} = \|V\|_1, \tag{173}$$

$$\|M_t\| \leq \beta_1 \|M_{t-1}\| + (1 - \beta_1)G_0, \tag{174}$$

$$\|V_t\|_1 \leq \beta_2 \|V_{t-1}\|_1 + (1 - \beta_2)G_0^2. \tag{175}$$

Induction and bias correction prove the first three lines. Also

$$\widehat{C}_t = G_t^\top G_t/(\|G_t\|^2 + \varepsilon_c), \tag{176}$$

$$\|Q_t\| = \|(I - P_t)\widehat{C}_t U_t\| \le \|\widehat{C}_t\| \le \mathrm{tr}(\widehat{C}_t) \le 1, \tag{177}$$

$$\|H_t\| \le \mu_H \|H_{t-1}\| + (1-\mu_H)\|Q_t\| \le 1. \tag{178}$$

The last line follows from Lemmas 2 and 5. None of these inequalities invokes $\lambda_r(C_t) - \lambda_{r+1}(C_t)$. □

Lemma 8 (A predictable denominator and its explicit current-sample defect)**.** *Define*

$$c_t = \frac{1-\beta_2}{1-\beta_2^t}, \qquad c_0 = 0, \tag{179}$$

$$\overline{V}_t = \frac{\beta_2 V_t^- + (1-\beta_2)\mathbb{E}_t Z_t^{\odot 2}}{1-\beta_2^t}, \qquad \overline{a}_t = (\sqrt{\overline{V}_t} + \varepsilon_a)^{\odot(-1)}, \tag{180}$$

$$\overline{\mathcal{A}}_t = \mathcal{A}(U_t, \overline{a}_t), \qquad m_t = \widehat{M}_t U_t^\top. \tag{181}$$

Then $\overline{\mathcal{A}}_t$ is predictable, its routed eigenvalues lie in $[a, b]$, and the actual adaptive direction satisfies

$$D_t^{\|} U_t^\top = \overline{\mathcal{A}}_t m_t + e_t, \tag{182}$$

$$\mathbb{E}_t\|e_t\|^2 \le 2G_0^4 \varepsilon_a^{-4} c_t. \tag{183}$$

For $\mathsf{H}_T = \sum_{t=1}^T t^{-1}$,

$$\frac{1}{T}\sum_{t=1}^T c_t \le (1-\beta_2) + \frac{\mathsf{H}_T}{T}. \tag{184}$$

Proof. For nonnegative scalars x, y,

$$(\sqrt{x} - \sqrt{y})^2 \le |x-y|, \qquad \left|\frac{1}{\sqrt{x}+\varepsilon_a} - \frac{1}{\sqrt{y}+\varepsilon_a}\right| \le \varepsilon_a^{-2}|\sqrt{x} - \sqrt{y}|. \tag{185}$$

Consequently,

$$\mathbb{E}_t\|\sqrt{\widehat{V}_t} - \sqrt{\overline{V}_t}\|^2 \le \mathbb{E}_t\|\widehat{V}_t - \overline{V}_t\|_1 \tag{186}$$

$$= c_t \mathbb{E}_t\|Z_t^{\odot 2} - \mathbb{E}_t Z_t^{\odot 2}\|_1 \le 2G_0^2 c_t, \tag{187}$$

$$\|e_t\|^2 \le \|\widehat{M}_t\|^2\|a_t - \overline{a}_t\|^2 \le G_0^2 \varepsilon_a^{-4}\|\sqrt{\widehat{V}_t} - \sqrt{\overline{V}_t}\|^2. \tag{188}$$

Let $u = 1-\beta_2$. Since $(1-u)^{-t} \ge 1 + tu$,

$$1 - \beta_2^t \ge \frac{tu}{1+tu}, \qquad c_t \le u + \frac{1}{t}. \tag{189}$$

Summation proves (184). The pointwise bounds on $\overline{V}_t$ follow by the same induction as for $\widehat{V}_t$. □

Lemma 9 (Variation of the transported predictable metric)**.** *Let $\mathcal{T}_t$ denote the isometric transport from U_{t-1} to U_t. For $t \ge 2$,*

$$\|\overline{\mathcal{A}}_t \mathcal{T}_t - \overline{\mathcal{A}}_{t-1}\|_{\mathrm{op,old}} \le K_\rho \rho + K_c(\sqrt{c_t} + \sqrt{c_{t-1}}), \tag{190}$$

where the domain is the old routed space and

$$K_\rho = (2\sqrt{3}+1)b + 2\sqrt{3}G_0\varepsilon_a^{-2}, \qquad K_c = 2G_0\varepsilon_a^{-2}. \tag{191}$$

Proof. For a nonnegative amplitude matrix A and an orthogonal R,

$$[\sqrt{A^{\odot 2}(R^\top)^{\odot 2}}]_{ij} = \|(A_{ik}R_{jk})_k\|_2, \tag{192}$$

$$\|\sqrt{A^{\odot 2}(R^\top)^{\odot 2}} - A\|^2 \le \sum_{ijk} A_{ik}^2(R_{jk}-\delta_{jk})^2 \le \|A\|^2\|R-I\|_{\mathrm{op}}^2. \tag{193}$$

For $\widehat{V}_{t-1}^{\mathrm{carried}} = \widehat{V}_{t-1}(R_{t-1}^\top)^{\odot 2}$,

$$\overline{V}_t = (1-c_t)\widehat{V}_{t-1}^{\mathrm{carried}} + c_t\mathbb{E}_t Z_t^{\odot 2}, \tag{194}$$

$$\|\sqrt{\overline{V}_t} - \sqrt{\widehat{V}_{t-1}^{\mathrm{carried}}}\| \le 2G_0\sqrt{c_t}, \tag{195}$$

$$\|\sqrt{\widehat{V}_{t-1}} - \sqrt{\overline{V}_{t-1}}\| \le 2G_0\sqrt{c_{t-1}}. \tag{196}$$

These last two bounds hold pathwise, using $\|Z^{\odot 2} - \mathbb{E}Z^{\odot 2}\|_1 \le 2G_0^2$. The triangle inequality gives

$$\|\sqrt{\overline{V}_t} - \sqrt{\overline{V}_{t-1}}\| \le 2\sqrt{3}G_0\rho + 2G_0(\sqrt{c_t}+\sqrt{c_{t-1}}). \tag{197}$$

The coordinate arrays here use the stored QR gauge. Lifting their inverse amplitudes into ambient operators gives

$$\|\overline{\mathcal{A}}_t - \overline{\mathcal{A}}_{t-1}\|_{\mathrm{op}} \le 2b\|U_t - U_{t-1}\| + \varepsilon_a^{-2}\|\sqrt{\overline{V}_t} - \sqrt{\overline{V}_{t-1}}\|. \tag{198}$$

Finally $\|\overline{\mathcal{A}}_t(\mathcal{T}_t - I)\|_{\mathrm{op,old}} \le b\rho$ by (60). □

Lemma 10 (Exact virtual-iterate cancellation of first-moment history)**.** *Set*

$$b_t^{(1)} = \frac{\beta_1(1-\beta_1^{t-1})}{1-\beta_1^t}, \qquad q_t = \frac{b_t^{(1)}}{1-b_t^{(1)}} = \frac{\beta_1(1-\beta_1^{t-1})}{1-\beta_1}, \tag{199}$$

$$q_0 = 0, \qquad q = \frac{\beta_1}{1-\beta_1}. \tag{200}$$

Then $m_t = b_t^{(1)}\mathcal{T}_t m_{t-1} + (1-b_t^{(1)})G_tP_t$. *For* $Y_1 = W_1$ *and* $Y_t = W_t - \eta q_{t-1}\overline{\mathcal{A}}_{t-1}m_{t-1}$,

$$Y_{t+1} - Y_t = -\eta(K_t + F_t), \tag{201}$$

$$K_t = \overline{\mathcal{A}}_t(G_tP_t) + s_tG_t(I-P_t), \tag{202}$$

$$F_t = e_t - J_t, \tag{203}$$

$$J_t = (q_{t-1}\overline{\mathcal{A}}_{t-1} - q_t\overline{\mathcal{A}}_t\mathcal{T}_t)m_{t-1}, \quad J_1 = 0, \tag{204}$$

$$\|W_t - Y_t\| \le \eta q b G_0. \tag{205}$$

Moreover, with $\overline{F}_T^2 = T^{-1}\sum_{t=1}^T \mathbb{E}\|F_t\|^2$,

$$\overline{F}_T^2 \le \frac{8G_0^2b^2}{T(1-\beta_1^2)} + 8G_0^2q^2K_\rho^2\rho^2 + (4G_0^4\varepsilon_a^{-4} + 16G_0^2q^2K_c^2)\left(1-\beta_2+\frac{\mathsf{H}_T}{T}\right). \tag{206}$$

Proof. Bias correction in (166) gives the displayed recursion for m_t. Since $(1+q_t)(1-b_t^{(1)}) = 1$,

$$Y_{t+1} = W_t - \eta(1+q_t)\overline{\mathcal{A}}_t m_t - \eta s_tG_t(I-P_t) - \eta e_t \tag{207}$$

$$= W_t - \eta\overline{\mathcal{A}}_tG_tP_t - \eta q_t\overline{\mathcal{A}}_t\mathcal{T}_t m_{t-1} - \eta s_tG_t(I-P_t) - \eta e_t. \tag{208}$$

Subtract Y_t. For $t \geq 2$, $q_t - q_{t-1} = \beta_1^{t-1}$, so

$$\|J_t\| \leq G_0\{b\beta_1^{t-1} + qK_\rho\rho + qK_c\sqrt{c_t} + qK_c\sqrt{c_{t-1}}\}, \tag{209}$$

$$\mathbb{E}\|F_t\|^2 \leq 2\mathbb{E}\|e_t\|^2 + 2\mathbb{E}\|J_t\|^2 \tag{210}$$

$$\leq 4G_0^4\varepsilon_a^{-4}c_t + 8G_0^2\{b^2\beta_1^{2(t-1)} + q^2K_\rho^2\rho^2 + q^2K_c^2(c_t + c_{t-1})\}. \tag{211}$$

The same upper bound is valid for $t = 1$ because $J_1 = 0$. Summing the geometric term and using (184) proves (206). □

Lemma 11 (Residual coercivity under radial unimodal noise)**.** *Under Assumption 3,*

$$\langle g_t(I - P_t), \mathbb{E}_t[s_tG_t(I - P_t)]\rangle \geq \|g_t(I - P_t)\|^2 - \varepsilon_r\|g_t(I - P_t)\| \tag{212}$$

$$\geq \tfrac{1}{2}\|g_t(I - P_t)\|^2 - \tfrac{1}{2}\varepsilon_r^2. \tag{213}$$

Thus, with $a_0 = \min(a, 1/2)$ *and* $B_0 = \max(b, s_{\max})$,

$$\langle g_t, \mathbb{E}_tK_t\rangle \geq a_0\|g_t\|^2 - \varepsilon_r^2/2, \qquad \|K_t\| \leq B_0G_0. \tag{214}$$

Proof. Condition additionally on the parallel component of ξ_t. In the residual space its conditional density remains radial and nonincreasing, and has mean zero. Denote the residual population gradient by v, the residual sample by $x = v + \zeta$, and the parallel sample norm by $a_\parallel$. For every nonnegative radial function h, reflection $\mathcal{R}$ across $v^\perp$ gives

$$\langle v, \mathbb{E}[h(\|x\|)x]\rangle = \int_{\langle v,x\rangle \geq 0} h(\|x\|)\langle v, x\rangle[p(x - v) - p(\mathcal{R}x - v)]\,dx \geq 0, \tag{215}$$

$$\|x - v\|^2 - \|\mathcal{R}x - v\|^2 = -4\langle v, x\rangle \leq 0. \tag{216}$$

Since $s_{\max} \geq 1$,

$$h(r) = \mathrm{clip}\left(\frac{\sqrt{a_\parallel^2 + r^2}}{r + \varepsilon_r}; 0, s_{\max}\right) - \frac{r}{r + \varepsilon_r} \geq 0. \tag{217}$$

Use (215) and then $\mathbb{E}x = v$:

$$\langle v, \mathbb{E}[sx]\rangle \geq \left\langle v, \mathbb{E}\left[x - \frac{\varepsilon_r x}{\|x\| + \varepsilon_r}\right]\right\rangle \geq \|v\|^2 - \varepsilon_r\|v\|. \tag{218}$$

The deterministic case follows pointwise. Young's inequality proves (213). Predictability of $\overline{\mathcal{A}}_t$ gives $\langle g_t, \mathbb{E}_t\overline{\mathcal{A}}_tG_tP_t\rangle \geq a\|g_tP_t\|^2$. The two branches have orthogonal row supports, which proves the norm bound and (214). □

Theorem 5 (Actual transported Adam: a quadratic-root bound)**.** *Under Assumption 3, define*

$$e_T = \left(T^{-1}\sum_{t=1}^{T}\mathbb{E}\|F_t\|^2\right)^{1/2}, \qquad K_0 = B_0G_0, \quad d_0 = L\eta qbG_0, \tag{219}$$

$$C_T^{\mathrm{virt}} = \frac{f(W_1) - f_\star}{\eta T} + \frac{\varepsilon_r^2}{2} + d_0(K_0 + e_T) + \frac{L\eta}{2}(K_0 + e_T)^2. \tag{220}$$

The implemented moments and current-batch denominator satisfy

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\|\nabla f(W_t)\|^2 \leq \left(\frac{e_T + \sqrt{e_T^2 + 4a_0C_T^{\mathrm{virt}}}}{2a_0}\right)^2. \tag{221}$$

One may use the closed budget (206) for e_T^2, while retaining a smaller rigorously established budget when available. For $T \geq 2$, $\eta = \eta_0T^{-1/2}$, $\rho = \rho_0T^{-1/2}$, $\beta_2 = 1 - T^{-1}$ and fixed β_1,

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\|\nabla f(W_t)\|^2 = O(T^{-1/2}) + O((\log T)/T) + O(\varepsilon_r^2). \tag{222}$$

These extra noise and schedule assumptions are not established properties of the benchmark runs. The bound closes the actual recursion but is not claimed optimal for all Adam trajectories.

Proof. Let $z_T^2 = T^{-1}\sum_t \mathbb{E}\|g_t\|^2$ and $u_t = \nabla f(Y_t)$. The virtual-iterate identity gives $Y_{t+1} = Y_t - \eta(K_t + F_t)$ and $\|u_t - g_t\| \le d_0$. Lemma 11 gives

$$\langle u_t, \mathbb{E}_t K_t\rangle \ge a_0\|g_t\|^2 - \varepsilon_r^2/2 - d_0 K_0. \tag{223}$$

Average the descent lemma *before* relaxing cross terms. Cauchy–Schwarz and Minkowski on the time–probability product space imply

$$T^{-1}\sum_t \mathbb{E}|\langle u_t, F_t\rangle| \le (z_T + d_0)e_T, \tag{224}$$

$$T^{-1}\sum_t \mathbb{E}\|K_t + F_t\|^2 \le (K_0 + e_T)^2. \tag{225}$$

Telescope $f(Y_t)$, use $Y_1 = W_1$, $f(Y_{T+1}) \ge f_\star$, and obtain

$$a_0 z_T^2 \le e_T z_T + C_T^{\mathrm{virt}}. \tag{226}$$

Its positive root is (221). Finally $e_T^2 = O(\rho^2) + O(1 - \beta_2 + (\log T)/T) + O(T^{-1})$. Using $z_T \le e_T/a_0 + \sqrt{C_T^{\mathrm{virt}}/a_0}$ gives the stated rate. This retains the averaged direction budget rather than inserting a per-step Young allocation or doubling both squared directions. □

Corollary 6 (Positive residual floor, fixed schedules, and the scope of the rate)**.** *If the residual rule in Assumption 3 is instead clipped to $[s_{\min}, s_{\max}]$ with $s_{\min} > 0$, the same radial-noise argument applied to $h(r) = s(r) - s_{\min}$ gives*

$$\langle g_t(I - P_t), \mathbb{E}_t[s_t G_t(I - P_t)]\rangle \ge s_{\min}\|g_t(I - P_t)\|^2. \tag{227}$$

Then Theorem 5 holds with $a_0 = \min(a, s_{\min})$ and the $\varepsilon_r^2/2$ term deleted from C_T^{virt}. For fixed $\beta_2 < 1$ and fixed $\rho > 0$, the explicit bound retains $O(1 - \beta_2) + O(\rho^2)$ terms. No claim that these are unavoidable lower bounds follows from this upper bound. For several routed matrices, a joint radial unimodality condition on the vectorized total gradient, block-diagonal $\overline{\mathcal{A}}_t$, and the sum of squared block residual offsets give the corresponding block result.

A.13. Compatibility of spectral tracking and parameter stationarity

Set $\mathcal{F}_t$ before sampling, $g_t = \mathbb{E}_t G_t$, and $\mathcal{C}_\epsilon(X) = X^\top X/(\|X\|^2 + \epsilon)$, $\epsilon > 0$. All derivatives below use the Frobenius metric. The symbol $\|\cdot\|_*$ denotes the nuclear norm.

Lemma 12 (Householder compression; sharp normalized-covariance envelope)**.** *For $s = \|X\|_F^2 + \epsilon$, $a = \langle X, E\rangle$,*

$$D\mathcal{C}_\epsilon(X)[E] = \frac{X^\top E + E^\top X}{s} - \frac{2aX^\top X}{s^2}, \tag{228}$$

$$D^2\mathcal{C}_\epsilon(X)[E,E] = \frac{2E^\top E}{s} - \frac{4a(X^\top E + E^\top X)}{s^2} - \frac{2\|E\|_F^2 X^\top X}{s^2} + \frac{8a^2 X^\top X}{s^3}. \tag{229}$$

Uniformly in X, E,

$$\|D\mathcal{C}_\epsilon(X)[E]\|_{\mathrm{op}} \le \epsilon^{-1/2}\|E\|_F, \tag{230}$$

$$\|D^2\mathcal{C}_\epsilon(X)[E,E]\|_{\mathrm{op}} \le \frac{2}{\epsilon}\|E\|_F^2. \tag{231}$$

The second constant is attained at $X = 0$ and a rank-one E.

Proof. The differential identities follow by the quotient rule. To avoid bounding its four terms separately, fix a unit $v \in \mathbb{R}^d$, augment $y = (\mathrm{vec}\, X, \sqrt{\epsilon})$, and define the orthogonal projector $B_v = \mathrm{Diag}(I_m \otimes vv^\top, 0)$. Then

$$f_v(X) = v^\top \mathcal{C}_\epsilon(X) v = \frac{y^\top B_v y}{y^\top y}. \tag{232}$$

For the Rayleigh quotient $R_B(y)$, the Householder involution $H_y = I - 2yy^\top/\|y\|^2$ gives the exact identity

$$\nabla_y^2 R_B(y) = \frac{2}{\|y\|^2} H_y(B - R_B(y)I)H_y. \tag{233}$$

This follows by differentiation and $y^\top(B - R_B I)y = 0$; $H_y^\top H_y = I$ is the standard Householder identity (Golub & Van Loan, 2013). Rayleigh–Ritz gives $0 \le R_{B_v} \le 1$, so $\|\nabla_y^2 R_{B_v}\|_{\mathrm{op}} \le 2/\|y\|^2 \le 2/\epsilon$. Compression to directions $(\mathrm{vec}\, E, 0)$ and supremum over v prove the second bound. Also, since $B_v^2 = B_v$,

$$\|\nabla_y R_{B_v}\|^2 = \frac{4}{\|y\|^2} R_{B_v}(1 - R_{B_v}) \le \frac{1}{\epsilon}, \tag{234}$$

which proves the first bound by the same compression. For $X = 0$, $D^2\mathcal{C}_\epsilon(0)[E, E] = 2E^\top E/\epsilon$. □

Theorem 6 (Spectral-gap collapse under spherical gradient noise)**.** *Conditional on $\mathcal{F}_t$, suppose $G_t = g_t + \xi_t$ and $\mathrm{vec}\,\xi_t$ is spherically symmetric, including the point mass at zero. No density or radial monotonicity is required. For $C_t = \mathbb{E}_t\mathcal{C}_{\varepsilon_c}(G_t)$, define*

$$a_t = \frac{1}{d}\mathbb{E}_t \frac{\|\xi_t\|^2}{\|\xi_t\|^2 + \varepsilon_c}. \tag{235}$$

For every $1 \le r < d$,

$$\|C_t - a_t I_d\|_{\mathrm{op}} \le \min\{1, \|g_t\|^2/\varepsilon_c\}, \tag{236}$$

$$0 \le \Delta_{r,t} := \lambda_r(C_t) - \lambda_{r+1}(C_t) \le \min\{r^{-1}, 2\|g_t\|^2/\varepsilon_c\}, \tag{237}$$

$$0 \le h_{C_t}(U) \le \min\{1/2, \|g_t\|^2/\varepsilon_c\} \qquad \textit{for every } U^\top U = I. \tag{238}$$

Consequently a fixed $\varepsilon_c > 0$ and a uniform gap $\Delta_{r,t} \ge \Delta_0 > 0$ imply

$$\frac{1}{T}\sum_{t=1}^{T} \mathbb{E}\|g_t\|^2 \ge \frac{\varepsilon_c \Delta_0}{2}. \tag{239}$$

Thus the positive-floor, vanishing-stationarity conclusion of Theorem 5 cannot simultaneously be combined with a fixed positive eigengap throughout an unbounded horizon under its spherical noise hypothesis. The parameter theorem itself requires no eigengap.

Proof. Spherical symmetry implies right orthogonal invariance, whence

$$\mathbb{E}_t\mathcal{C}_{\varepsilon_c}(\xi_t) = a_t I_d, \tag{240}$$

$$D\mathcal{C}_{\varepsilon_c}(-X)[E] = -D\mathcal{C}_{\varepsilon_c}(X)[E], \tag{241}$$

$$\mathbb{E}_t D\mathcal{C}_{\varepsilon_c}(\xi_t)[g_t] = 0. \tag{242}$$

The derivatives are uniformly bounded for positive ε_c, so expectation and the integral remainder may be interchanged:

$$C_t - a_t I_d = \mathbb{E}_t \int_0^1 (1-u) D^2\mathcal{C}_{\varepsilon_c}(\xi_t + u g_t)[g_t, g_t]\, du, \tag{243}$$

$$\|C_t - a_t I_d\|_{\mathrm{op}} \le \int_0^1 (1-u)\frac{2\|g_t\|^2}{\varepsilon_c}\, du. \tag{244}$$

Weyl's inequality bounds the difference between any two eigenvalues by $2\|C_t - a_t I\|_{\mathrm{op}}$, proving (237). For the energy, use the traceless test matrix $B = P_t^\star - P$, $\|B\|_{\mathrm{op}} \le 1$. Its augmented Rayleigh matrix has spectrum in $[-1, 1]$, so (233) bounds its Hessian by $4/\varepsilon_c$. The isotropic and linear Taylor terms cancel, giving

$$h_{C_t}(P) = \tfrac{1}{2}\mathbb{E}_t\, \mathrm{tr}[B\mathcal{C}_{\varepsilon_c}(g_t + \xi_t)] \le \|g_t\|^2/\varepsilon_c. \tag{245}$$

Finally $\mathrm{tr}\, C_t \le 1$ gives $\Delta_{r,t} \le 1/r$ and $h_{C_t} \le 1/2$; the operator deviation is also at most one. The coefficient $1/\varepsilon_c$ in (236) is sharp: take $m = 1$, zero noise, and $g = av^\top$, then let $a \downarrow 0$. This proves (238). Average the pointwise lower bound obtained from (237) to prove (239). □

Proposition 9 (Exact covariance structure under spherical noise)**.** *Fix a nonzero $g \in \mathbb{R}^{m\times d}$, let $D = md > 1$, $e = \mathrm{vec}(g)/\|g\|$, $x = \mathrm{vec}(g+\xi)$, and let ξ be spherically symmetric. Put*

$$t_\parallel = \mathbb{E}\frac{(e^\top x)^2}{\|x\|^2+\varepsilon_c}, \tag{246}$$

$$\alpha = \frac{1}{D-1}\mathbb{E}\frac{\|x\|^2-(e^\top x)^2}{\|x\|^2+\varepsilon_c}, \qquad \beta = \frac{t_\parallel-\alpha}{\|g\|^2}. \tag{247}$$

Then the normalized matrix covariance and raw covariance satisfy

$$C(g) = m\alpha I_d + \beta g^\top g, \tag{248}$$

$$S(g) = g^\top g + m\sigma_\xi^2 I_d, \qquad \sigma_\xi^2 = D^{-1}\mathbb{E}\|\xi\|^2, \tag{249}$$

where the second equality requires a finite second moment. For zero noise,

$$\Delta_r(C(g)) = \frac{\sigma_r(g)^2-\sigma_{r+1}(g)^2}{\|g\|^2+\varepsilon_c} \le \frac{\|g\|^2}{r\varepsilon_c}. \tag{250}$$

Proof. Every orthogonal map O fixing e preserves the law of x and therefore commutes with $J = \mathbb{E}[xx^\top/(\|x\|^2+\varepsilon_c)]$. The invariant splitting $\mathbb{R}^D = \mathrm{span}(e)\oplus e^\perp$ gives $J = \alpha I_D + (t_\parallel-\alpha)ee^\top$. Summing its m diagonal $d\times d$ blocks gives (248). Similarly $\mathbb{E}\xi = 0$ and $\mathbb{E}[\mathrm{vec}\,\xi\,\mathrm{vec}\,\xi^\top] = \sigma_\xi^2 I_D$ give (249). For zero noise, diagonalize $g^\top g$ and use $r\sigma_r(g)^2 \le \|g\|^2$. Formula (248) does not presume a sign for β; eigenspace ordering must be checked separately. □

Proposition 10 (Exact Gaussian normalization integrals and the collapse coefficient)**.** *Let $\mathrm{vec}\,\xi \sim \mathcal{N}(0,\sigma^2 I_D)$, $\sigma > 0$, $D = md$, $N = \|g\|^2$, and define, for $u \ge 0$,*

$$s_u = 1+2\sigma^2 u, \qquad M_u = s_u^{-D/2}\exp(-uN/s_u), \tag{251}$$

$$\alpha(g) = \int_0^\infty e^{-\varepsilon_c u} M_u \frac{\sigma^2}{s_u}\,du, \qquad \beta(g) = \int_0^\infty e^{-\varepsilon_c u} M_u s_u^{-2}\,du. \tag{252}$$

Then $C(g) = m\alpha(g)I_d + \beta(g)g^\top g$, with

$$0 < \beta(g) \le \min\{\varepsilon_c^{-1}, [\sigma^2(D+2)]^{-1}\}, \tag{253}$$

$$\Delta_r(C(g)) = \beta(g)[\sigma_r(g)^2 - \sigma_{r+1}(g)^2]. \tag{254}$$

For any fixed g_0 and $a \downarrow 0$,

$$\frac{\Delta_r(C(ag_0))}{a^2} \longrightarrow [\sigma_r(g_0)^2-\sigma_{r+1}(g_0)^2]\int_0^\infty e^{-\varepsilon_c u}(1+2\sigma^2 u)^{-D/2-2}\,du. \tag{255}$$

For nonzero Gaussian noise, the integral also extends to $\varepsilon_c = 0$; then its small-signal coefficient is exactly $\beta(0) = 1/[\sigma^2(D+2)]$. Thus fixed isotropic Gaussian noise alone can collapse the normalized gap quadratically, even without an offset. This covariance calculation does not invoke the bounded-gradient hypothesis of Theorem 5; untruncated Gaussian gradients do not meet it.

Proof. Use $1/(\|x\|^2+\varepsilon_c) = \int_0^\infty e^{-u(\|x\|^2+\varepsilon_c)}\,du$. For the Gaussian density $\varphi_{h,V}$, completing the square gives

$$e^{-u\|x\|^2}\varphi_{\mathrm{vec}\,g,\sigma^2 I_D}(x) = M_u\varphi_{\mathrm{vec}\,g/s_u,(\sigma^2/s_u)I_D}(x). \tag{256}$$

Consequently,

$$\mathbb{E}e^{-u\|x\|^2} = M_u, \tag{257}$$

$$\mathbb{E}[xx^\top e^{-u\|x\|^2}] = M_u\left(\frac{\sigma^2}{s_u}I_D + \frac{\mathrm{vec}(g)\,\mathrm{vec}(g)^\top}{s_u^2}\right). \tag{258}$$

Integrate and sum the diagonal row blocks. The integrand for β is strictly positive and bounded both by $e^{-\varepsilon_c u}$ and by $(1+2\sigma^2 u)^{-D/2-2}$. Their integrals give the two envelopes. Diagonalization of $g^\top g$ gives the gap identity, and dominated convergence proves (255). □

A.14. A stopped-process guarantee for staying inside the tracking basin

Theorem 7 (Finite-horizon basin certificate)**.** *Fix T, $R \in (0, \pi/2)$, $\delta \in (0,1)$, and a predictable covariance sequence with $0 \preceq C_t \preceq C_0 I$, gap at least $\Delta_0 > 0$, and $\|C_{t+1} - C_t\| \le v$ for $0 \le t < T$. Let $h_t = h_{C_t}(U_t)$ and $\tau = \inf\{t : \theta_{\max}(U_t, U_t^\star) \ge R\}$. Before τ, suppose the actual QR update has the representation (30), with*

$$\mathbb{E}_t \zeta_t = 0, \quad \|\zeta_t\| \le \sigma, \quad \|b_t\| \le B, \quad \|\operatorname{grad} \Phi_t(U_t)\| \le \Gamma, \quad \mathbb{E}_t \langle \operatorname{grad} \Phi_t(U_t), \zeta_t\rangle^2 \le \nu_0. \tag{259}$$

Here ν_0 is a deterministic directional-variance budget; $\nu_0 = \Gamma^2\sigma^2$ is always valid. Use the sharp QR pullback constant L_R, a constant step ρ, and set

$$\mu_R = \Delta_0 \cos^2 R, \qquad \alpha = \rho\mu_R, \tag{260}$$

$$q = (1 + \alpha/2)(1 - \alpha), \qquad \ell = \log(T/\delta), \tag{261}$$

$$A_\rho = 2(1+\alpha/2)(L_R\rho^2\sigma^2 + \rho B^2) + (1/(2\alpha) + 1/4)v^2/\Delta_0, \tag{262}$$

$$S_\rho = (1+\alpha/2)\rho\Gamma\sigma, \qquad H_R = \tfrac{1}{2}\Delta_0 \sin^2 R, \tag{263}$$

$$V_\rho = \frac{(1+\alpha/2)^2\rho^2\nu_0}{1-q^2}, \tag{264}$$

$$Z_\rho = \min\left\{ S_\rho\sqrt{\frac{2\ell}{1-q^2}}, \frac{S_\rho\ell}{3} + \sqrt{2V_\rho\ell + \frac{S_\rho^2\ell^2}{9}} \right\}. \tag{265}$$

If $L_R\rho \le 1/4$, $\alpha \le 1/2$, and

$$h_0 + \frac{A_\rho}{1-q} + Z_\rho < H_R, \tag{266}$$

then $\Pr(\tau \le T) \le \delta$. Moreover,

$$\mathbb{E}h_T \le q^T h_0 + \frac{A_\rho(1-q^T)}{1-q} + \frac{rC_0}{2}\delta. \tag{267}$$

This theorem supplies a sufficient neighborhood-invariance condition, rather than assuming it. Its gap condition is finite-horizon and is subject to Theorem 6 when the noise is spherical.

Proof. The pathwise smoothness and excess-drift estimates used in Theorem 2, applied before τ, give

$$h_{t+1} \le q h_t + A_\rho + M_t, \qquad M_t = -(1+\alpha/2)\rho\langle \operatorname{grad}\Phi_t(U_t), \zeta_t\rangle. \tag{268}$$

The stopped increment $\widetilde{M}_t = \mathbf{1}\{t < \tau\}M_t$ obeys $\mathbb{E}_t\widetilde{M}_t = 0$, $|\widetilde{M}_t| \le S_\rho$. For each fixed $1 \le j \le T$, conditional Hoeffding and Chernoff give

$$\Pr\left(\sum_{t<j} q^{j-1-t}\widetilde{M}_t > u\right) \le \exp\left(-\frac{u^2}{2S_\rho^2\sum_{t<j} q^{2(j-1-t)}}\right) \tag{269}$$

$$\le \exp\left(-\frac{(1-q^2)u^2}{2S_\rho^2}\right). \tag{270}$$

For the same weighted, stopped martingale, predictable quadratic variation is at most V_ρ and every increment is at most S_ρ. Freedman's Bernstein inequality (Freedman, 1975; Tropp, 2011) therefore gives the second tail estimate

$$\Pr\left(\sum_{t<j} q^{j-1-t}\widetilde{M}_t > u\right) \le \exp\left[-\frac{u^2}{2(V_\rho + S_\rho u/3)}\right]. \tag{271}$$

Solving its quadratic exponent at ℓ gives the second term defining Z_ρ. The first is the preceding Azuma–Hoeffding threshold. All quantities are deterministic, so choose the smaller threshold before observing the process. A union bound over $j \le T$ then fails with probability at most δ. On $\{\tau = j \le T\}$, unroll (268) only to j:

$$h_j \le q^j h_0 + A_\rho\sum_{k<j} q^k + \sum_{t<j} q^{j-1-t}\widetilde{M}_t < H_R, \tag{272}$$

$$h_j \ge \tfrac{1}{2}\Delta_0\|\sin\Theta_j\|^2 \ge \tfrac{1}{2}\Delta_0\sin^2 R = H_R, \tag{273}$$

a contradiction. Finally, positivity and conditional expectation imply

$$\mathbb{E}[h_{t+1}\mathbf{1}\{\tau > t+1\}] \le q\mathbb{E}[h_t\mathbf{1}\{\tau > t\}] + A_\rho. \tag{274}$$

Unroll this killed-process bound and use $0 \le h_T \le rC_0/2$ on $\{\tau \le T\}$ to obtain (267). The zero-noise case follows by the same deterministic argument with $S_\rho = 0$. □

Corollary 7 (Explicit coupling budget for the implemented tracker)**.** *For normalized covariance samples and zero-initialized tracker momentum, $\|H_t\|, \|\widehat{Q}_t\|, \|Q_t\| \le 1$. Its descent representation has*

$$\zeta_t = -(1-\mu_H)(\widehat{Q}_t - Q_t), \tag{275}$$

$$b_t = -\mu_H(\Pi_{U_t}H_{t-1} - Q_t), \qquad \|\zeta_t\| \le 2(1-\mu_H), \quad \|b_t\| \le 2\mu_H. \tag{276}$$

For equal branch step η, bounded corrected gradients and moments,

$$v \le L_C\eta D_0, \qquad D_0 = G_0\sqrt{\varepsilon_a^{-2} + s_{\max}^2}. \tag{277}$$

Consequently, with $\chi_\alpha = (1+\alpha/2)/(1+\alpha)$,

$$\frac{A_\rho}{1-q} \le \chi_\alpha\left[\frac{16L_R\rho(1-\mu_H)^2}{\mu_R} + \frac{16\mu_H^2}{\mu_R} + \frac{L_C^2\eta^2D_0^2}{\rho^2\mu_R^2\Delta_0}\right]. \tag{278}$$

Dropping $\chi_\alpha \le 1$ gives an envelope of the form $A\rho + B_* + D\eta^2/\rho^2$. Its unconstrained optimum is $\rho_* = (2D\eta^2/A)^{1/3}$. It must also satisfy (266) and the step constraints. The μ_H^2 term is a conservative bias budget, not a lower bound on the fixed-momentum tracker error.

Proof. Insert the implemented momentum into $-H_t = g_t + \zeta_t + b_t$. Orthogonal branch supports give $\|D_t\|^2 \le \varepsilon_a^{-2}G_0^2 + s_{\max}^2G_0^2$; Lipschitz continuity of $C(W)$ gives the drift bound. Substitute $\sigma = 2(1-\mu_H)$, $B = 2\mu_H$, and $1-q = \alpha(1+\alpha)/2$. □

A.15. Fixed momentum: exact modal dynamics and stationary tracking error

Let $\delta > 0$ denote one selected–unselected eigenvalue difference. The additive-noise tangent model is

$$v_t = \mu v_{t-1} - (1-\mu)\delta x_t + (1-\mu)\zeta_t, \tag{279}$$

$$x_{t+1} = x_t + \rho v_t - b, \qquad \mathbb{E}_t\zeta_t = 0, \quad \mathbb{E}_t\zeta_t^2 = \sigma^2. \tag{280}$$

Here b is a constant target displacement per step. This model specifies an additive horizontal oracle. General covariance samples can additionally produce multiplicative noise; no equivalence with that oracle is assumed.

Theorem 8 (Exact fixed-momentum tracking covariance)**.** *Let $0 \le \mu < 1$ and*

$$0 < \rho(1-\mu)\delta < 2(1+\mu). \tag{281}$$

The equilibrium mean, transition matrix, and characteristic polynomial are

$$\bar{x} = -b/(\rho\delta), \qquad \bar{v} = b/\rho, \tag{282}$$

$$T_\delta = \begin{pmatrix} 1-\rho(1-\mu)\delta & \rho\mu \\ -(1-\mu)\delta & \mu \end{pmatrix}, \qquad B = (1-\mu)\begin{pmatrix}\rho \\ 1\end{pmatrix}, \tag{283}$$

$$p_\delta(z) = z^2 - [1+\mu-\rho(1-\mu)\delta]z + \mu. \tag{284}$$

For finite initial second moment, the limiting centered covariance of (x_t, v_{t-1}) exists. With $D_\delta = 2(1+\mu) - \rho(1-\mu)\delta$, its entries are

$$\mathrm{Var}_\infty(x) = \frac{\rho(1+\mu)\sigma^2}{\delta D_\delta}, \tag{285}$$

$$\mathrm{Var}_\infty(v) = \frac{2(1-\mu)\sigma^2}{D_\delta}, \tag{286}$$

$$\mathrm{Cov}_\infty(x_t, v_{t-1}) = \frac{\rho(1-\mu)\sigma^2}{D_\delta}. \tag{287}$$

Thus

$$\lim_{t\to\infty} \mathbb{E}x_t^2 = \frac{b^2}{\rho^2\delta^2} + \frac{\rho(1+\mu)\sigma^2}{\delta D_\delta}. \tag{288}$$

For $0 < \mu < 1$, the two roots have modulus $\sqrt{\mu}$ whenever $|1+\mu-\rho(1-\mu)\delta| < 2\sqrt{\mu}$.

Proof. The equilibrium equations give $\rho\bar{v} = b$ and $\bar{v} = -\delta\bar{x}$. Subtract them to obtain $z_{t+1} = T_\delta z_t + B\zeta_t$. The second-order Schur–Cohn stability conditions are

$$1-\mu > 0, \quad \rho(1-\mu)\delta > 0, \quad 2(1+\mu) - \rho(1-\mu)\delta > 0. \tag{289}$$

For $S_t = \mathbb{E}[z_t z_t^\top]$,

$$S_t = T_\delta^t S_0 (T_\delta^\top)^t + \sigma^2 \sum_{k=0}^{t-1} T_\delta^k B B^\top (T_\delta^\top)^k, \tag{290}$$

$$S_\infty = T_\delta S_\infty T_\delta^\top + \sigma^2 B B^\top. \tag{291}$$

The Yule–Walker reduction for an AR(2) process (Brockwell & Davis, 1991) gives an independent derivation. Eliminate v:

$$y_{t+1} = a y_t - \mu y_{t-1} + s\zeta_t, \qquad a = 1+\mu-s\delta, \quad s = \rho(1-\mu), \quad y_t = x_t - \bar{x}. \tag{292}$$

The Yule–Walker equations for stationary autocovariances γ_k are

$$\gamma_1 = a\gamma_0 - \mu\gamma_1 = \frac{a\gamma_0}{1+\mu}, \tag{293}$$

$$\gamma_2 = a\gamma_1 - \mu\gamma_0, \tag{294}$$

$$\gamma_0 = a\gamma_1 - \mu\gamma_2 + s^2\sigma^2, \tag{295}$$

$$\gamma_0 = \frac{s^2\sigma^2(1+\mu)}{(1-\mu)[(1+\mu)^2 - a^2]} = \frac{\rho(1+\mu)\sigma^2}{\delta D_\delta}. \tag{296}$$

Since $v_{t-1} - \bar{v} = (y_t - y_{t-1})/\rho$,

$$\mathrm{Var}(v) = 2(\gamma_0 - \gamma_1)/\rho^2, \tag{297}$$

$$\mathrm{Cov}(x_t, v_{t-1}) = (\gamma_0 - \gamma_1)/\rho. \tag{298}$$

Substitution gives the remaining entries. Complex conjugate roots have product μ, which proves the modulus statement. □

Corollary 8 (Several modes and the momentum/noise distinction)**.** *For modal gaps $\delta_j \in [\Delta, \Lambda]$ and corresponding drift and noise coordinates b_j, σ_j^2, the quadratic routing excess satisfies*

$$\lim_t \mathbb{E}h_{\mathrm{quad}}(x_t) = \sum_j \frac{b_j^2}{2\rho^2\delta_j} + \sum_j \frac{\rho(1+\mu)\sigma_j^2}{2[2(1+\mu) - \rho(1-\mu)\delta_j]}. \tag{299}$$

In particular, as $\rho \downarrow 0$ at fixed $\mu < 1$,

$$\mathrm{Var}_\infty(x_j) = \frac{\rho\sigma_j^2}{2\delta_j} + O(\rho^2), \qquad \mathrm{Var}_\infty(v_j) = \frac{1-\mu}{1+\mu}\sigma_j^2 + O(\rho). \tag{300}$$

The EMA's reduced velocity variance does not multiply the leading subspace-position variance by the same factor. For $0 < \Delta < \Lambda$, Polyak's optimal quadratic tuning (Polyak, 1964) gives the choices

$$\mu_* = \left(\frac{\sqrt{\Lambda} - \sqrt{\Delta}}{\sqrt{\Lambda} + \sqrt{\Delta}}\right)^2, \qquad \rho_* = \frac{1}{\sqrt{\Delta\Lambda}} \tag{301}$$

give $\mathrm{spr}(T_\delta) \le (\sqrt{\Lambda} - \sqrt{\Delta})/(\sqrt{\Lambda} + \sqrt{\Delta})$ for every $\delta \in [\Delta, \Lambda]$. This is a linear-model spectral statement.

Proof. Sum $\delta_j \mathbb{E}x_j^2/2$ and expand the denominators. For the last statement put $q = (\sqrt{\Lambda} - \sqrt{\Delta})/(\sqrt{\Lambda} + \sqrt{\Delta})$. Then $\mu_* = q^2$ and $1 + \mu_* - \rho_*(1-\mu_*)\delta$ ranges from $2q$ to $-2q$. All roots therefore have modulus q, including the repeated endpoint roots. No diagonalizability or global accelerated QR guarantee is inferred. □

A.16. An invariant neighborhood for the actual rank-one QR momentum map

Theorem 9 (Nonlinear local convergence with an explicit Lyapunov radius)**.** *Let $r = 1$, $C = \mathrm{Diag}(\lambda_1, \dots, \lambda_d)$, $\delta_j = \lambda_1 - \lambda_j > 0$ for $j > 1$, and $\Lambda = \lambda_1 - \lambda_d$. Use the deterministic implemented tracker*

$$H_+ = \mu(I - uu^\top)H + (1-\mu)(I - uu^\top)Cu, \tag{302}$$

$$u_+ = \frac{u + \rho H_+}{\|u + \rho H_+\|}, \qquad 0 \le \mu < 1. \tag{303}$$

In the positive first-coordinate chart $u(x) = (1, x^\top)^\top / \sqrt{1 + \|x\|^2}$, put $z = (x, H)$ and write its full state map as $F(z)$. Suppose $0 < \rho(1-\mu)\Lambda < 2(1+\mu)$. Its derivative $A = DF(0)$ consists, up to permutation, of the blocks T_{δ_j} and one zero block for the first component of H. Let

$$P = \sum_{k=0}^{\infty} (A^\top)^k A^k, \qquad A^\top P A - P = -I, \tag{304}$$

$$c_H = \mu/2 + \tfrac{3}{2}(1-\mu)\Lambda, \qquad c_L = \mu + (1-\mu)\Lambda, \tag{305}$$

$$\epsilon_0 = \min\{1/4, (4\rho c_H)^{-1/2}\}, \tag{306}$$

$$c_2 = c_H + 2\rho\{\tfrac{5}{4}c_H + \tfrac{1}{4}(\tfrac{1}{2} + \rho c_H)c_L\}. \tag{307}$$

Choose

$$0 < \epsilon \le \min\left\{\epsilon_0, \frac{1}{8\|A^\top P\|_{\mathrm{op}} c_2}, \frac{1}{2c_2\sqrt{\|P\|_{\mathrm{op}}}}\right\}, \tag{308}$$

where a zero denominator in the second entry imposes no restriction. Then the ellipsoid $\mathcal{E} = \{z : z^\top P z \le \lambda_{\min}(P)\epsilon^2\}$ is invariant and

$$F(z)^\top P F(z) \le \left(1 - \frac{1}{2\lambda_{\max}(P)}\right) z^\top P z \qquad (z \in \mathcal{E}). \tag{309}$$

Thus $u_t \to e_1$ and $H_t \to 0$ geometrically in this neighborhood.

Proof. A scalar shift of C leaves the tracker unchanged; take $C' = C - \lambda_1 I = \mathrm{Diag}(0, -D)$, $D = \mathrm{Diag}(\delta_2, \dots, \delta_d)$. For $t = \|z\| \le 1/4$,

$$\|u(x) - e_1 - (0,x)^\top\| \le \|x\|^2, \tag{310}$$

$$\|uu^\top - e_1 e_1^\top\|_{\mathrm{op}} \le \|x\|, \tag{311}$$

$$\|(I - uu^\top)C'u + (0, Dx)^\top\| \le \tfrac{3}{2}\Lambda\|x\|^2. \tag{312}$$

Consequently, for $H_{\mathrm{lin}} = (0, \mu H_{2:d} - (1-\mu)Dx)^\top$,

$$\|H_+ - H_{\mathrm{lin}}\| \le \mu\|x\|\|H\| + \tfrac{3}{2}(1-\mu)\Lambda\|x\|^2 \le c_H t^2, \tag{313}$$

$$\|H_{\mathrm{lin}}\| \le c_L t, \quad |(H_+)_1| \le c_H t^2. \tag{314}$$

For $t \le \epsilon_0$, the new graph denominator $a = u_1 + \rho(H_+)_1$ satisfies

$$a \ge \frac{1}{\sqrt{1+t^2}} - \rho c_H t^2 > \frac{1}{2}, \qquad |1 - a| \le (1/2 + \rho c_H)t^2. \tag{315}$$

The exact graph update is $x_+ = x + \rho[(H_+)_{2:d} - x(H_+)_1]/a$. Hence

$$\|x_+ - x - \rho(H_{\mathrm{lin}})_{2:d}\| \le 2\rho\{\tfrac{5}{4}c_H + \tfrac{1}{4}(\tfrac{1}{2} + \rho c_H)c_L\}t^2, \tag{316}$$

$$F(z) = Az + r(z), \qquad \|r(z)\| \le c_2\|z\|^2. \tag{317}$$

Differentiating the linear terms gives A and its stated modal blocks. They are Schur stable by (281), so P converges and $P \succeq I$. For $V(z) = z^\top P z$,

$$V(F(z)) - V(z) = -\|z\|^2 + 2\langle Az, Pr(z)\rangle + r(z)^\top P r(z) \tag{318}$$

$$\leq -\{1 - 2\|A^\top P\|_{\mathrm{op}} c_2 \|z\| - \|P\|_{\mathrm{op}} c_2^2 \|z\|^2\}\|z\|^2 \tag{319}$$

$$\leq -\tfrac{1}{2}\|z\|^2 \leq -\frac{V(z)}{2\lambda_{\max}(P)}. \tag{320}$$

Condition (308) bounds the two positive corrections by $1/4$ each. If $z \in \mathcal{E}$, then $\|z\| \leq \epsilon$; the inequality keeps $F(z)$ in the same ellipsoid. Induction proves invariance and geometric convergence. This theorem is for rank one; within-subspace gauge directions for $r > 1$ require an orbital analysis. □

A.17. Generalization through the full optimizer state

Generalization requires a sampling model beyond smooth optimization. In this subsection $S = (z_1, \ldots, z_n)$ is an i.i.d. training set, $S^{(i)}$ replaces one observation independently, $L(W) = \mathbb{E}_z \ell(W, z)$, and $\widehat{L}_S(W) = n^{-1}\sum_i \ell(W, z_i)$. These assumptions do not directly describe transductive node classification on a single dependent graph or nonstationary continual-learning evaluation. Applying the bounds to such settings requires a separate statistical model.

The augmented stored state is

$$X_t = (W_t, U_t, M_t, A_t, H_t), \qquad A_t = \sqrt{V_t}. \tag{321}$$

We retain the *basis* U_t with the algorithm's QR gauge, rather than identifying all bases on the quotient. Lemma 3 shows that coordinatewise Adam does not define a gauge-invariant state transition on the quotient alone. Distances below compare stored frames in a common regular chart. For fixed positive weights ω_j, let

$$d_\omega(X, X') = \|W - W'\| + \omega_U\|U - U'\| + \omega_M\|M - M'\| + \omega_A\|A - A'\| + \omega_H\|H - H'\|. \tag{322}$$

Multiple routed layers are handled by adding their terms. The metric must include all persistent fallback states as well; omitting a full Adam fallback from the sensitivity calculation would analyze a different method.

Assumption 4 (Regular full-state update maps)**.** *For every possible training example and every pair of reachable states, the one-example update Ψ_t satisfies*

$$d_\omega(\Psi_t(X, z), \Psi_t(X', z)) \leq L_t d_\omega(X, X'), \tag{323}$$

$$d_\omega(\Psi_t(X, z), \Psi_t(X, z')) \leq b_t. \tag{324}$$

The constants L_t, b_t are deterministic and uniform over the data sets under comparison, and initialization is data independent and coupled. The loss is L_ℓ-Lipschitz in W on the reachable parameter region.

This is a fixed-configuration analysis. If hyperparameters are chosen by validation, the entire selection procedure becomes part of the learning algorithm. A neighboring observation can change its discrete choice; fixed-configuration stability does not automatically establish stability of that selection pipeline. A separate selection argument or uniform control of its branches is needed for that stronger conclusion.

Theorem 10 (Full-state stability and expected generalization)**.** *Under Assumption 4, couple the sampling indices of the two runs so that the replaced observation is used with marginal probability $1/n$ at each step. Then*

$$\mathbb{E} d_\omega(X_T, X_T^{(i)}) \leq \frac{1}{n}\sum_{t<T} b_t \prod_{j=t+1}^{T-1} L_j, \tag{325}$$

$$\left|\mathbb{E}_{S,\Omega}\{L(W_T) - \widehat{L}_S(W_T)\}\right| \leq \beta_T := \frac{L_\ell}{n}\sum_{t<T} b_t \prod_{j=t+1}^{T-1} L_j. \tag{326}$$

Here Ω denotes the algorithm's randomness. If the loss belongs to $[0, M_\ell]$, with probability at least $1 - \delta$ over S,

$$\left|\mathbb{E}_\Omega[L(W_T) - \widehat{L}_S(W_T)]\right| \leq \beta_T + (2n\beta_T + M_\ell)\sqrt{\frac{\log(2/\delta)}{2n}}. \tag{327}$$

The high-probability statement concerns the randomness-averaged output; it is not automatically a guarantee for one realized seed.

Proof. Use the same sampled index and all other random choices in both runs. The triangle inequality, (323), and (324) imply pathwise

$$d_\omega(X_{t+1}, X_{t+1}^{(i)}) \le L_t d_\omega(X_t, X_t^{(i)}) + b_t \mathbf{1}\{I_t = i\}. \tag{328}$$

Taking expectation and using the marginal sampling probability gives the recursion with additive term b_t/n. Uniformity of L_t, b_t is what allows this unconditional step; it does not require independence between the past state and the indicator. Starting from zero distance and iterating proves (325). Lipschitzness in W gives a loss stability bound β_T for every evaluation example.

For completeness, introduce an independent ghost observation z_i' and the replacement set $S^{(i)}$. Exchangeability gives

$$\mathbb{E}\{L(W_T(S)) - \widehat{L}_S(W_T(S))\} = \frac{1}{n}\sum_i \mathbb{E}\{\ell(W_T(S), z_i') - \ell(W_T(S^{(i)}), z_i')\}. \tag{329}$$

Each summand has absolute value at most β_T, proving the expected bound. This is the standard replacement argument of algorithmic stability (Bousquet & Elisseeff, 2002; Hardt et al., 2016).

Let $F(S) = \mathbb{E}_\Omega[L(W_T) - \widehat{L}_S(W_T)]$. Replacing one example changes its population term by at most β_T. Its empirical term changes by at most $\beta_T + M_\ell/n$: unchanged examples contribute stability, and the replaced loss is bounded in an interval of length M_ℓ. Thus each bounded difference is at most $2\beta_T + M_\ell/n$. McDiarmid's inequality applied to $F(S)$, together with $|\mathbb{E}F(S)| \le \beta_T$, proves (327). □

For a mini-batch of B sampled indices the probability of touching the replaced observation is at most B/n. The corresponding update difference must be computed for one changed member of the averaged batch. It cannot be assumed to equal b_t/B after nonlinear Adam normalization. The same proof applies with that batch-specific injection constant. Likewise, random reshuffling has a uniform marginal index but does not give the conditional unbiased-gradient assumption of Theorem 4 without further analysis. The two theorems have deliberately different sampling requirements.

A.18. Checking the stability constants for the implemented primitives

Assumption 4 is meaningful only if its constants are connected to the algorithm. The following bounds provide that connection and make the sources of potentially vacuous constants explicit.

Lemma 13 (Lipschitz building blocks)**.** *Suppose single-example gradients are L_g-Lipschitz and bounded by G_0, stored moments are bounded, $\varepsilon_c, \varepsilon > 0$, and all compared QR/polar operations stay in a fixed regular region. The normalized covariance map obeys*

$$\left\| \frac{G^\top G}{\|G\|^2 + \varepsilon_c} - \frac{G'^\top G'}{\|G'\|^2 + \varepsilon_c} \right\| \le \frac{2}{\sqrt{\varepsilon_c}} \|G - G'\|. \tag{330}$$

For orthonormal U, U' and bounded C, C',

$$\|\Pi_U(CU) - \Pi_{U'}(C'U')\| \le 4C_0\|U - U'\| + \|C - C'\|. \tag{331}$$

The elementwise root-EMA map is jointly Lipschitz:

$$\|\sqrt{\beta A^{\odot 2} + (1-\beta)Z^{\odot 2}} - \sqrt{\beta A'^{\odot 2} + (1-\beta)Z'^{\odot 2}}\| \le \sqrt{\beta}\|A - A'\| + \sqrt{1-\beta}\|Z - Z'\|. \tag{332}$$

For a fixed orthogonal Q, diagonal covariance transport is nonexpansive in root amplitudes:

$$\|\sqrt{A^{\odot 2}Q^{\odot 2}} - \sqrt{A'^{\odot 2}Q^{\odot 2}}\| \le \|A - A'\|. \tag{333}$$

Together with (149), these inequalities yield finite full-state constants on the stated region.

Proof. Differentiating $G^\top G/(\|G\|^2 + \varepsilon_c)$ in direction E bounds its derivative by

$$\left\{ \frac{2\|G\|}{\|G\|^2 + \varepsilon_c} + \frac{2\|G\|^3}{(\|G\|^2 + \varepsilon_c)^2} \right\} \|E\| \le \frac{2}{\sqrt{\varepsilon_c}} \|E\|. \tag{334}$$

Integrating along the line segment proves (330). For (331), expand $CU - UU^\top CU$ into differences of its factors and use $\|UU^\top - U'U'^\top\|_{\mathrm{op}} \le 2\|U - U'\|_{\mathrm{op}}$. The C difference has coefficient one after grouping $(I - P)(C - C')U$; the U differences have total coefficient at most $4C_0$. To prove (332), view each coordinate as the Euclidean norm of $(\sqrt{\beta}A_{ij}, \sqrt{1-\beta}Z_{ij})$ and apply the reverse triangle inequality followed by Minkowski. Similarly, the transported amplitude in coordinate j of a row is the norm of $(A_kQ_{kj})_k$. Squaring the reverse-triangle bound and summing over j uses $\sum_j Q_{kj}^2 = 1$ and proves (333).

QR with a fixed positive-diagonal convention is smooth on full-column-rank matrices. The polar factor is smooth when its input singular values stay away from zero. Bounded derivatives on the stipulated compact regular region give their finite constants. Multiplication and projection are Lipschitz on bounded sets; the residual scale uses Lipschitz norms, a positive denominator offset, and nonexpansive scalar clipping. Finally, the Adam division bound is (149). □

One can compose these bounds as a nonnegative sensitivity matrix K_t. Its entries bound the output block differences $(\delta W, \delta U, \delta M, \delta A, \delta H)$ in terms of their input differences. For example, the fresh routed-coordinate difference is at most $L_g\delta W + G_0\delta U$, the first-moment EMA contributes $\beta_1\delta M + (1-\beta_1)(L_g\delta W + G_0\delta U)$ before alignment, and the parameter update contributes $\delta W + \eta_t\delta D$. Projection transport adds $2H_0\delta U$ to the transported momentum comparison. QR and polar Lipschitz factors then propagate these differences through basis and moment alignment. If $D_\omega = \mathrm{Diag}(1, \omega_U, \omega_M, \omega_A, \omega_H)$, a sufficient choice is

$$L_t = \|D_\omega K_t D_\omega^{-1}\|_1. \tag{335}$$

The changed-example constant is obtained by repeating this composition with zero initial state differences and gradient difference at most $2G_0$. This procedure includes the moment states rather than treating Adam as a memoryless gradient map.

Very small denominator offsets can make these constants large. Moreover, if $L_t \le 1 + c(\eta_t + \rho_t)$ is established in a chosen weighted metric, the product bound is at most $\exp\{c\sum_t(\eta_t + \rho_t)\}$. A constant tracking step over a long horizon can therefore yield a vacuous bound. We report the product form within this scope; we do not infer a numerical certificate for the experiments without measuring valid constants. Identical deterministic QR, transport, and reset operations in coupled runs do not create an additive error independent of the sample change. Such a term would incorrectly predict different trajectories even when the two data sets and initializations are identical. This does not make a thresholded reset Lipschitz: crossing its threshold can create a discontinuity. Such a reset requires an additional branch analysis or a restriction that both coupled runs use the same branch. The corrected diagonal-transport update avoids that particular switch.

Proposition 11 (A nonvacuous contractive regime and the coupling condition)**.** *Suppose, after including or validly eliminating contractive moment states, the coupled same-example distances obey*

$$\begin{pmatrix} w_+ \\ u_+ \end{pmatrix} \le \begin{pmatrix} 1-\eta a & \eta b \\ \rho c & 1-\rho d \end{pmatrix} \begin{pmatrix} w \\ u \end{pmatrix}, \qquad a, d > 0,\ b, c \ge 0,\ ad > bc, \tag{336}$$

with nonnegative diagonal entries. For $b, c > 0$ choose

$$\frac{\eta b}{\rho d} < \omega < \frac{\eta a}{\rho c}, \qquad \kappa = \min\{\eta a - \omega\rho c,\ \rho d - \eta b/\omega\} > 0. \tag{337}$$

If replacing the sampled example adds at most $(\eta B_W, \rho B_U)$ to these two blocks, then

$$\beta_T \le \frac{L_\ell(\eta B_W + \omega\rho B_U)}{n\kappa}\{1 - (1-\kappa)^T\}. \tag{338}$$

The cases $b = 0$ or $c = 0$ follow by the corresponding one-sided choice of ω or by a limit.

Proof. Multiplying the second row of (336) by ω and adding the first gives $w_+ + \omega u_+ \le (1-\kappa)(w + \omega u)$. The expected changed-example injection is $(\eta B_W + \omega\rho B_U)/n$. Summing the geometric recursion and applying the loss Lipschitz bound proves (338). □

This proposition identifies a sufficient nonvacuous regime: each block must contract and their cross-coupling must satisfy $ad > bc$. A routing eigengap can improve d in a genuinely locally convex neighborhood, but PL over the larger normal neighborhood is insufficient to assert (336). Strong convexity of the parameter objective in the relevant metric is also an additional condition, seldom globally true in deep learning. For explicit moment blocks, the same argument uses a positive weight vector with weighted column sums of their full sensitivity matrix strictly below one. These statements give checkable sufficient conditions, rather than claiming all empirical runs satisfy them.

A.19. A complementary PAC-Bayes guarantee for data-dependent routing

Stability is algorithm dependent and can become loose over long horizons. A second route uses a posterior over the final weights, with no contractivity assumption. It establishes a statistical guarantee for a randomized predictor and separates optimizer-state dimension from model capacity. We provide the elementary exponential-moment proof so that its statistical content and its assumptions are explicit.

Proposition 12 (PAC-Bayes bound and Grassmann-shaped perturbations)**.** *Let $0 \le \ell(w,z) \le 1$, let $\mathcal{P}$ be any data-independent prior on weights, and fix $\lambda > 0$ before observing S. With probability at least $1-\delta$, simultaneously for every posterior $\mathcal{Q}$ absolutely continuous with respect to $\mathcal{P}$,*

$$L(\mathcal{Q}) \le \widehat{L}_S(\mathcal{Q}) + \frac{\mathrm{KL}(\mathcal{Q}\|\mathcal{P}) + \log(1/\delta)}{\lambda} + \frac{\lambda}{8n}. \tag{339}$$

For one matrix layer, use a data-independent center W_0 and

$$\mathcal{P} = \mathcal{N}(\mathrm{vec}\, W_0, s_0^2 I_{md}), \tag{340}$$

$$\mathcal{Q}_S = \mathcal{N}(\mathrm{vec}\, W_T, I_m \otimes (s_\parallel^2 P_T + s_\perp^2 (I - P_T))), \tag{341}$$

where vectorization is rowwise and all variances are strictly positive. The final, data-dependent projector P_T is allowed. Then

$$\mathrm{KL}(\mathcal{Q}_S\|\mathcal{P}) = \frac{\|W_T - W_0\|^2}{2s_0^2} + \frac{m}{2}\left[r\, k\!\left(\frac{s_\parallel^2}{s_0^2}\right) + (d-r)\, k\!\left(\frac{s_\perp^2}{s_0^2}\right)\right], \tag{342}$$

$$k(x) = x - 1 - \log x. \tag{343}$$

Proof. For fixed w, Hoeffding's exponential-moment inequality for the independent bounded losses gives

$$\mathbb{E}_S \exp\{\lambda(L(w) - \widehat{L}_S(w))\} \le \exp(\lambda^2/(8n)). \tag{344}$$

Integrating over the prior and applying Markov's inequality shows that, with probability $1-\delta$,

$$\log \mathbb{E}_{w\sim\mathcal{P}} e^{\lambda(L(w) - \widehat{L}_S(w))} \le \lambda^2/(8n) + \log(1/\delta). \tag{345}$$

For any posterior, Jensen applied after change of measure gives

$$\lambda \mathbb{E}_{\mathcal{Q}}(L - \widehat{L}_S) \le \mathrm{KL}(\mathcal{Q}\|\mathcal{P}) + \log \mathbb{E}_{\mathcal{P}} e^{\lambda(L - \widehat{L}_S)}. \tag{346}$$

Combining the last two displays proves (339); the high-probability event is independent of the posterior, giving simultaneous validity. This is a standard PAC-Bayesian change-of-measure argument (McAllester, 1999; Seeger, 2002).

For the Gaussian specialization, the covariance in each row has eigenvalue $s_\parallel^2$ with multiplicity r and $s_\perp^2$ with multiplicity $d-r$. Insert its trace and determinant in the Gaussian KL identity

$$\mathrm{KL} = \tfrac{1}{2}\left[\frac{\|W_T - W_0\|^2 + \mathrm{tr}\,\Sigma_Q}{s_0^2} - md + md\log s_0^2 - \log\det\Sigma_Q\right]. \tag{347}$$

Collecting the two eigenvalue groups proves (342). For independent layerwise Gaussian factors, KL adds across layers. □

The displacement term can be related to the optimizer path without assuming the final parameter is low rank. Orthogonality of the two implemented branches gives

$$\|D_t\|^2 = \|D_t^{\parallel}\|^2 + \gamma_t^2 \|G_t(I - P_t)\|^2, \tag{348}$$

$$\|W_T - W_0\| \le \sum_{t<T} \eta_t \|D_t\|. \tag{349}$$

For adaptive branch-to-residual scaling, γ_t here includes that actual scalar. Equations (14) and (349) connect raw rank tails and routing quality to an upper bound on the path's squared length, under the requisite conditional measurability. They do not force a smaller path or smaller generalization gap than AdamW.

Several qualifications prevent misleading rank-based interpretations. First, $k(x) \geq 0$ and diverges as $x \downarrow 0$; concentrating posterior variance in only a few directions can *increase* the KL cost in the $m(d-r)$ residual coordinates. Second, reducing those variances may improve the empirical perturbation loss, producing a trade-off which must be measured. Third, the prior cannot be centered at a data-dependent fitted weight without an additional valid prior-selection argument. The isotropic prior above allows a data-dependent posterior orientation precisely because (339) holds for all posteriors; no assumption that P_T is independent of the training set has been made.

The empirical term is the loss of the Gaussian-perturbed predictor, not the unperturbed network's training loss. A certificate requires evaluating that expectation, with a separate Monte Carlo error allowance if sampled. Unbounded cross-entropy must be replaced by a specified bounded surrogate or analyzed with an appropriate unbounded-loss theorem. Also, optimizing λ after observing the posterior is not justified by a bound proved for one fixed value. A finite grid of J predeclared values is valid after replacing $\log(1/\delta)$ by $\log(J/\delta)$ and taking a union bound. Neither the stability bound nor PAC-Bayes is added to the other as a spurious improvement; they are alternative guarantees with distinct assumptions and output objects.

A.20. Implications that can be tested, and claims that do not follow

The analysis suggests four concrete measurements. First, estimate the full covariance spectrum and the distance to an *independent* oracle subspace; GU alone does not reveal λ_{r+1}, and consecutive bases do not reveal the oracle radius. Second, separate normalized and raw covariances when comparing tracking and residual energy. Third, log actual parameter-direction alignment or its comparison-metric defect before claiming the rate in Theorem 4. Fourth, estimate neighboring-data sensitivity with shared initialization and sampling, including moment states, before interpreting a train–test gap as a stability measurement. Small single-seed changes in accuracy cannot establish any of these theoretical hypotheses.

The four main result families provide a conditional chain from spectral separation and angles to routing quality, from a controlled tracking process to an explicit perturbation budget, from that budget to stationarity, and from full-state sensitivity or posterior complexity to generalization. They do not imply universal optimizer superiority, a compulsory rank-tail error floor with a nonzero residual branch, full gauge invariance of Adam, or a nonvacuous numerical generalization certificate for the present deep networks. These boundaries are part of the mathematical conclusions. Theorem 6 additionally shows why the separated tracking and stationarity regimes need not hold simultaneously throughout training; Theorem 15 supplies an explicit finite-window statistical connection with a valid independent prior.

A.21. An explicit sensitivity matrix for every stored state

The following construction replaces unspecified composition constants by finite algebraic expressions. Norms compare stored QR frames, not merely projectors. We specify one routed matrix; further routed layers or fallback states require additional rows and columns. Moment states start at zero, so fresh bias-corrected first moments are bounded by G_0 as in Lemma 7. Let $e_W, e_U, e_M, e_A, e_H, e_g$ be the coordinate row vectors of $\mathbb{R}^6$ and

$$d = (\delta W, \delta U, \delta M, \delta A, \delta H, \delta g)^\top \geq 0, \qquad \|G - G'\| \leq L_g \delta W + \delta g. \tag{350}$$

For a common example $\delta g = 0$; for one replaced example at a common state $\delta g \leq 2G_0$. Assume all reachable gradients are bounded by G_0, all raw first moments and root second moments by G_0, and all tracker momenta by H_0. The QR comparison segments have minimum singular value at least $s_Q > 0$; Procrustes inputs have minimum singular value at least $s_R > 0$. Define $\kappa_Q = \sqrt{3}/s_Q$, $\kappa_R = 1/s_R$, and the following nonnegative rows:

$$k_G = L_g e_W + e_g, \tag{351}$$

$$k_Z = k_G + G_0 e_U, \qquad k_E = k_G + 2G_0 e_U, \tag{352}$$

$$k_M = \beta_1 e_M + (1-\beta_1) k_Z, \tag{353}$$

$$k_A = \sqrt{\beta_2} e_A + \sqrt{1-\beta_2} k_Z, \tag{354}$$

$$k_D = \frac{\varepsilon_a^{-1}}{1-\beta_1^t} k_M + \frac{G_0 \varepsilon_a^{-2}}{\sqrt{1-\beta_2^t}} k_A + G_0 \varepsilon_a^{-1} e_U, \tag{355}$$

$$k_{\mathrm{res}} = (s_{\max} + G_0^2 \varepsilon_r^{-2}) k_E + G_0 \varepsilon_r^{-1} k_G, \tag{356}$$

$$k_Q = 4e_U + 2\varepsilon_c^{-1/2} k_G, \tag{357}$$
$$k_H = \mu_H e_H + 2\mu_H H_0 e_U + (1-\mu_H) k_Q, \tag{358}$$
$$k_{U+} = \kappa_Q (e_U + \rho_t k_H), \tag{359}$$
$$k_R = \kappa_R (k_{U+} + e_U), \tag{360}$$
$$k_{M+} = k_M + G_0 k_R, \qquad k_{A+} = k_A + G_0 k_R, \tag{361}$$
$$k_{W+} = |1 - \eta_t \lambda_{\mathrm{wd}}| e_W + \eta_t k_D + \eta_{\perp,t} k_{\mathrm{res}}. \tag{362}$$

Here $k_{W+}, k_{U+}, k_{M+}, k_{A+}, k_H$ are stacked into $\mathbb{K}_t \in \mathbb{R}_+^{5\times 6}$; write $\mathbb{K}_t = (K_t \mid v_t)$.

Proposition 13 (Constructive full-state Lipschitz and injection constants)**.** *Under the preceding bounds, the one-step state differences obey*

$$\begin{pmatrix} \delta W_+ \\ \delta U_+ \\ \delta M_+ \\ \delta A_+ \\ \delta H_+ \end{pmatrix} \le K_t \begin{pmatrix} \delta W \\ \delta U \\ \delta M \\ \delta A \\ \delta H \end{pmatrix} + v_t \delta g. \tag{363}$$

For $\omega = (1, \omega_U, \omega_M, \omega_A, \omega_H) > 0$, valid constants in Assumption 4 are

$$L_t = \max_j \frac{\sum_i \omega_i (K_t)_{ij}}{\omega_j}, \qquad b_t = 2G_0 \sum_i \omega_i (v_t)_i. \tag{364}$$

In particular, identical input states and identical examples imply zero output difference exactly.

Proof. The routed and residual gradient estimates are

$$\|GU - G'U'\| \le \|G - G'\| + G_0 \|U - U'\|, \tag{365}$$
$$\|G(I-P) - G'(I-P')\| \le \|G - G'\| + 2G_0 \|U - U'\|. \tag{366}$$

The first-moment EMA gives k_M. Reverse triangle inequalities for the coordinatewise two-dimensional Euclidean norms give k_A. Bias correction and (149), followed by lifting, give k_D. For $n = \|G\|$, $e = \|G(I-P)\|$, scalar clipping is nonexpansive, so

$$|s - s'| \le \varepsilon_r^{-1} |n - n'| + G_0 \varepsilon_r^{-2} |e - e'|, \tag{367}$$
$$\|sE - s'E'\| \le s_{\max} \|E - E'\| + G_0 |s - s'|. \tag{368}$$

These are precisely the coefficients in k_{res}. Lemma 13 gives k_Q, while

$$\|(I-P)H - (I-P')H'\| \le \|H - H'\| + 2H_0 \|U - U'\| \tag{369}$$

gives k_H. Lemma 5 gives k_{U+}. The polar perturbation inequality gives

$$\|R - R'\| \le s_R^{-1} \|U_+^\top U - {U'_+}^\top U'\| \le \kappa_R (\delta U_+ + \delta U). \tag{370}$$

Finally,

$$\|\widetilde{M} R^\top - \widetilde{M}' R'^\top\| \le \|\widetilde{M} - \widetilde{M}'\| + G_0 \|R - R'\|, \tag{371}$$
$$\|\sqrt{\widetilde{A}^{\odot 2} (R^\top)^{\odot 2}} - \sqrt{\widetilde{A}'^{\odot 2} (R'^\top)^{\odot 2}}\| \le \|\widetilde{A} - \widetilde{A}'\| + G_0 \|R - R'\|. \tag{372}$$

The second inequality follows by writing each output amplitude as the norm of a row weighted by a column of the rotation, then using its orthogonality. These prove every row of (362). Multiplying (363) by $\omega^\top$ yields (364). All operations are deterministic functions of states and examples; there is no sample-independent additive error. □

Corollary 9 (Matrix-valued replacement stability)**.** *For independent uniform sampling with replacement and common data-independent initialization, define* $u_t = \mathbb{E}(\delta W_t, \delta U_t, \delta M_t, \delta A_t, \delta H_t)^\top$*. If the constants above are uniform over the two coupled runs, then*

$$u_{t+1} \le K_t u_t + \frac{2G_0}{n} v_t, \tag{373}$$

$$u_T \le \frac{2G_0}{n} \sum_{t=0}^{T-1} K_{T-1} \cdots K_{t+1} v_t, \tag{374}$$

$$|\mathbb{E}[L(W_T) - \widehat{L}_S(W_T)]| \le \frac{2L_\ell G_0}{n} e_W^{(5)} \sum_{t=0}^{T-1} K_{T-1} \cdots K_{t+1} v_t. \tag{375}$$

Here $e_W^{(5)} = (1, 0, 0, 0, 0)$ *and an empty product is the identity.*

Proof. The probability of drawing the replaced example is $1/n$; iterate (363) from $u_0 = 0$ and use the ghost-sample identity in Theorem 10. □

A.22. Spectral sensitivity, feedback, and a verifiable contractive specialization

Proposition 14 (Optimal weights and the Schur feedback condition)**.** *For a nonnegative time-independent sensitivity matrix* K*,*

$$\inf_{\omega > 0} \|D_\omega K D_\omega^{-1}\|_1 = \mathrm{spr}(K). \tag{376}$$

Partition a valid sensitivity bound as

$$K = \begin{pmatrix} k & b^\top \\ c & A \end{pmatrix}, \qquad k \ge 0, \quad A, b, c \ge 0, \quad \mathrm{spr}(A) < 1. \tag{377}$$

Then

$$\mathrm{spr}(K) < 1 \quad \iff \quad k + b^\top (I - A)^{-1} c < 1. \tag{378}$$

For an injection $h = (h_W, h_S)^\top \ge 0$ *and* $u_{t+1} \le K u_t + h/n$*,* $u_0 = 0$*, the resulting uniform parameter sensitivity satisfies*

$$\sup_T (u_T)_W \le \frac{h_W + b^\top (I - A)^{-1} h_S}{n\{1 - k - b^\top (I - A)^{-1} c\}}. \tag{379}$$

The triangle-inequality construction (362) *usually has* $k \ge 1$ *when weight decay is absent. It does not by itself establish* (378)*; that requires a sharper signed contraction estimate for the parameter transition.*

Proof. Every induced norm bounds the spectral radius from below. For any $q > \mathrm{spr}(K)$,

$$\omega = (qI - K^\top)^{-1} \mathbf{1} = \sum_{j=0}^{\infty} q^{-j-1} (K^\top)^j \mathbf{1} > 0, \tag{380}$$

$$K^\top \omega = q\omega - \mathbf{1} < q\omega. \tag{381}$$

Hence the norm in (376) is less than q; let q decrease to $\mathrm{spr}(K)$. If (378) holds, choose

$$x = \begin{pmatrix} 1 \\ (I - A)^{-1} c + \epsilon (I - A)^{-1} \mathbf{1} \end{pmatrix}, \quad \epsilon > 0. \tag{382}$$

For sufficiently small ϵ, $Kx < x$, proving $\mathrm{spr}(K) < 1$. Conversely, if $\mathrm{spr}(K) < 1$, then $(I - K)^{-1} = \sum_{j \ge 0} K^j \ge 0$. Its top-left entry, computed by Schur complementation, is $[1 - k - b^\top (I - A)^{-1} c]^{-1}$ and is positive, proving the converse. Finally $u_T \le n^{-1} \sum_{j=0}^{T-1} K^j h \le n^{-1} (I - K)^{-1} h$; the top component is (379). □

Proposition 15 (A signed, frozen-metric quadratic specialization)**.** *Let $\ell(w,z) = w^\top H w/2 - b_z^\top w$, $H \succ 0$, $\|b_z\| \le B_z$, and use a fixed positive metric $B \succ 0$ and momentum $\beta \in [0,1)$:*

$$m_t = \beta m_{t-1} + (1-\beta)(Hw_t - b_{I_t}), \qquad w_{t+1} = w_t - \eta B m_t. \tag{383}$$

Set $\widetilde{H} = B^{1/2} H B^{1/2}$, with eigenvalues $0 < \gamma_{\min} \le \gamma_j \le \gamma_{\max}$. The signed two-state transition is asymptotically stable if

$$0 < \eta < \frac{2(1+\beta)}{(1-\beta)\gamma_{\max}}. \tag{384}$$

For any $q \in (\max_j \operatorname{spr}(T_{\gamma_j}), 1)$ there is a finite C_q such that the neighboring-data parameter difference obeys

$$\sup_T \mathbb{E}\|w_T - w_T'\| \le \frac{2C_q(1-\beta)B_z\|B^{1/2}\|_{\mathrm{op}}^2\sqrt{1+\eta^2}}{n(1-q)}. \tag{385}$$

This is a frozen-metric specialization, not a contraction theorem for unrestricted Adam or a moving routing basis.

Proof. Use coordinates $x = B^{-1/2}w$, $y = B^{1/2}m$ and diagonalize $\widetilde{H}$. Each eigencoordinate has transition

$$T_\gamma = \begin{pmatrix} 1-\eta(1-\beta)\gamma & -\eta\beta \\ (1-\beta)\gamma & \beta \end{pmatrix}, \qquad \det(zI - T_\gamma) = z^2 - [1+\beta-\eta(1-\beta)\gamma]z + \beta. \tag{386}$$

The real second-order Schur conditions reduce to

$$1-\beta > 0, \tag{387}$$

$$1 - [1+\beta-\eta(1-\beta)\gamma] + \beta = \eta(1-\beta)\gamma > 0, \tag{388}$$

$$1 + [1+\beta-\eta(1-\beta)\gamma] + \beta = 2(1+\beta) - \eta(1-\beta)\gamma > 0. \tag{389}$$

These prove (384). In finite dimension, $\|T^k\| \le C_q q^k$ for any q strictly larger than its spectral radius, including nondiagonalizable blocks. A replaced example injects a vector of norm at most $2(1-\beta)B_z\|B^{1/2}\|_{\mathrm{op}}\sqrt{1+\eta^2}$ into (x,y) with probability $1/n$. Sum the geometric response and multiply by $\|B^{1/2}\|_{\mathrm{op}}$ to return to w. □

A.23. Validation selection is a separate statistical operation

Theorem 11 (Finite-candidate validation selection)**.** *Let $\theta_1(S), \dots, \theta_K(S)$ be any predictors constructed without access to an independent validation set V of size n_v. For losses in $[0,1]$, let $\widehat{k} \in \arg\min_k \widehat{L}_V(\theta_k)$. With probability at least $1-\delta$ over V, conditional on S and all candidate-generation randomness,*

$$\max_{k \le K} |L(\theta_k) - \widehat{L}_V(\theta_k)| \le \epsilon_v := \sqrt{\frac{\log(2K/\delta)}{2n_v}}, \tag{390}$$

$$L(\theta_{\widehat{k}}) \le \min_{k \le K} L(\theta_k) + 2\epsilon_v. \tag{391}$$

Every inspected checkpoint, learning rate, rank, and restart must be included among the candidates. Repeated adaptive use of V to generate new candidates requires a different argument.

Proof. Conditional on the candidates, Hoeffding and the union bound yield

$$\Pr\left(\max_k |L_k - \widehat{L}_{V,k}| > u \mid S, \Omega \right) \le 2Ke^{-2n_v u^2}. \tag{392}$$

On its complementary event, for every k,

$$L_{\widehat{k}} \le \widehat{L}_{V,\widehat{k}} + \epsilon_v \le \widehat{L}_{V,k} + \epsilon_v \le L_k + 2\epsilon_v. \tag{393}$$

This proof neither assumes nor transfers training-set stability of the selected learning-rate map. □

A.24. PAC-Bayes: anisotropic posterior optimization and a margin specialization

Let Π be data independent and $\ell \in [0,1]$. For fixed $\lambda > 0$, Hoeffding's lemma and independence imply

$$\mathbb{E}_S \exp\{\lambda[L(\theta) - \widehat{L}_S(\theta)]\} \le e^{\lambda^2/(8n)}, \tag{394}$$

$$\mathbb{E}_S \mathbb{E}_{\theta\sim\Pi} \exp\{\lambda[L(\theta) - \widehat{L}_S(\theta)] - \lambda^2/(8n)\} \le 1. \tag{395}$$

Markov's inequality and the Gibbs–Donsker–Varadhan variational principle (Donsker & Varadhan, 1975) give, on a single event of probability at least $1-\delta$, simultaneously for all $Q \ll \Pi$,

$$L(Q) \le \widehat{L}_S(Q) + \frac{\mathrm{KL}(Q\|\Pi) + \log(1/\delta)}{\lambda} + \frac{\lambda}{8n}. \tag{396}$$

Indeed, for $F = \lambda(L - \widehat{L}_S)$,

$$\mathbb{E}_Q F - \mathrm{KL}(Q\|\Pi) = \mathbb{E}_Q \log\left(e^F \frac{d\Pi}{dQ}\right) \le \log \mathbb{E}_\Pi e^F. \tag{397}$$

For a countable data-independent set $\{\lambda_j\}$ with masses $\pi_j > 0$, $\sum_j \pi_j = 1$, replace $\log(1/\delta)$ by $\log[1/(\delta\pi_j)]$ to obtain simultaneous validity and permit selection of j after training.

Proposition 16 (Optimal two-variance Gaussian certificate)**.** *Let $P = UU^\top$ be any data-dependent rank-r projector, $0 < r < d$, and $\Pi = \mathcal{N}(\mathrm{vec}\, W_0, s_0^2 I_{md})$, with W_0, s_0 data independent. Let*

$$Q = \mathcal{N}(\mathrm{vec}\, W, I_m \otimes [s_\parallel^2 P + s_\perp^2 (I-P)]), \quad s_\parallel, s_\perp > 0, \tag{398}$$

using row-wise vectorization. Suppose for all E and $u \in [0,1]$,

$$\langle E, \nabla^2 \widehat{L}_S(W + uE)[E]\rangle \le h_\parallel \|EP\|^2 + h_\perp \|E(I-P)\|^2, \quad h_\parallel, h_\perp \ge 0. \tag{399}$$

Then the variances minimizing the sum of the quadratic empirical-loss upper bound and KL/λ are

$$s_\parallel^2 = (s_0^{-2} + \lambda h_\parallel)^{-1}, \qquad s_\perp^2 = (s_0^{-2} + \lambda h_\perp)^{-1}. \tag{400}$$

On the event (396), this posterior satisfies

$$L(Q) \le \widehat{L}_S(W) + \frac{\|W - W_0\|^2}{2\lambda s_0^2} + \frac{m}{2\lambda}\{r \log(1 + \lambda h_\parallel s_0^2) + (d-r)\log(1 + \lambda h_\perp s_0^2)\} + \frac{\log(1/\delta)}{\lambda} + \frac{\lambda}{8n}. \tag{401}$$

No derivative of the learned projector with respect to the sample is needed for this PAC-Bayes statement; the prior must remain data independent.

Proof. Put $\Sigma = s_\parallel^2 P + s_\perp^2 (I-P)$. Its eigenvalues and determinant are

$$\mathrm{spec}\,\Sigma = \{s_\parallel^2 \ (r \text{ times}), s_\perp^2 \ (d-r \text{ times})\}, \tag{402}$$

$$\mathrm{tr}\,\Sigma = r s_\parallel^2 + (d-r) s_\perp^2, \tag{403}$$

$$\log\det\Sigma = r \log s_\parallel^2 + (d-r)\log s_\perp^2. \tag{404}$$

The Gaussian density ratio integrates to

$$2\,\mathrm{KL}(Q\|\Pi) = s_0^{-2}\|W - W_0\|^2 + m\{s_0^{-2}\,\mathrm{tr}\,\Sigma - d + d\log s_0^2 - \log\det\Sigma\} \tag{405}$$

$$= s_0^{-2}\|W - W_0\|^2 + mr\,k(s_\parallel^2/s_0^2) + m(d-r)k(s_\perp^2/s_0^2), \quad k(x) = x - 1 - \log x. \tag{406}$$

Taylor's integral formula and $\mathbb{E}E = 0$ imply

$$\widehat{L}_S(Q) \le \widehat{L}_S(W) + \tfrac{1}{2} m[r h_\parallel s_\parallel^2 + (d-r) h_\perp s_\perp^2]. \tag{407}$$

For a subspace of multiplicity p, minimize over $v > 0$:

$$\phi_h(v) = \tfrac{mp}{2}\{hv + \lambda^{-1}k(v/s_0^2)\}, \tag{408}$$

$$\phi_h'(v) = \tfrac{mp}{2}\{h + \lambda^{-1}(s_0^{-2} - v^{-1})\}, \tag{409}$$

$$\phi_h''(v) = \frac{mp}{2\lambda v^2} > 0, \tag{410}$$

$$\min_{v>0} \phi_h(v) = \frac{mp}{2\lambda}\log(1 + \lambda h s_0^2). \tag{411}$$

Apply this separately to $p = r, d - r$ and substitute into (396). The Hessian condition is global along Gaussian perturbation segments; a local empirical Hessian estimate alone does not verify it for an untruncated Gaussian posterior. □

Corollary 10 (A deterministic center under a loss Lipschitz condition)**.** *If $\ell(\cdot, z)$ is L_ℓ-Lipschitz for every z, then*

$$L(W) \le L(Q) + L_\ell\sqrt{m[rs_\parallel^2 + (d-r)s_\perp^2]}. \tag{412}$$

Proof. $L(W) - L(Q) \le L_\ell \mathbb{E}_Q\|W - \widetilde{W}\| \le L_\ell\sqrt{\mathbb{E}_Q\|W - \widetilde{W}\|^2}$. □

Proposition 17 (Gaussian conditioning and a sharp center-to-Gibbs conversion)**.** *Let $x = \phi(z) \neq 0$ be a feature map fixed before observing S, and let $W \in \mathbb{R}^{m\times d}$, $m \ge 2$, be a multiclass linear head. Use the row-independent Gaussian posterior $Q = \mathcal{N}(\operatorname{vec} W, I_m \otimes \Sigma)$, $\Sigma \succ 0$, and a legal data-independent prior Π. Write φ, Φ for the standard normal density and distribution function. For each training pair,*

$$\nu_i^2 = x_i^\top \Sigma x_i, \qquad \Delta_{ij} = (Wx_i)_{y_i} - (Wx_i)_j, \tag{413}$$

$$p_i = \int_{\mathbb{R}} \varphi(t)\left[1 - \prod_{j\neq y_i} \Phi(t + \Delta_{ij}/\nu_i)\right] dt, \qquad p_Q = n^{-1}\sum_i p_i. \tag{414}$$

The empirical Gibbs classification error is exactly p_Q. With probability at least $1 - \delta$,

$$L_{0/1}(W) \le \min\left\{1, 2\,\mathrm{kl}_+^{-1}\left(p_Q, \frac{\mathrm{KL}(Q\|\Pi) + \log(\Xi_n/\delta)}{n}\right)\right\}. \tag{415}$$

For a simpler empirical margin summary, define

$$\widehat{L}_\gamma(W) = n^{-1}\sum_i \mathbf{1}\{\min_{j\neq y_i} \Delta_{ij} \le \gamma\}, \tag{416}$$

$$B_m(h) = \int_{\mathbb{R}} \varphi(t)[1 - \Phi(t+h)^{m-1}]\, dt, \qquad \nu_{\max} = \max_{i\le n} \nu_i. \tag{417}$$

For $\gamma > 0$, p_Q in (415) may be replaced by

$$\overline{p}_\gamma = \widehat{L}_\gamma(W) + (1 - \widehat{L}_\gamma(W))B_m(\gamma/\nu_{\max}). \tag{418}$$

In particular $B_2(h) = 1 - \Phi(h/\sqrt{2})$, and $B_m(h) \le \min\{1, (m-1)[1 - \Phi(h/\sqrt{2})]\}$ for $h \ge 0$. No global feature-norm bound is needed for this empirical summary.

Proof. Conditional on the true-class perturbation $E_{y_i}x_i = \nu_i t$, the other logit perturbations are independent $\mathcal{N}(0, \nu_i^2)$. The conditional probability of correct classification is $\prod_{j\neq y_i}\Phi(t + \Delta_{ij}/\nu_i)$; integrate to get p_i. If all mean margins exceed γ, monotonicity gives $p_i \le B_m(\gamma/\nu_i) \le B_m(\gamma/\nu_{\max})$. For bad-margin examples use $p_i \le 1$, retaining the exact fraction of good examples. The binary and pairwise bounds use $E_j x - E_y x \sim \mathcal{N}(0, 2\nu(x)^2)$ and the union bound; no coordinatewise-infinity-norm relaxation is needed.

For every test input misclassified by W, select a competitor with mean logit at least the true-class logit. Its centered symmetric Gaussian logit difference makes a Gibbs error with probability at least $1/2$. Hence $L_{0/1}(W) \le 2L_{0/1}(Q)$. This factor two is attained by a binary zero-mean-margin example when ties are counted as center errors. Theorem 12 and monotonicity of the upper KL inverse prove (415). The identities use an independent fixed feature map; an encoder learned from S must be included in the hypothesis space and prior construction. Numerical evaluation of the integrals requires a controlled quadrature error if used for a certificate. □

A.25. Residual geometry instead of full perturbation norms

Lemma 14 (Davis–Kahan via the Sylvester–Rosenblum integral)**.** *Let $A, \widehat{A} = A + E$ be symmetric, let $U_\star$ span the leading r-space of A, and let $\widehat{U}$ span the leading r-space of $\widehat{A}$. Set $k_r = \min(r, d-r)$ and $\delta_{\mathrm{mix}} = \lambda_r(\widehat{A}) - \lambda_{r+1}(A) > 0$. Then*

$$\|\sin\Theta(\widehat{U}, U_\star)\|_F \le \frac{\|U_\perp^\top E \widehat{U}\|_F}{\delta_{\mathrm{mix}}} \tag{419}$$

$$\le \frac{\min\{\|E\|_F, \sqrt{k_r}\|E\|_{\mathrm{op}}\}}{\delta_{\mathrm{mix}}}. \tag{420}$$

If A has gap Δ and $e = \|E\|_{\mathrm{op}} < \Delta$, the denominator may be replaced by $\Delta - e$. The residual prefactor one is optimal.

Proof. This is the separated-cluster Davis–Kahan theorem (Davis & Kahan, 1970; Yu et al., 2015); its short integral proof also explains which perturbations matter. With $X = U_\perp^\top \widehat{U}$ and $R = U_\perp^\top E \widehat{U}$,

$$X\widehat{\Lambda}_1 - \Lambda_2 X = R, \tag{421}$$

$$X = \int_0^\infty e^{t\Lambda_2} R e^{-t\widehat{\Lambda}_1}\, dt, \tag{422}$$

$$\|X\|_F \le \int_0^\infty e^{-t\delta_{\mathrm{mix}}} \|R\|_F\, dt = \|R\|_F/\delta_{\mathrm{mix}}. \tag{423}$$

The integral is the ordered-self-adjoint case of the Sylvester–Rosenblum resolvent construction (Bhatia & Rosenthal, 1997). Compression gives $\|R\|_F \le \|E\|_F$ and $\|R\|_F \le \sqrt{k_r}\|E\|_{\mathrm{op}}$. Weyl gives $\delta_{\mathrm{mix}} \ge \Delta - e$. For a two-dimensional rank-one problem the Sylvester equation has one entry, so the first inequality is equality. □

A.26. Concentrating the normalized covariance in two norms

Lemma 15 (McDiarmid–Tropp covariance envelopes)**.** *Let $X_1, \dots, X_N$ be i.i.d. positive semidefinite $d \times d$ matrices with $\operatorname{tr} X_i \le 1$, mean M, and $\widehat{M} = N^{-1}\sum_i X_i$. Define deterministic population variance budgets*

$$v_F \ge \mathbb{E}\|X_1 - M\|_F^2, \qquad v \ge \|\mathbb{E}(X_1 - M)^2\|_{\mathrm{op}}. \tag{424}$$

The universal choices $v_F = 1$, $v = 1/4$ are valid. Put

$$h_N(\delta; v_F) = \sqrt{v_F/N} + \sqrt{\log(2/\delta)/N}, \tag{425}$$

$$b_N(\delta; v) = \frac{\ell_N}{3N} + \sqrt{\frac{2v\ell_N}{N} + \frac{\ell_N^2}{9N^2}}, \qquad \ell_N = \log(4d/\delta). \tag{426}$$

With probability at least $1 - \delta$, both

$$\|\widehat{M} - M\|_F \le h_N(\delta; v_F), \qquad \|\widehat{M} - M\|_{\mathrm{op}} \le b_N(\delta; v). \tag{427}$$

Smaller verified population variance budgets may be used; empirical variances cannot be substituted without an empirical-Bernstein argument.

Proof. Jensen and independence give $\mathbb{E}\|\widehat{M} - M\|_F \le \sqrt{v_F/N}$. For PSD trace-one contractions,

$$\|X - X'\|_F^2 = \operatorname{tr} X^2 + \operatorname{tr} X'^2 - 2\operatorname{tr}(XX') \le 2. \tag{428}$$

Each bounded difference is therefore $\sqrt{2}/N$; McDiarmid's inequality (McDiarmid, 1989) gives $\Pr(\|\widehat{M} - M\|_F > \sqrt{v_F/N} + t) \le e^{-Nt^2}$. For $Y_i = X_i - M$, $\|Y_i\|_{\mathrm{op}} \le 1$ and

$$\mathbb{E}Y_i^2 = \mathbb{E}X_i^2 - M^2 \preceq M - M^2 \preceq \tfrac{1}{4}I. \tag{429}$$

Tropp's matrix Bernstein theorem (Tropp, 2012, Theorem 1.4) gives

$$\Pr(\|\widehat{M} - M\|_{\mathrm{op}} > t) \le 2d \exp\left(-\frac{Nt^2}{2(v + t/3)}\right). \tag{430}$$

Solve the quadratic exponent exactly at ℓ_N and allocate failure probability $\delta/2$ to each norm. This yields (427), without a power-law dependence on δ^{-1}. □

A.27. Seeger–Maurer inversion instead of a quadratic loss relaxation

Write $\mathrm{kl}(p\|q) = p\log(p/q) + (1-p)\log[(1-p)/(1-q)]$, with the usual continuous endpoint conventions. Define

$$\Xi_n = \sum_{k=0}^{n} \binom{n}{k} (k/n)^k (1-k/n)^{n-k}, \qquad 0^0 = 1, \tag{431}$$

$$\mathrm{kl}_+^{-1}(p, c) = \sup\{q \in [p, 1] : \mathrm{kl}(p\|q) \le c\}. \tag{432}$$

Theorem 12 (Seeger–Maurer PAC-Bayes-kl)**.** *For i.i.d. data, loss in $[0,1]$, and a data-independent prior Π, with probability at least $1-\delta$, simultaneously for all $Q \ll \Pi$,*

$$\mathrm{kl}(\widehat{L}_S(Q)\|L(Q)) \le \frac{\mathrm{KL}(Q\|\Pi) + \log(\Xi_n/\delta)}{n}, \tag{433}$$

$$L(Q) \le \mathrm{kl}_+^{-1}\left(\widehat{L}_S(Q), \frac{\mathrm{KL}(Q\|\Pi) + \log(\Xi_n/\delta)}{n}\right). \tag{434}$$

For $n \ge 8$, $\Xi_n \le 2\sqrt{n}$. If $\widehat{L}_S(Q) = 0$, the displayed inverse is exactly

$$1 - \exp\left[-\frac{\mathrm{KL}(Q\|\Pi) + \log(\Xi_n/\delta)}{n}\right]. \tag{435}$$

This preserves the binary Chernoff geometry and has no temperature to tune. It is not claimed to dominate every separately prespecified-temperature bound at every empirical loss.

Proof. Seeger's PAC-Bayes-kl argument and Maurer's convex-extremal moment bound (Seeger, 2002; Maurer, 2004) give, for each fixed predictor,

$$\mathbb{E}_S \exp\{n\,\mathrm{kl}(\widehat{L}_S(w)\|L(w))\} \le \Xi_n. \tag{436}$$

For a nondegenerate Bernoulli loss the expectation equals Ξ_n exactly: its probability factor $p^k(1-p)^{n-k}$ cancels the denominator inside the exponential. For losses in $[0,1]$, the separately convex envelope used by Maurer reduces the exponential moment to the Bernoulli case. Fubini and Markov give, on one event,

$$\log \mathbb{E}_\Pi e^{n\,\mathrm{kl}(\widehat{L}_S(w)\|L(w))} \le \log(\Xi_n/\delta). \tag{437}$$

The Gibbs–Donsker–Varadhan entropy variational inequality (Donsker & Varadhan, 1975), followed by joint convexity of binary relative entropy yields

$$n\,\mathrm{kl}(\widehat{L}_S(Q)\|L(Q)) \le n\mathbb{E}_Q\,\mathrm{kl}(\widehat{L}_S(w)\|L(w)) \tag{438}$$

$$\le \mathrm{KL}(Q\|\Pi) + \log(\Xi_n/\delta). \tag{439}$$

This is the standard change-of-measure step, not a new PAC-Bayes theorem. The inverse and its $p = 0$ expression follow by monotonicity for $q \ge p$. Maurer's moment estimate supplies $\Xi_n \le 2\sqrt{n}$ for $n \ge 8$. □

A.28. Generalization with neutral routing states and summable feedback

Theorem 13 (Replacement stability without strict routing contraction)**.** *Suppose a valid common-randomness neighboring-data coupling admits two expected distances $w_t, u_t \ge 0$, initially zero, satisfying*

$$w_{t+1} \le q w_t + b_t u_t + B_W/n, \qquad 0 \le q < 1, \tag{440}$$

$$u_{t+1} \le (1 + d_t)u_t + c_t w_t + B_{U,t}/n. \tag{441}$$

Here $c_t, d_t, B_{U,t}, B_W \ge 0$. The first block controls parameter distance; any eliminated moment states must be accounted for in these inequalities. Assume deterministic uniform bounds

$$0 \le b_t \le B, \quad C_* := \sum_{t\ge0} c_t < \infty, \quad D_* := \sum_{t\ge0} d_t < \infty, \quad J_* := \sum_{t\ge0} B_{U,t} < \infty. \tag{442}$$

Then, with $A_* = D_* + BC_*/(1-q)$,

$$\sup_t u_t \le \frac{e^{A_*}}{n}\left(J_* + \frac{C_* B_W}{1-q}\right) =: U_*, \tag{443}$$

$$\sup_t w_t \le \frac{B_W}{n(1-q)} + \frac{B}{1-q} U_*. \tag{444}$$

For L_ℓ*-Lipschitz evaluation losses, the corresponding expected replacement-stability bound is* $\beta_t \le L_\ell w_t$ *uniformly in time. The routing diagonal in* (441) *may equal one or exceed one; no uniform positive routing eigengap is assumed in this theorem.*

Proof. Unroll the first recurrence:

$$w_t \le \frac{B_W}{n(1-q)} + \sum_{j<t} q^{t-1-j} b_j u_j. \tag{445}$$

For $V_t = \max_{j \le t} u_j$,

$$w_t \le \frac{B_W}{n(1-q)} + \frac{B}{1-q} V_t, \tag{446}$$

$$V_{t+1} \le \left(1 + d_t + \frac{B c_t}{1-q}\right) V_t + \frac{1}{n}\left(B_{U,t} + \frac{c_t B_W}{1-q}\right). \tag{447}$$

The right side is at least V_t, so it also bounds the new maximum. Iteration and $\prod_j (1+a_j) \le e^{\sum_j a_j}$ imply

$$V_T \le \frac{1}{n} \sum_{t<T} \left(B_{U,t} + \frac{c_t B_W}{1-q}\right) \prod_{j=t+1}^{T-1} \left(1 + d_j + \frac{B c_j}{1-q}\right), \tag{448}$$

which proves (444). Apply loss Lipschitzness. □

Corollary 11 (Geometrically decaying feedback budgets)**.** *If* $c_t \le C_1 a^t$, $d_t \le D_1 a^t$, $B_{U,t} \le J_1 a^{2t}$ *with* $0 < a < 1$, *one may take*

$$C_* = \frac{C_1}{1-a}, \qquad D_* = \frac{D_1}{1-a}, \qquad J_* = \frac{J_1}{1-a^2}. \tag{449}$$

For per-example gradients bounded by G_t *and* L_g*-Lipschitz, the normalized-covariance primitives satisfy*

$$\|\mathcal{C}_{\varepsilon_c}(G_z(W_t))\|_{\mathrm{op}} \le G_t^2/\varepsilon_c, \tag{450}$$

$$\|D_W \mathcal{C}_{\varepsilon_c}(G_z(W_t))\|_{\mathrm{op}} \le L_g(2G_t/\varepsilon_c + 2G_t^3/\varepsilon_c^2), \tag{451}$$

$$\|\mathcal{C}_{\varepsilon_c}(G_z(W_t)) - \mathcal{C}_{\varepsilon_c}(G_{z'}(W_t))\| \le 2G_t^2/\varepsilon_c. \tag{452}$$

These quantify how covariance feedback and sample injection can vanish near an interpolating solution. A signed contraction for the parameter block and a valid reduction of all moment states are still separate requirements of (441).

Proof. Sum the geometric series. Bound the two first-differential terms in Lemma 12 by $2\|G\|\|E\|/\varepsilon_c$ and $2\|G\|^3\|E\|/\varepsilon_c^2$; use the triangle inequality for the changed-example term. □

A.29. Anisotropic prior geometry: an exact angle-dependent KL identity

Theorem 14 (Prior–posterior orientation cost)**.** *Let* P_0 *be a rank-*r *projector independent of the training set, and* P *any data-dependent rank-*r *projector. Independence may hold conditional on an auxiliary pilot sample. Define full-rank row covariances*

$$\Sigma_0 = a_0 P_0 + b_0(I - P_0), \qquad a_0, b_0 > 0, \tag{453}$$

$$\Sigma = aP + b(I - P), \qquad a, b > 0, \tag{454}$$

and Gaussian laws $\Pi = \mathcal{N}(\operatorname{vec} W_0, I_m \otimes \Sigma_0)$, $Q = \mathcal{N}(\operatorname{vec} W, I_m \otimes \Sigma)$ *with row-wise vectorization. Put* $D = W - W_0$, $s^2 = \|\sin\Theta(P, P_0)\|^2$ *and* $k(x) = x - 1 - \log x$. *Then*

$$\mathrm{KL}(Q\|\Pi) = \frac{1}{2}\left(\frac{\|DP_0\|^2}{a_0} + \frac{\|D(I-P_0)\|^2}{b_0}\right) + \frac{m}{2}\left[rk(a/a_0) + (d-r)k(b/b_0) + (a-b)(b_0^{-1} - a_0^{-1})s^2\right]. \tag{455}$$

In the matched-variance case $a = a_0$, $b = b_0$, $\kappa = a/b$,

$$\mathrm{KL}(Q\|\Pi) = \frac{1}{2}\left(\frac{\|DP_0\|^2}{a} + \frac{\|D(I-P_0)\|^2}{b}\right) + \frac{m}{2}\frac{(\kappa-1)^2}{\kappa}s^2. \tag{456}$$

For an isotropic prior $a_0 = b_0$, *the orientation coefficient is exactly zero, consistently with Proposition 12.*

Proof. Projector trace identities give

$$\operatorname{tr}(P_0 P) = r - s^2, \qquad \operatorname{tr}(P_0(I-P)) = s^2, \tag{457}$$

$$\operatorname{tr}((I-P_0)P) = s^2, \qquad \operatorname{tr}((I-P_0)(I-P)) = d - r - s^2. \tag{458}$$

Since $\Sigma_0^{-1} = a_0^{-1}P_0 + b_0^{-1}(I - P_0)$,

$$\operatorname{tr}(\Sigma_0^{-1}\Sigma) = ra/a_0 + (d-r)b/b_0 + (a-b)(b_0^{-1} - a_0^{-1})s^2, \tag{459}$$

$$\log\frac{\det\Sigma_0}{\det\Sigma} = r\log(a_0/a) + (d-r)\log(b_0/b). \tag{460}$$

The Gaussian density ratio therefore integrates to

$$2\,\mathrm{KL}(Q\|\Pi) = \operatorname{tr}(D\Sigma_0^{-1}D^\top) + m\{\operatorname{tr}(\Sigma_0^{-1}\Sigma) - d + \log(\det\Sigma_0/\det\Sigma)\}. \tag{461}$$

Expand the mean term using $P_0(I - P_0) = 0$ and collect the variance terms. For matched variances, $(a-b)(b^{-1} - a^{-1}) = (a-b)^2/(ab) = (\kappa-1)^2/\kappa$. □

A.30. From a tracking certificate to a valid pilot-prior generalization bound

Let $S = (z_i)_{i=1}^n$ and $S_0 = (z_j^0)_{j=1}^{n_0}$ be independent samples, and fix W_0 independently of both. This subsection uses one-example uniform sampling for its covariance definition:

$$\psi(W, z) = \frac{G_z(W)^\top G_z(W)}{\|G_z(W)\|^2 + \varepsilon_c}, \tag{462}$$

$$C_S(W) = n^{-1}\sum_i \psi(W, z_i), \qquad \widehat{C}_0 = n_0^{-1}\sum_j \psi(W_0, z_j^0), \tag{463}$$

$$C_{\mathrm{pop}}(W_0) = \mathbb{E}_z\psi(W_0, z), \qquad P_0 = \mathrm{Top}_r(\widehat{C}_0). \tag{464}$$

Choose a fixed measurable tie convention for Top_r. Conditional on S_0, a Gaussian prior with this P_0 and fixed mean W_0 is independent of the training set S.

Lemma 16 (A two-norm, fixed-initialization concentration bridge)**.** *Assume* G_z *is* L_g*-Lipschitz on the comparison domain and put* $L_F = 2L_g/\sqrt{\varepsilon_c}$, $L_{\mathrm{op}} = L_g/\sqrt{\varepsilon_c}$. *For* $\delta_S, \delta_0 \in (0,1)$, *define*

$$E_F(W) = \min\{\sqrt{2},\ L_F\|W - W_0\| + h_n(\delta_S; 1) + h_{n_0}(\delta_0; 1)\}, \tag{465}$$

$$E_{\mathrm{op}}(W) = \min\{1,\ L_{\mathrm{op}}\|W - W_0\| + b_n(\delta_S; 1/4) + b_{n_0}(\delta_0; 1/4)\}. \tag{466}$$

With probability at least $1 - \delta_S - \delta_0$, *simultaneously for every* W *in that domain,*

$$\|C_S(W) - \widehat{C}_0\|_F \le E_F(W), \qquad \|C_S(W) - \widehat{C}_0\|_{\mathrm{op}} \le E_{\mathrm{op}}(W). \tag{467}$$

The envelopes are those of Lemma 15.

Proof. Apply that lemma separately to the training and pilot averages at the fixed, data-independent W_0. Then use, in each norm,

$$C_S(W) - \widehat{C}_0 = [C_S(W) - C_S(W_0)] + [C_S(W_0) - C_{\mathrm{pop}}(W_0)] + [C_{\mathrm{pop}}(W_0) - \widehat{C}_0]. \tag{468}$$

Lemma 13 supplies L_F; Lemma 12 supplies L_{op}. All terms are PSD trace-one contractions, so their difference has operator norm at most one and Frobenius norm at most $\sqrt{2}$. This justifies the caps. Only the initial point is concentrated; the Lipschitz bridge is pointwise in every W, including a fitted W_T. □

Theorem 15 (Exact PAC-Bayes certificate with a residual-based tracking envelope)**.** *Use the pilot prior and Gaussian posterior of Theorem 14, with a_0, b_0 fixed before observing S conditional on S_0, and let K_T^{obs} be the exact right side of (455) with $P = P_T$, $W = W_T$. No tracking or eigengap assumption is needed for*

$$L(Q_T) \le \mathrm{kl}_+^{-1}\left(\widehat{L}_S(Q_T), \frac{K_T^{\mathrm{obs}} + \log(\Xi_n/\delta_{\mathrm{PB}})}{n}\right), \tag{469}$$

which holds with probability at least $1 - \delta_{\mathrm{PB}}$.

For a structural tracking envelope, additionally suppose an event of probability at least $1 - \delta_{\mathrm{tr}}$ gives

$$\Delta_r(C_S(W_T)) \ge \Delta_0 > 0, \quad h_{C_S(W_T)}(P_T) \le H_T, \quad E_{\mathrm{op}}(W_T) < \Delta_0. \tag{470}$$

Let $\gamma_0 = (a-b)(b_0^{-1} - a_0^{-1}) \ge 0$, $k_r = \min(r, d-r)$, and

$$B_T = \frac{\min\{E_F(W_T), \sqrt{k_r} E_{\mathrm{op}}(W_T)\}}{\Delta_0 - E_{\mathrm{op}}(W_T)}, \tag{471}$$

$$J_T = \min\{k_r, (\sqrt{2H_T/\Delta_0} + B_T)^2\}, \tag{472}$$

$$K_T^{\mathrm{str}} = \frac{1}{2}\left(\frac{\|(W_T - W_0)P_0\|^2}{a_0} + \frac{\|(W_T - W_0)(I - P_0)\|^2}{b_0}\right) + \frac{m}{2}\{rk(a/a_0) + (d-r)k(b/b_0) + \gamma_0 J_T\}. \tag{473}$$

With probability at least $1 - \delta_S - \delta_0 - \delta_{\mathrm{tr}}$, $K_T^{\mathrm{obs}} \le K_T^{\mathrm{str}}$. Thus (469) remains valid with K_T^{str} after adding those failure probabilities. For reporting a certificate, the observed-angle cost is preferable: it is exact and does not pay the additional concentration or tracking failures. The structural envelope explains how tracking controls that cost.

Proof. Conditional on every pilot sample, Theorem 12 holds simultaneously for all posteriors. Use the exact Gaussian identity of Theorem 14 and integrate over S_0. For the structural statement put $P_T^\star = \mathrm{Top}_r(C_S(W_T))$. On the concentration and tracking events,

$$s(P_T, P_T^\star) \le \sqrt{2H_T/\Delta_0}, \tag{474}$$

$$s(P_T^\star, P_0) \le B_T. \tag{475}$$

The second line is Lemma 14, with $A = C_S(W_T)$ and $\widehat{A} = \widehat{C}_0$. Its mixed gap is at least $\Delta_0 - E_{\mathrm{op}}(W_T) > 0$. Chordal triangle inequality and $s(P,Q)^2 \le k_r$ give $s(P_T, P_0)^2 \le J_T$. Insert this in the KL identity, using $\gamma_0 \ge 0$, and union bound. No independence between the auxiliary good events is needed. In particular, gap collapse invalidates the structural tracking envelope, not the exact PAC-Bayes certificate with an independent prior. □

Corollary 12 (Clopper–Pearson evaluation of the empirical Gibbs loss)**.** *For a $0/1$ evaluation loss, after selecting Q_T and before inspecting its Monte Carlo evaluations, fix M and draw independent pairs $\theta_j \sim Q_T$, $I_j \sim \mathrm{Unif}\{1, \dots, n\}$. Let $K = \sum_{j=1}^M \ell(\theta_j, z_{I_j})$ and*

$$p_{\mathrm{CP}} = \begin{cases} \mathrm{Beta}^{-1}(1 - \delta_{\mathrm{MC}}; K+1, M-K), & K < M, \\ 1, & K = M. \end{cases} \tag{476}$$

With joint probability at least $1 - \delta_{\mathrm{PB}} - \delta_{\mathrm{MC}}$,

$$L(Q_T) \le \mathrm{kl}_+^{-1}\left(p_{\mathrm{CP}}, \frac{K_T^{\mathrm{obs}} + \log(\Xi_n/\delta_{\mathrm{PB}})}{n}\right). \tag{477}$$

For $K = 0$, $p_{\mathrm{CP}} = 1 - \delta_{\mathrm{MC}}^{1/M}$, not zero.

Proof. Conditional on S, S_0, Q_T, the indicators are i.i.d. Bernoulli with mean $\widehat{L}_S(Q_T)$. The one-sided Clopper–Pearson construction (Clopper & Pearson, 1934) gives $\Pr(\widehat{L}_S(Q_T) > p_{\mathrm{CP}} \mid S, S_0, Q_T) \le \delta_{\mathrm{MC}}$. The upper binary-KL inverse is nondecreasing in its first argument: for $p < q$, $\partial_p \operatorname{kl}(p\|q) \le 0$ and $\partial_q \operatorname{kl}(p\|q) > 0$. Apply this monotonicity to (469) and union bound the two failures. □

A.31. Orientation complexity from Haar measure and Gibbs duality

Let $\mu_{r,d}$ be normalized Haar measure on $\mathrm{Gr}(r,d)$ (Chikuse, 2003). Fix $a \ge b > 0$ and a data-independent W_0. For $P_0 \sim \mu_{r,d}$, write

$$\Pi_{P_0} = \mathcal{N}(\operatorname{vec} W_0, I_m \otimes [aP_0 + b(I - P_0)]), \tag{478}$$

$$\Pi_H = \int \Pi_{P_0}\, d\mu_{r,d}(P_0), \qquad Q = \mathcal{N}(\operatorname{vec} W, I_m \otimes [aP + b(I-P)]). \tag{479}$$

The prior Π_H is independent of the fitted orientation P.

Theorem 16 (Haar mixture and the Gibbs variational principle)**.** *Put $D = W - W_0$, $\kappa = a/b$, $c = m(\kappa-1)^2/(2\kappa)$, and*

$$V(P_0) = \tfrac{1}{2}\left(\frac{\|DP_0\|^2}{a} + \frac{\|D(I-P_0)\|^2}{b}\right) + c\, s(P,P_0)^2, \tag{480}$$

$$Z_{d,r}(c) = \int e^{-c\, s(P,P_0)^2}\, d\mu_{r,d}(P_0), \qquad \Phi_{d,r}(c) = -\log Z_{d,r}(c). \tag{481}$$

Then

$$\mathrm{KL}(Q\|\Pi_H) \le -\log \int e^{-V(P_0)}\, d\mu_{r,d}(P_0) \tag{482}$$

$$\le \frac{\|D\|^2}{2b} + \Phi_{d,r}(c). \tag{483}$$

The partition function $Z_{d,r}$ is independent of P. For $D_r = r(d-r)$,

$$\Phi_{d,r}(0) = 0, \qquad 0 \le \Phi_{d,r}(c) \le cD_r/d, \tag{484}$$

$$\Phi'_{d,r}(c) = \mathbb{E}_{\nu_c} s(P,P_0)^2, \qquad \Phi''_{d,r}(c) = -\operatorname{Var}_{\nu_c}(s(P,P_0)^2) \le 0, \tag{485}$$

$$\frac{d\nu_c}{d\mu_{r,d}} = Z_{d,r}(c)^{-1} e^{-cs(P,P_0)^2}. \tag{486}$$

The first bound retains coupling between mean displacement and orientation; the second is a simpler closed envelope. Neither is asserted to equal the actual marginal KL in general. For $a = b$, both equal the Gaussian mean-shift KL exactly and the orientation term vanishes.

Proof. For any $\nu \ll \mu_{r,d}$, compare the joint laws $\nu(dP_0)Q(dW)$ and $\mu_{r,d}(dP_0)\Pi_{P_0}(dW)$. The KL chain rule and data-processing inequality (Cover & Thomas, 2006) give

$$\mathrm{KL}(Q\|\Pi_H) \le \mathrm{KL}(\nu\|\mu_{r,d}) + \mathbb{E}_\nu \mathrm{KL}(Q\|\Pi_{P_0}) \tag{487}$$

$$= \mathrm{KL}(\nu\|\mu_{r,d}) + \mathbb{E}_\nu V(P_0). \tag{488}$$

The equality is Theorem 14. Minimize over ν using the Gibbs variational principle (Donsker & Varadhan, 1975):

$$\inf_\nu \{\mathbb{E}_\nu V + \mathrm{KL}(\nu\|\mu_{r,d})\} = -\log \mathbb{E}_{\mu_{r,d}} e^{-V}. \tag{489}$$

Bounding the mean term by $\|D\|^2/(2b)$ gives the second bound. Haar invariance eliminates P from Z and gives $\mathbb{E}_{\mu_{r,d}} P_0 = (r/d)I$, so Jensen yields $\Phi(c) \le c\mathbb{E}s^2 = cD_r/d$. Differentiate the compact-domain integral to obtain the exact first and second derivatives. □

Corollary 13 (Exact weak-anisotropy curvature)**.** *For $d > 1$ and $D_r = r(d-r)$,*

$$\Phi_{d,r}(c) = \frac{D_r}{d}c - \frac{D_r^2}{d^2(d-1)(d+2)}c^2 + O(c^3) \qquad (c \downarrow 0). \tag{490}$$

Proof. A diagonal entry of a Haar rank-r projector has distribution $\mathrm{Beta}(r/2,(d-r)/2)$, with variance $2r(d-r)/[d^2(d+2)]$. Exchangeability and the fixed trace give covariance $-1/(d-1)$ times this variance between distinct diagonal entries. Therefore $\mathrm{Var}(\mathrm{tr}(PP_0)) = 2D_r^2/[d^2(d-1)(d+2)]$. Insert this into $\Phi'(0) = \mathbb{E}s^2$, $\Phi''(0) = -\mathrm{Var}(s^2)$. □

Proposition 18 (Wishart–Jacobi evaluation and the exact entropy exponent)**.** *Put $k = \min(r, d-r)$, $u = k/2$, $v = (d-k)/2$, and $D_r = k(d-k)$. For the multivariate gamma and beta functions,*

$$\Gamma_k(z) = \pi^{k(k-1)/4} \prod_{j=1}^{k} \Gamma(z-(j-1)/2), \tag{491}$$

$$B_k(u,v) = \Gamma_k(u)\Gamma_k(v)/\Gamma_k(u+v), \tag{492}$$

both $u, v > (k-1)/2$, so the matrix-beta integral is finite:

$$Z_{d,r}(c) = \frac{1}{B_k(u,v)} \int_{0 \prec Y \prec I} e^{-c(k-\mathrm{tr}\, Y)} |Y|^{-1/2} |I-Y|^{(d-2k-1)/2}\, dY. \tag{493}$$

As $c \to \infty$ at fixed d, r,

$$Z_{d,r}(c) \sim \frac{\Gamma_k(d/2)}{\Gamma_k(k/2)} c^{-D_r/2}, \tag{494}$$

$$\Phi_{d,r}(c) = \tfrac{1}{2} D_r \log c + \log \Gamma_k(k/2) - \log \Gamma_k(d/2) + o(1). \tag{495}$$

For $k = 1$, a scalar form is

$$Z_{d,1}(c) = {}_1F_1((d-1)/2; d/2; -c). \tag{496}$$

Thus the orbital term automatically balances concentration and entropy, without a chart-count factor or a chosen covering resolution.

Proof. Complementation preserves chordal distance, so reduce to rank $k \le d/2$. Generate a Haar frame by normalizing a $d \times k$ standard Gaussian matrix. Split it into k and $d-k$ rows. Their Gram matrices are independent Wishart matrices, and the normalized upper Gram has the matrix-beta law $\mathrm{Beta}_k(k/2,(d-k)/2)$ (Muirhead, 1982, Chapter 3). Its eigenvalues are the squared canonical cosines; $s^2 = k - \mathrm{tr}\, Y$ gives (493). For completeness, the asymptotic constant follows directly. Set $S = I - Y$, then $T = cS$. On $\mathrm{tr}\, S < 1/2$, the rescaled integrand is

$$c^{-D_r/2} e^{-\mathrm{tr}\, T} |T|^{v-(k+1)/2} |I - T/c|^{-1/2}. \tag{497}$$

The last factor is at most $2^{k/2}$; dominated convergence gives the matrix-gamma integral $\Gamma_k(v)$. On $\mathrm{tr}\, S \ge 1/2$, the original normalized integral is at most $e^{-c/2}$, negligible after multiplication by $c^{D_r/2}$. Therefore $Z(c)c^{D_r/2} \to \Gamma_k(v)/B_k(u,v) = \Gamma_k(d/2)/\Gamma_k(k/2)$. The $k=1$ formula is the scalar beta Laplace transform, equivalently Kummer's confluent hypergeometric integral. □

Corollary 14 (Rank and covariance selection with a fixed mixture prior)**.** *For a predeclared countable mixture $\Pi = \sum_{r,j} \pi_{r,j} \Pi_{H,r,a_j,b_j}$, $\pi_{r,j} > 0$, $\sum_{r,j} \pi_{r,j} = 1$, either bound in Theorem 16 remains valid after adding $\log(1/\pi_{r,j})$. Rank and covariance scale may then be selected after training. The parameter displacement cost remains present; low-rank optimizer state is not, by itself, a low-capacity model assumption.*

Proof. The mixture density is at least $\pi_{r,j}$ times that component density. Integrate its log ratio under Q and apply Theorem 16. □

A.32. Closing the risk decomposition under an empirical PL condition

Theorem 17 (Stationarity plus state stability implies an excess-risk bound)**.** *Let S be i.i.d., $\widehat{L}_S = n^{-1}\sum_i \ell(\cdot, z_i)$, and let τ be independent uniform on $\{1,\dots,T\}$. Suppose all quantities below are integrable, replacement stability at step t is β_t, and*

$$\mathbb{E}\|\nabla \widehat{L}_S(W_\tau)\|^2 \le R_T, \tag{498}$$

$$\widehat{L}_S(W_t) - \inf_{w \in \mathcal{W}} \widehat{L}_S(w) \le \frac{\|\nabla \widehat{L}_S(W_t)\|^2}{2\mu_f} \quad \text{at every reachable iterate, for every } S, \qquad \mu_f > 0. \tag{499}$$

Then

$$\mathbb{E}L(W_\tau) - \inf_{w\in\mathcal{W}} L(w) \le \frac{1}{T}\sum_{t=1}^{T}\beta_t + \frac{R_T}{2\mu_f}. \tag{500}$$

One may substitute the full-state stability product, the contractive feedback bound, or Theorem 13 for β_t. Substituting Theorem 5 for R_T additionally requires its stochastic-gradient hypotheses for the empirical *objective, uniformly over S. This theorem does not assert empirical PL or those noise hypotheses for the deep-network benchmarks.*

Proof. Couple the two runs using the same τ as well as all training randomness. Their loss difference is at most $\bar{\beta}_T = T^{-1}\sum_t \beta_t$. For every fixed data-independent comparator $w \in \mathcal{W}$,

$$\begin{aligned}\mathbb{E}[L(W_\tau) - L(w)] &= \mathbb{E}[L(W_\tau) - \widehat{L}_S(W_\tau)] \\ &\quad + \mathbb{E}[\widehat{L}_S(W_\tau) - \inf_v \widehat{L}_S(v)] \\ &\quad + \mathbb{E}[\inf_v \widehat{L}_S(v) - \widehat{L}_S(w)] + \mathbb{E}[\widehat{L}_S(w) - L(w)] \qquad (501)\\ &\le \bar{\beta}_T + \frac{1}{2\mu_f}\mathbb{E}\|\nabla\widehat{L}_S(W_\tau)\|^2. \qquad (502)\end{aligned}$$

The third term is nonpositive and the fourth is zero by i.i.d. sampling. Take the infimum over fixed comparators. No existence of an attained population minimizer is required. □

B. Experimental Coverage, Provenance, and Reproducibility

B.1. What was recovered and what is being rerun

The historical archive contains optimizer sources, tuning summaries, epoch trajectories, resource probes, and several incomplete training batches. It was recovered from the original three-GPU training host rather than reconstructed from the paper's rounded table. The original environment reports PyTorch 2.10.0 with CUDA 12.8 on three RTX 6000 Ada GPUs with 48 GB each. Existing data caches for MNIST, CIFAR-10/100, Planetoid, ogbn-arxiv, and MovieLens were accessed and their resolved locations recorded. Additional public datasets are materialized before their queued runs. The machine-readable inventories are `revision/hpc_inventory.json` and `revision/remote_status/data_access.json`.

The extension uses an isolated source and result tree. Each task–method pair has three screening learning rates, selected by its final validation score, followed by a fresh, full-budget formal run. Formal runs use seed 42 only, with no early stopping. A classification epoch is a complete pass over the designated training split, not a capped number of mini-batches. Graphs and PINNs use one full-graph or freshly sampled collocation update per epoch. Generated regression uses 2000 full-batch epochs. Screening runs are never pooled with formal endpoints.

Every formal run stores its configuration before training, source hashes, routed parameter names and actual ranks, full epoch records, checkpointed optimizer/RNG states, validation scores, final test score, persistent state bytes, and peak allocated GPU memory. Test scores are evaluated only after formal training; no test-best checkpoint is selected. Tables are regenerated from completed result files after checking the epoch count and ordering. The provenance ledger is `revision/status.json`. Missing, failed, or interrupted runs are not numerical results.

B.2. Representative configurations for all proposal tasks

Table 5. Proposal coverage. The experiment manifest contains 27 representative configurations and six optimizer methods per configuration. Listing a configuration does not imply its training has finished; completion is reported in Table 17 and the provenance ledger.

Family	Datasets or generated tasks	Representative model / epochs
Regression / MLP	Matrix and planted-low-rank-target regression at input dimensions 128, 256, 512; MNIST	Linear matrix model / 2000; deep MLP / 2000
CNN	CIFAR-10, CIFAR-100, Tiny-ImageNet, ImageNet-100	ResNet-18 / 1000
ViT	CIFAR-100, Tiny-ImageNet, ImageNet-100	ViT-Tiny / 1000
Graph	Cora, CiteSeer, PubMed, ogbn-arxiv	two-layer GraphSAGE / 1000
Scientific learning	Poisson, forced Burgers, variable-coefficient Darcy, periodic Burgers solution operator	PINN, DeepONet, FNO / 2000
Recommendation	MovieLens-1M, Amazon Books subset, Yelp subset	MF-BPR / 1000
Continual learning	Rotated MNIST; Split CIFAR-100	MLP; ResNet-18 / 1000 total

Data and split definitions. Classification reserves a deterministic 10% validation split from training data. The official test set, or official validation images for ImageNet-style data, supplies the held-out test endpoint. CIFAR inputs are 32×32; Tiny-ImageNet and ImageNet-100 use 64×64 in this extension. The latter is a stated resolution-specific experiment, not a standard-resolution ImageNet result. ViT-Tiny uses patch size 4 at 32 pixels and 8 at 64 pixels. ImageNet-100 is the public `https://huggingface.co/datasets/clane9/imagenet-100` release. Tiny-ImageNet uses the Stanford release. Planetoid and ogbn-arxiv retain their public splits; every graph model in the extension is GraphSAGE, so the archived Cora GCN comparison is not silently reused as a SAGE result.

Recommendation uses the original MovieLens rating file and the author-released Amazon Books/Yelp2018 interaction lists at `https://github.com/gusye1234/LightGCN-PyTorch/tree/master/data`. Amazon/Yelp subsets select up to 2000 users by hashing their identifiers with the fixed seed string. MovieLens retains all users with at least five positive interactions from the complete input rating file. For each user the last two interactions in the input order are held out for validation and testing; MovieLens input is time sorted, while the author-preprocessed Amazon/Yelp order is not claimed to be chronological. MF has 64-dimensional embeddings and is trained with BPR and sampled unobserved negatives. Evaluation ranks the entire subset catalog, masking observed training items and the validation item for test evaluation. With one held-out target per user, Recall@20 is the hit rate. The Amazon/Yelp catalogs are the subset catalogs; MovieLens uses the full eligible catalog. This differs from the archived explicit-rating MF, whose Recall denominator was capped by 20.

The Poisson PINN solves $-u_{xx} = \pi^2 \sin(\pi x)$ on $[0, 1]$ with zero Dirichlet values. The new forced Burgers task uses the exact solution $u(x, t) = -e^{-t} \sin(\pi x)$, the corresponding manufactured forcing, zero spatial boundary values, and its exact initial condition. It differs from the archived unforced PINN. Both new PINNs draw 1024 interior collocation points per training update and use independent fixed validation/test coordinates. Their reported metric is actual relative solution error against the exact solution, with PDE residual reported separately.

Darcy data solve $-\nabla \cdot (a \nabla u) = 1$ on a 32-by-32 interior grid with zero Dirichlet values. Permeability is the exponential of a random smooth cosine field. Harmonic face conductivities define the sparse finite-difference operator, solved independently for each realization. DeepONet receives 64 permeability sensors and predicts the solution grid. The FNO task uses a periodic one-dimensional viscous Burgers solution operator on 64 points, $\nu = 0.01$, final time 0.1, with a dealiased pseudospectral RK4 solver at step 0.0005. Its real/imaginary spectral parameters are stored as paired real coordinates for compatibility with all optimizers. Each operator dataset has 256 training, 64 validation, and 64 test functions. These small generated operator benchmarks are not claims of state-of-the-art PDE accuracy.

Continual tasks have no replay and a single classifier head without a task identifier at inference. The 1000 total epochs are distributed equally over four MNIST rotations or five disjoint 20-class CIFAR tasks. A global learning rate schedule is used. Task-boundary validation matrices permit a separate forgetting calculation; final held-out average test accuracy remains distinct.

B.3. Methods and hyperparameter selection

All methods receive the same training/validation/test split and a fresh paired model initialization. The optimizer's random basis initialization is isolated from subsequent data, dropout, and collocation random streams. The extension compares SGD with momentum 0.9, AdamW, fixed random routing, periodic SVD at interval 25, GrassRoute, and the official SubTrack++ implementation pinned to commit `f85bfb369a4a73b26f2e46459abc1e20863185f0`. The official optimizer is imported from its source; a shape-only wrapper flattens routed convolution tensors during its update and restores their parameter/gradient shapes. Its own projection-aware moments and recovery scaling are retained. It uses interval 25 and an iteratively decreasing raw-gradient subspace step, initialized at 1.0. This is not the archive's unofficial geodesic surrogate. Its projection side may differ from the right-sided GrassRoute convention, following its upstream implementation.

SGD screens learning rates $\{0.01, 0.1, 0.3\}$. For regression, graph, and scientific tasks the adaptive methods screen $\{0.001, 0.003, 0.01\}$; other adaptive tasks screen $\{0.0003, 0.001, 0.003\}$. Screening lasts 100 epochs for regression, graph, and scientific tasks, and 30 for others. These budgets are equal across methods within each task. The GrassRoute tracking rate is fixed at 3.0 on graph tasks and 0.3 elsewhere; it is not claimed to be exhaustively optimized. Main and residual parameter rates share the cosine schedule with a 1% floor; ViT adds 5% warmup. Vision uses weight decay 0.01; other extension tasks use zero except the explicit BPR embedding penalty. Divergent screen configurations are recorded and excluded from selection; a failed formal run is not replaced by a test-informed alternative.

Routed ranks start from 32 for CNN, 16 for ViT/MLP/recommendation, and 8 for graph/scientific tasks, capped at half the output dimension and below the input dimension. Small parameters use full AdamW fallback. The actual per-layer values, which can differ from these starting values, are recorded in each configuration. In contrast to the historical driver, these values are set directly in the parameter groups rather than merely passed as an optimizer default. GrassRoute uses the diagonal transport in Table 2. The fixed variant does not change its basis. An implementation audit found that the local periodic comparator transports its first moment at refreshes but retains its second-moment diagonal in the old coordinates, despite accepting the same configuration flag. The original extension endpoints therefore describe that specific periodic implementation; they do not isolate the effect of the subspace update under identical moment transport. They and their source hashes are preserved. The independent follow-up in Section B.10 includes a corrected PERIODICDIAG comparator, which applies $V \leftarrow V(R^\top \odot R^\top)$ after each refresh while retaining its moment age. No running extension checkpoint is silently converted to a different algorithm. GrassRoute uses $\beta_1 = 0.9$, $\beta_2 = 0.999$, subspace momentum $\mu_H = 0.9$, Adam offset 10^{-8}, covariance/residual offsets 10^{-12}, residual clipping $[0, 10]$, and FP32 routing states. The official SubTrack++ Adam offset remains its upstream default 10^{-6}; other Adam branches use 10^{-8}. In particular, the positive residual-floor condition of Corollary 4 is a sufficient regime condition, not a property asserted for these zero-lower-clip configurations.

Numerical recovery of the independent graph follow-up. Numerically invalid screening candidates are retained as failures and excluded from selection while the remaining candidates continue. Two searches initially stopped on a non-finite SVD input; the recovery policy records those failures and resumes the unchanged candidate grids. If a selected full run fails

numerically, the next candidate in screening-validation order is attempted, with every failed attempt preserved and no test score used for this fallback. The CPU SubTrack++ follow-up additionally adapts six hardcoded CUDA allocation targets to the current gradient device at runtime. The pinned source files and all projection/moment arithmetic remain unchanged; both projector orientations are checked through repeated CPU refreshes. This device adapter is recorded separately from the numerical-failure policy. For CiteSeer's periodic search, the same failed candidate completed in a fresh diagnostic process. A separately recorded recovery stage therefore permits one fresh-process retry of each numerically failed candidate, preserving its initial failure and keeping the nine unique hyperparameters unchanged. Full training still uses validation ranking and never test-driven retries. This execution sensitivity is reported rather than interpreted as a reliable method advantage. The machine-readable protocols are `revision/strength_results/graph_rescue_protocol.json`, `revision/strength_results/graph_cpu_device_adapter.json`, and `revision/strength_results/graph_process_retry_protocol.json`. A full-horizon diagnostic subsequently identified a more specific backend failure: at epoch 275, the gradient, covariance, old frame, moments and parameters were finite, but the CPU eigensolver returned non-finite new eigenvectors. A recorded numerical repair therefore finishes only such failed refreshes using the mathematically equivalent thin SVD of the same gradient. It does not repeat the already applied parameter update or accept non-finite training gradients. An injected-eigensolver-failure check matches the thin-SVD parameter and moment updates exactly. The original failed runs remain preserved; repaired endpoints identify this backend policy in their source manifest.

Table 6. Generated regression extension at 2000 full epochs. Low-rank tasks have an explicitly planted rank-eight target matrix. The dimension in the task label is input dimension; output dimension is 32, with 2048 training and two independent 256-example validation/test sets. Covariance and observation-noise definitions are recorded in the configuration. – denotes an unfinished formal comparison.

Task / model	Metric	SGD-M	AdamW	Fixed	Periodic	GrassRoute	SubTrack++
Matrix reg. / 128	MSE ↓	$\mathbf{1.07\times10^{-4}}$	1.07×10^{-4}	0.3347	1.07×10^{-4}	0.4840	1.07×10^{-4}
Matrix reg. / 256	MSE ↓	1.12×10^{-4}	1.12×10^{-4}	0.3410	$\mathbf{1.12\times10^{-4}}$	0.3413	1.12×10^{-4}
Matrix reg. / 512	MSE ↓	1.31×10^{-4}	1.31×10^{-4}	0.3662	$\mathbf{1.31\times10^{-4}}$	0.3664	1.31×10^{-4}
Low-rank reg. / 128	MSE ↓	$\mathbf{1.04\times10^{-4}}$	1.04×10^{-4}	0.0734	1.04×10^{-4}	0.0230	1.04×10^{-4}
Low-rank reg. / 256	MSE ↓	1.19×10^{-4}	1.19×10^{-4}	0.0710	$\mathbf{1.19\times10^{-4}}$	9.62×10^{-3}	1.19×10^{-4}
Low-rank reg. / 512	MSE ↓	1.45×10^{-4}	$\mathbf{1.32\times10^{-4}}$	0.0653	1.32×10^{-4}	0.0194	1.32×10^{-4}

The planted-target tasks explicitly construct a rank-eight ground-truth matrix and check its numerical rank before training. An earlier pilot only concentrated the input covariance while keeping a full-rank target. Its completed results are retained as an additional control below, but do not count toward the planted-low-rank task requirement. Likewise, a preliminary 2000-user MovieLens run was archived and superseded by the all-user protocol. The initial source snapshots and pilot trajectories remain available under `revision/exploratory_archive/initial_coverage_pilots` and `revision/source_versions/initial`.

Table 7. Additional covariance-concentration pilot, retained separately from the proposal-completion count. These 2000-epoch runs have full-rank target matrices and concentrated input covariance; they do not satisfy the planted-low-rank-target benchmark. Values are final test MSE.

d	SGD-M	AdamW	Fixed	Periodic	GrassRoute	SubTrack++
128	1.07×10^{-4}	1.07×10^{-4}	0.0719	1.07×10^{-4}	8.98×10^{-3}	1.07×10^{-4}
256	1.12×10^{-4}	1.12×10^{-4}	0.0681	1.12×10^{-4}	0.0410	1.12×10^{-4}
512	1.45×10^{-4}	1.31×10^{-4}	0.0667	1.31×10^{-4}	0.0676	1.31×10^{-4}

Table 8. Completed-run resource and selection audit. Persistent state is in decimal MB. Wall time includes validation, diagnostics and checkpointing on a shared host; it is not an isolated speed benchmark.

Task	Method	LR	State MB	Seconds	Epochs
Matrix reg. / 128	SGD-M	0.3	0.016	30.1	2000
Matrix reg. / 128	AdamW	0.01	0.033	34.9	2000
Matrix reg. / 128	Fixed	0.01	0.020	50.2	2000
Matrix reg. / 128	Periodic SVD	0.01	0.020	40.6	2000
Matrix reg. / 128	**GrassRoute**	0.003	0.020	46.8	2000
Matrix reg. / 128	SubTrack++	0.01	0.025	35.0	2000
Matrix reg. / 256	SGD-M	0.3	0.033	35.1	2000
Matrix reg. / 256	AdamW	0.01	0.066	34.8	2000
Matrix reg. / 256	Fixed	0.01	0.037	42.3	2000

Task	Method	LR	State MB	Seconds	Epochs
Matrix reg. / 256	Periodic SVD	0.01	0.037	37.8	2000
Matrix reg. / 256	**GrassRoute**	0.01	0.037	42.6	2000
Matrix reg. / 256	SubTrack++	0.01	0.041	36.7	2000
Matrix reg. / 512	SGD-M	0.3	0.066	46.0	2000
Matrix reg. / 512	AdamW	0.01	0.131	40.6	2000
Matrix reg. / 512	Fixed	0.01	0.070	63.9	2000
Matrix reg. / 512	Periodic SVD	0.01	0.070	52.2	2000
Matrix reg. / 512	**GrassRoute**	0.01	0.070	58.8	2000
Matrix reg. / 512	SubTrack++	0.01	0.074	54.7	2000
Low-rank reg. / 128	SGD-M	0.3	0.016	29.6	2000
Low-rank reg. / 128	AdamW	0.01	0.033	30.9	2000
Low-rank reg. / 128	Fixed	0.01	0.020	43.2	2000
Low-rank reg. / 128	Periodic SVD	0.01	0.020	36.6	2000
Low-rank reg. / 128	**GrassRoute**	0.01	0.020	47.0	2000
Low-rank reg. / 128	SubTrack++	0.01	0.025	33.5	2000
Low-rank reg. / 256	SGD-M	0.3	0.033	32.6	2000
Low-rank reg. / 256	AdamW	0.01	0.066	30.8	2000
Low-rank reg. / 256	Fixed	0.01	0.037	46.7	2000
Low-rank reg. / 256	Periodic SVD	0.01	0.037	36.8	2000
Low-rank reg. / 256	**GrassRoute**	0.003	0.037	44.4	2000
Low-rank reg. / 256	SubTrack++	0.01	0.041	34.2	2000
Low-rank reg. / 512	SGD-M	0.3	0.066	32.2	2000
Low-rank reg. / 512	AdamW	0.01	0.131	32.7	2000
Low-rank reg. / 512	Fixed	0.01	0.070	47.2	2000
Low-rank reg. / 512	Periodic SVD	0.01	0.070	37.1	2000
Low-rank reg. / 512	**GrassRoute**	0.003	0.070	47.5	2000
Low-rank reg. / 512	SubTrack++	0.01	0.074	34.9	2000
MNIST / MLP	SGD-M	0.1	3.194	15153.4	2000
MNIST / MLP	AdamW	0.003	6.388	11618.1	2000
MNIST / MLP	Fixed	0.003	0.416	18554.2	2000
MNIST / MLP	Periodic SVD	0.003	0.416	13858.7	2000
MNIST / MLP	**GrassRoute**	0.003	0.416	26244.9	2000
MNIST / MLP	SubTrack++	0.0003	2.776	12787.8	2000
ResNet-18 / CIFAR-10	SGD-M	0.01	44.696	19073.2	1000
ResNet-18 / CIFAR-10	AdamW	0.001	89.392	16410.2	1000
ResNet-18 / CIFAR-10	Fixed	0.003	9.254	91476.8	1000
ResNet-18 / CIFAR-10	Periodic SVD	0.003	9.254	39569.7	1000
ResNet-18 / CIFAR-10	**GrassRoute**	0.003	9.254	63174.5	1000
ResNet-18 / CIFAR-10	SubTrack++	0.001	15.169	28167.4	1000
ResNet-18 / CIFAR-100	SGD-M	0.01	44.881	18169.7	1000
ResNet-18 / CIFAR-100	AdamW	0.001	89.761	24274.0	1000
ResNet-18 / CIFAR-100	Periodic SVD	0.003	9.390	51469.9	1000
Cora / SAGE	SGD-M	0.3	1.475	5.2	1000
Cora / SAGE	AdamW	0.01	2.950	9.0	1000
Cora / SAGE	Fixed	0.01	0.207	24.9	1000
Cora / SAGE	Periodic SVD	0.01	0.207	19.4	1000
Cora / SAGE	**GrassRoute**	0.01	0.207	25.3	1000
Cora / SAGE	SubTrack++	0.003	0.339	12.7	1000
CiteSeer / SAGE	SGD-M	0.3	3.799	9.0	1000
CiteSeer / SAGE	AdamW	0.003	7.597	9.1	1000
CiteSeer / SAGE	Fixed	0.01	0.498	25.8	1000
CiteSeer / SAGE	Periodic SVD	0.01	0.498	24.7	1000
CiteSeer / SAGE	**GrassRoute**	0.003	0.498	26.3	1000
CiteSeer / SAGE	SubTrack++	0.003	0.629	12.6	1000
PubMed / SAGE	SGD-M	0.3	0.516	9.0	1000
PubMed / SAGE	AdamW	0.01	1.031	6.2	1000
PubMed / SAGE	Fixed	0.001	0.088	30.0	1000
PubMed / SAGE	Periodic SVD	0.01	0.088	10.3	1000
PubMed / SAGE	**GrassRoute**	0.01	0.088	17.2	1000
PubMed / SAGE	SubTrack++	0.01	0.219	11.1	1000
ogbn-arxiv / SAGE	SGD-M	0.3	0.173	17.7	1000
ogbn-arxiv / SAGE	AdamW	0.01	0.345	22.3	1000
ogbn-arxiv / SAGE	Fixed	0.01	0.056	38.7	1000
ogbn-arxiv / SAGE	Periodic SVD	0.01	0.056	32.5	1000
ogbn-arxiv / SAGE	**GrassRoute**	0.01	0.056	38.6	1000
ogbn-arxiv / SAGE	SubTrack++	0.01	0.200	26.6	1000
Poisson / PINN	SGD-M	0.01	0.134	8.3	2000
Poisson / PINN	AdamW	0.01	0.267	13.4	2000
Poisson / PINN	Fixed	0.003	0.038	70.3	2000
Poisson / PINN	Periodic SVD	0.01	0.038	10.7	2000

Task	Method	LR	State MB	Seconds	Epochs
Poisson / PINN	**GrassRoute**	0.01	0.038	22.1	2000
Poisson / PINN	SubTrack++	0.01	0.169	10.4	2000
Forced Burgers / PINN	SGD-M	0.01	0.134	10.2	2000
Forced Burgers / PINN	AdamW	0.01	0.268	14.7	2000
Forced Burgers / PINN	Fixed	0.01	0.039	17.0	2000
Forced Burgers / PINN	Periodic SVD	0.01	0.039	20.6	2000
Forced Burgers / PINN	**GrassRoute**	0.01	0.039	17.1	2000
Forced Burgers / PINN	SubTrack++	0.003	0.170	22.0	2000
Darcy / DeepONet	SGD-M	0.3	0.233	79.5	2000
Darcy / DeepONet	AdamW	0.003	0.466	36.5	2000
Darcy / DeepONet	Fixed	0.01	0.077	743.7	2000
Darcy / DeepONet	Periodic SVD	0.01	0.077	70.4	2000
Darcy / DeepONet	**GrassRoute**	0.003	0.077	494.1	2000
Darcy / DeepONet	SubTrack++	0.01	0.257	59.5	2000
Burgers operator / FNO	SGD-M	0.3	0.419	68.9	2000
Burgers operator / FNO	AdamW	0.01	0.838	85.2	2000
Burgers operator / FNO	Fixed	0.01	0.230	488.0	2000
Burgers operator / FNO	Periodic SVD	0.01	0.230	345.8	2000
Burgers operator / FNO	**GrassRoute**	0.01	0.230	1458.0	2000
Burgers operator / FNO	SubTrack++	0.01	0.267	241.7	2000
MovieLens-1M / MF	SGD-M	0.3	2.449	4301.9	1000
MovieLens-1M / MF	AdamW	0.003	4.898	4039.9	1000
MovieLens-1M / MF	Fixed	0.003	1.241	7321.2	1000
MovieLens-1M / MF	Periodic SVD	0.003	1.241	5004.4	1000
MovieLens-1M / MF	**GrassRoute**	0.003	1.241	6852.2	1000
MovieLens-1M / MF	SubTrack++	0.003	1.274	5245.6	1000
Amazon Books subset / MF	SGD-M	0.01	12.350	584.0	1000
Amazon Books subset / MF	AdamW	0.003	24.699	628.8	1000
Amazon Books subset / MF	Fixed	0.003	6.191	932.0	1000
Amazon Books subset / MF	Periodic SVD	0.003	6.191	734.0	1000
Amazon Books subset / MF	**GrassRoute**	0.003	6.191	936.3	1000
Amazon Books subset / MF	SubTrack++	0.003	6.224	719.6	1000
Yelp subset / MF	SGD-M	0.1	7.190	493.3	1000
Yelp subset / MF	AdamW	0.003	14.379	520.9	1000
Yelp subset / MF	Fixed	0.003	3.611	871.9	1000
Yelp subset / MF	Periodic SVD	0.003	3.611	556.3	1000
Yelp subset / MF	**GrassRoute**	0.003	3.611	832.9	1000
Yelp subset / MF	SubTrack++	0.003	3.644	614.2	1000

Table 9. Follow-up selections and resources. Shared local CPU wall times include evaluation; persistent state is in decimal MB. All selected endpoints are reported, irrespective of their ranking.

Task	Method	LR	State MB	Seconds	Other selection
Poisson / PINN	AdamW	0.01471	0.267	6.6	–
Poisson / PINN	Fixed	0.01	0.038	8.2	residual ratio=0.3
Poisson / PINN	PeriodicDiag	0.01	0.038	10.3	period=25
Poisson / PINN	**GrassRoute**	0.01	0.038	11.5	ρ=1
Forced Burgers / PINN	AdamW	0.007208	0.268	11.5	–
Forced Burgers / PINN	Fixed	0.01	0.039	13.1	residual ratio=0.3
Forced Burgers / PINN	PeriodicDiag	0.01	0.039	9.5	period=5
Forced Burgers / PINN	**GrassRoute**	0.01	0.039	12.3	ρ=1

B.4. Corrections to the archived evidence

The archive is retained for provenance. Several qualifications change its interpretation and are not repaired by renaming a diagnostic.

1. The logged `principal_angle_R` compares consecutive epoch bases, not U_t and a population oracle. The logged eigengap comes from the routed $r \times r$ Gram matrix and cannot identify $\lambda_r(C) - \lambda_{r+1}(C)$. The historical win map therefore does not establish an empirical predictor based on $\Delta \cos^2 R$.
2. Rank arguments in the old driver did not override ranks already stored in parameter groups. Identical state sizes and endpoints across several nominal rank sweeps cannot support rank robustness. New configurations record their actual per-layer ranks.

Table 10. Independent graph follow-up selections and persistent optimizer state. The local comparator correction and enlarged selection budgets define a separate comparison from the original extension.

Task	Method	LR	State MB	Other selection
Cora / SAGE	SGD-M	0.3	1.475	–
Cora / SAGE	AdamW	0.02	2.950	–
Cora / SAGE	Fixed	0.01	0.207	residual ratio=3
Cora / SAGE	PeriodicDiag	0.01	0.207	period=25
Cora / SAGE	**GrassRoute**	0.003	0.207	ρ=3
Cora / SAGE	SubTrack++	0.01	0.339	ρ=0.1
CiteSeer / SAGE	SGD-M	0.2	3.799	–
CiteSeer / SAGE	AdamW	0.002	7.597	–
CiteSeer / SAGE	Fixed	0.01	0.498	residual ratio=1
CiteSeer / SAGE	PeriodicDiag	0.01	0.498	period=5
CiteSeer / SAGE	**GrassRoute**	0.003	0.498	ρ=3
CiteSeer / SAGE	SubTrack++	0.01	0.629	ρ=0.1
PubMed / SAGE	SGD-M	0.2	0.516	–
PubMed / SAGE	AdamW	0.03	1.031	–
PubMed / SAGE	Fixed	0.003	0.088	residual ratio=0.3
PubMed / SAGE	PeriodicDiag	0.01	0.088	period=5
PubMed / SAGE	**GrassRoute**	0.01	0.088	ρ=3
PubMed / SAGE	SubTrack++	0.01	0.219	ρ=0.1

Table 11. **Controlled geometry checks.** Random pairs cover $(d, r) = (6, 1), (8, 3), (8, 6), (32, 8)$; ratios at least one verify the sampled lower bounds.

Check	Value
Random covariance/subspace cases	600
Minimum measured / PL lower bound	1.131
Minimum measured / quadratic-growth bound	1.111
Maximum directional-derivative error	2.17×10^{-10}

3. The historical second-moment amplitude-rotation heuristic is not exact diagonal covariance transport. Hard reset and no-reset runs are different variants; an ablation label cannot reconcile their state rules. The proof now distinguishes these errors explicitly.

4. Archived SciML best training loss is distinct from a relative PDE solution error. The old evaluation's `rel_l2_proxy` sums PDE residual and initial-condition errors; the training loss omits an explicit spatial boundary penalty. The old SGD value 0.427 belongs to this proxy, so it is not placed in the training-loss comparison column.

5. Some historical accuracies are the best observed test score rather than final held-out scores. In particular, the rounded tuned summaries and default full trajectories are separate sources. Neither is substituted for the other. The new extension uses final test endpoints after validation selection.

6. The archival B6 is a local geodesic surrogate with different recovery and moment rules from official SubTrack++. Its numbers may be used as a historical mechanism comparison but not as a faithful implementation claim.

The historical win-map source and figure remain in the archive for provenance; their uncalibrated proxy axes are not reused as quantitative evidence here.

B.5. Controlled theory diagnostics

The numerical check suite samples 600 covariance/subspace pairs across $(d, r) = (6, 1), (8, 3), (8, 6), (32, 8)$, including a rank larger than the complement dimension. It checks the PL and quadratic-growth inequalities and the directional derivative, and tests the QR cubic bound over 16 step sizes. The tracking bed crosses three eigengaps, three drift rates, and three tracking rates, using the same seeded Gaussian gradient stream for each method. It compares online QR, interval-25 SVD, and fixed routing against the known population target. Momentum and covariance normalization are excluded in this auxiliary bed to isolate the stated mechanism.

Table 12. **Principal-angle diagnostic.** Eleven evenly spaced stored angles from the 200-point sweep at fixed eigengap; all original records remain in the diagnostic archive. The angular PL lower bound coincides with the exact rank-one squared gradient.

θ	$\|\mathrm{grad}\,\Phi\|^2$	**PL bound**	**Hessian**
0.0050	2.50×10^{-5}	2.50×10^{-5}	1
0.1619	0.0253	0.0253	0.9481
0.3187	0.08855	0.08855	0.8036
0.4756	0.1657	0.1657	0.5807
0.6325	0.2273	0.2273	0.3011
0.7893	0.25	0.25	-0.007843
0.9462	0.225	0.225	-0.3161
1.1030	0.162	0.162	-0.5934
1.2599	0.08482	0.08482	-0.8128
1.4168	0.02298	0.02298	-0.9529
1.5658	2.50×10^{-5}	2.50×10^{-5}	-1

Table 13. **QR versus exponential-map projectors.** All 16 tested tangent norms; the measured discrepancy is compared with the cubic bound.

Tangent norm	**Projector error**	**Cubic bound**
0.001	1.42×10^{-10}	4.71×10^{-10}
0.001445	4.28×10^{-10}	1.42×10^{-9}
0.002088	1.29×10^{-9}	4.29×10^{-9}
0.003017	3.90×10^{-9}	1.29×10^{-8}
0.00436	1.18×10^{-8}	3.91×10^{-8}
0.0063	3.55×10^{-8}	1.18×10^{-7}
0.009103	1.07×10^{-7}	3.56×10^{-7}
0.01315	3.23×10^{-7}	1.07×10^{-6}
0.01901	9.74×10^{-7}	3.24×10^{-6}
0.02746	2.94×10^{-6}	9.77×10^{-6}
0.03969	8.86×10^{-6}	2.95×10^{-5}
0.05734	2.67×10^{-5}	8.89×10^{-5}
0.08286	8.06×10^{-5}	2.68×10^{-4}
0.1197	2.43×10^{-4}	8.09×10^{-4}
0.173	7.30×10^{-4}	0.002441
0.25	0.00219	0.007366

The auxiliary sensitivity study compares AdamW, corrected GrassRoute, fixed routing, and hard reset, while logging parameter distance, bounded-loss sensitivity, and a full-batch covariance oracle. It does not estimate the supremum over all neighboring datasets required for uniform stability. Likewise, measured sample-gradient alignment or a train–test gap alone would not verify the conditional direction-defect assumption of the convergence theorem. We keep these distinctions when interpreting the diagnostics. In this auxiliary probe, final mean bounded-loss sensitivity is 0.00392 (AdamW), 0.00282 (GrassRoute), 0.11066 (HardReset), 0.00053 (Fixed). AdamW and GrassRoute obtain 9.0% and 9.0% test accuracy, respectively; hard reset obtains 2.4% and fixed routing 5.4%. GrassRoute's training cross-entropy is 2.812 versus AdamW's 0.051; its lower sensitivity is therefore accompanied by substantially weaker fitting. The fixed router is less sensitive still, but also less accurate. These measurements illustrate a fit–sensitivity trade-off and reset instability, not superior population generalization.

B.6. Evaluation and resource conventions

All formal comparisons use seed 42 and a predefined validation budget. The dataset, resolution, model and training horizon are recorded with each configuration. Test metrics refer to the final selected endpoint; shared-host resource times include evaluation and checkpointing. Completion counts refer to verified full trajectories, with queued and partial runs listed separately.

Table 14. **Raw-covariance tracking diagnostic.** All 27 gap–drift–step configurations; mean squared sine error on the final 500 of 2000 steps. Each method observes the same stream. No momentum or covariance normalization is used in this auxiliary bed.

Eigengap	Drift	ρ	Online QR	Periodic / 25	Fixed
0.05	0	0.03	0.004577	0.4787	0.009967
0.05	0	0.15	0.01449	0.4787	0.009967
0.05	0	0.5	0.03044	0.4787	0.009967
0.05	0.001	0.03	0.4247	0.4804	0.9735
0.05	0.001	0.15	0.04543	0.4804	0.9735
0.05	0.001	0.5	0.03566	0.4804	0.9735
0.05	0.01	0.03	0.4021	0.4962	0.4067
0.05	0.01	0.15	0.4801	0.4962	0.4067
0.05	0.01	0.5	0.1734	0.4962	0.4067
0.2	0	0.03	0.001159	0.2176	0.009967
0.2	0	0.15	0.003768	0.2176	0.009967
0.2	0	0.5	0.01587	0.2176	0.009967
0.2	0.001	0.03	0.03462	0.2192	0.9735
0.2	0.001	0.15	0.005958	0.2192	0.9735
0.2	0.001	0.5	0.01618	0.2192	0.9735
0.2	0.01	0.03	0.5615	0.243	0.4067
0.2	0.01	0.15	0.1175	0.243	0.4067
0.2	0.01	0.5	0.02536	0.243	0.4067
0.5	0	0.03	3.86×10^{-4}	0.05117	0.009967
0.5	0	0.15	0.001931	0.05117	0.009967
0.5	0	0.5	0.008031	0.05117	0.009967
0.5	0.001	0.03	0.005592	0.05166	0.9735
0.5	0.001	0.15	0.002225	0.05166	0.9735
0.5	0.001	0.5	0.008069	0.05166	0.9735
0.5	0.01	0.03	0.5085	0.07187	0.4067
0.5	0.01	0.15	0.01792	0.07187	0.4067
0.5	0.01	0.5	0.009177	0.07187	0.4067

B.7. Archived and original proposal results

The clear archived joint gain is the Burgers PINN: GrassRoute improves best training loss relative to both AdamW and tuned periodic SVD, with a reported 84.1% reduction in persistent optimizer state. That is one scientific-learning family, not evidence from several independent PDE tasks. Its reported 1.78× step cost is a compute penalty accompanying the state saving. The old PDE evaluation is a residual-plus-initial-condition proxy rather than an actual solution relative error, and the boundary-condition omission limits physical interpretation (Appendix B).

Cora is a qualified positive result against AdamW, but tuned periodic SVD is slightly better. The CNN partial variant improves AdamW by 1.96 percentage points and tuned SGD by only 0.19 points in the archived single run. Its state is 88.19 MB versus AdamW's 89.76 MB, saving only about 1.8%. Full routing reduces state to 8.68 MB but loses 11.91 accuracy points relative to AdamW. These are distinct accuracy–memory trade-offs, not a CNN joint win. The graph-arxiv, MLP, recommendation, and continual-learning results do not establish a robust GrassRoute advantage. Archived ViT has complete AdamW and periodic-SVD trajectories but no complete GrassRoute endpoint.

The old broad claims about a single geometric predictor are not retained. Captured energy is not an eigengap, the recorded consecutive-basis angle is not oracle error, and nominal rank sweeps did not always change the parameter-group ranks. These issues motivate new controlled diagnostics and explicit provenance rather than retrospective reclassification of negative results.

B.8. Validation-selected original proposal matrix

The current extension contains 117 completed formal task–method runs out of 162, with 19 configurations complete across all six methods. The original queue was stopped after 117 completed runs; unfinished cells are marked – and have not been

Table 15. **Neighboring-data probe at epoch 1000.** Frozen random ResNet-18 features on CIFAR-100; one replacement in 2048 training examples and a 1000-image test probe. Sensitivity uses cross-entropy clipped at 10 and divided by 10. These measurements are conditional on the specified dataset pair.

Method	Train CE	Test acc.	Parameter distance	Mean sensitivity	Max. sensitivity
AdamW	0.05073	9.00	2.336	0.003923	0.2061
GrassRoute	2.812	9.00	0.8386	0.002825	0.1039
HardReset	371.6	2.40	461.1	0.1107	1
Fixed	3.663	5.40	0.5005	5.33×10^{-4}	0.03581

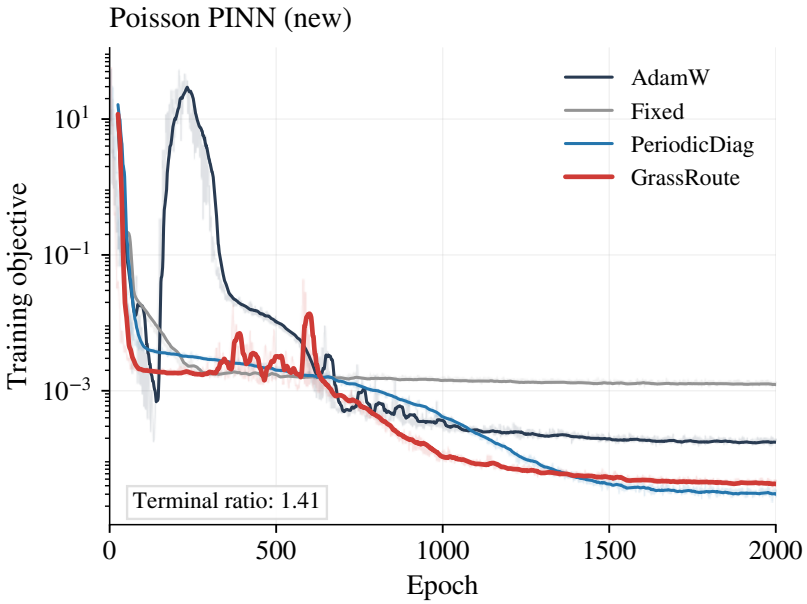


Figure 3. Poisson loss control from the independent nine-candidate validation search, with all four selected 2000-epoch trajectories retained. Faint curves are raw objectives and solid curves are 25-epoch trailing means. The terminal ratio is the ratio of the final-100-epoch mean losses, GrassRoute / PeriodicDiag. GrassRoute has lower training loss than the selected AdamW run, whereas PeriodicDiag has the lowest loss. As with the held-out endpoints in Table 28, this comparison is sensitive to finite-horizon validation selection; it does not supersede the stronger AdamW endpoint from the original search. This control accompanies the core loss figure in the main text without treating the protocols as interchangeable.

used to declare a winner. The completed extension comparisons do not yet establish a new case that improves on both AdamW and periodic SVD. Against AdamW alone, the completed paired scores favor GrassRoute on PubMed / SAGE; this is a qualified comparison, not an all-baseline win. All values are final held-out test scores of the validation-selected configuration; classification and Recall@20 are percentages.

The original manifest spans every proposed task/dataset with representative models, rather than the full dataset–architecture–rank Cartesian product. Its 27 configurations include three input dimensions for both matrix and low-rank regression, MNIST, four CNN datasets, three ViT datasets, four graph datasets, four scientific tasks, three recommendation datasets, and two continual-learning protocols. Completed runs use full epochs and save restartable checkpoints and all epoch records. Three learning rates per method are screened on validation data before a fresh formal run. Dataset sources, actual ranks, hyperparameters, and source hashes are recorded with each result. The high-cost queue was stopped after 117 completed cells; the rest remain visibly pending. Appendix B defines the tasks and selection budgets precisely.

New task coverage is evidence only when the formal comparison completes. A lower parameter-state cost can remain useful without a test-metric win; conversely, an isolated lower error does not establish a practical advantage without accounting for state and time. We therefore retain all completed baselines, including strong SGD and fixed-routing outcomes, and report shared-host resource measurements separately from accuracy. The current finite hyperparameter search and one-seed protocol do not support statistical-significance claims or exhaustive optimizer rankings.

Table 16. Reported tuned historical summaries. Accuracy and Recall@20 are percentages. SciML reports best training loss; its historical SGD proxy is a different metric and is omitted. Values are selected historical observations, often best observed test accuracy, and are not final held-out endpoints from the extension.

Task	Metric	SGD-M	AdamW	Periodic SVD	GrassRoute
Burgers / PINN	train loss ↓	–	.0110	.00327	.00197
Cora / GCN	acc. ↑	–	78.3	80.4	80.0
ogbn-arxiv / SAGE	acc. ↑	54.8	55.1	53.4	52.7
CIFAR-100 / partial	acc. ↑	73.13	71.36	–	73.32
CIFAR-100 / full	acc. ↑	–	71.36	65.6	59.45
MNIST / MLP	acc. ↑	98.65	98.53	98.36	98.25
Rotated MNIST	avg. acc. ↑	–	61.6	59.66	58.25
MovieLens / MF	R@20 ↑	17.1	6.1	6.2	7.5

Table 17. Validation-selected extension, seed 42. Formal budgets are 1000 epochs for classification and recommendation, 2000 for MLP and scientific learning. Accuracy and Recall@20 are percentages; scientific results report solution relative error. Bold marks a best score only after all six methods have completed. – denotes an unfinished formal comparison. GrassRoute uses diagonal moment transport; the periodic comparator retains its old second-moment diagonal at refreshes. A transport-corrected follow-up is reported separately. SubTrack++ is the pinned official implementation.

Task / model	Metric	SGD-M	AdamW	Fixed	Periodic	GrassRoute	SubTrack++
MNIST / MLP	Acc. ↑	**98.65**	98.13	97.25	97.82	97.90	97.18
ResNet-18 / CIFAR-10	Acc. ↑	**95.06**	94.50	91.43	92.85	89.99	93.25
ResNet-18 / CIFAR-100	Acc. ↑	77.02	73.12	–	68.35	–	–
ResNet-18 / Tiny-ImageNet	Acc. ↑	–	–	–	–	–	–
ResNet-18 / ImageNet-100	Acc. ↑	–	–	–	–	–	–
ViT-Tiny / CIFAR-100	Acc. ↑	–	–	–	–	–	–
ViT-Tiny / Tiny-ImageNet	Acc. ↑	–	–	–	–	–	–
ViT-Tiny / ImageNet-100	Acc. ↑	–	–	–	–	–	–
Cora / SAGE	Acc. ↑	**79.50**	78.30	43.10	79.10	78.00	76.70
CiteSeer / SAGE	Acc. ↑	**69.70**	68.10	30.80	68.70	67.10	67.10
PubMed / SAGE	Acc. ↑	**77.40**	74.20	59.50	77.10	76.00	75.60
ogbn-arxiv / SAGE	Acc. ↑	53.78	55.05	32.57	53.16	51.86	**55.46**
Poisson / PINN	Rel.L_2 ↓	1.0000	$\mathbf{9.28\times10^{-5}}$	9.95×10^{-4}	2.99×10^{-4}	1.26×10^{-4}	4.03×10^{-4}
Forced Burgers / PINN	Rel.L_2 ↓	0.0132	$\mathbf{1.49\times10^{-3}}$	6.41×10^{-3}	1.90×10^{-3}	2.08×10^{-3}	1.55×10^{-3}
Darcy / DeepONet	Rel.L_2 ↓	0.0867	**0.0199**	0.0223	0.0370	0.0380	0.1087
Burgers operator / FNO	Rel.L_2 ↓	0.0145	$\mathbf{7.82\times10^{-3}}$	0.0177	0.0148	0.0201	0.0141
MovieLens-1M / MF	R@20 ↑	10.59	**12.36**	11.77	12.35	12.25	12.33
Amazon Books subset / MF	R@20 ↑	0.00	**1.25**	0.85	0.90	0.95	0.75
Yelp subset / MF	R@20 ↑	0.20	**2.30**	1.90	2.00	1.45	1.95
Rotated MNIST / MLP	Acc. ↑	–	–	–	–	–	–
Split CIFAR-100 / ResNet-18	Acc. ↑	–	–	–	–	–	–

B.9. Focused exploratory follow-up after scope reduction

The original 162-cell plan was stopped after 117 completed runs. The remaining cells are retained as unfinished, including partial CIFAR-100 trajectories; they are not interpreted as full-budget endpoints. To test the proposed tracking mechanism with less computation, we froze a separate three-condition rotating rank-four regression stream before inspecting its formal outcomes. Each method used four equal-budget validation candidates and a new 1500-step run; Table 19 includes all conditions and comparators. GrassRoute's smaller subspace-angle error in the drifting conditions did not yield a lower prediction MSE than the strongest baselines. The Cora GCN check is motivated by the prior Cora GraphSAGE result, so it is an exploratory architecture replication on the same dataset. Its full six-method matrix is in Table 20.

We additionally froze the seed-42 validation-selected hyperparameters from the original Cora GraphSAGE and SMS studies, then reran three new initializations (43–45) per method on each unchanged public split. These replications use four prespecified core comparators and retain all endpoints in Tables 21 and 22; they do not add independent datasets. A numerical failure at a frozen hyperparameter is kept as such and does not trigger a test-selected replacement.

Table 18. **Scientific-learning validation and test endpoints.** Complete 2000-epoch formal runs from the original proposal extension, selected on validation data. All entries are solution relative error. Initial validation denotes the first stored validation checkpoint; models and tasks follow the versioned protocol.

Task	Method	Initial val.	Final val.	Final test
Poisson / PINN	SGD-M	0.4309	1	1
Poisson / PINN	AdamW	2.481	9.19×10^{-5}	9.28×10^{-5}
Poisson / PINN	Fixed	0.7925	9.79×10^{-4}	9.95×10^{-4}
Poisson / PINN	Periodic SVD	0.4188	2.92×10^{-4}	2.99×10^{-4}
Poisson / PINN	**GrassRoute**	0.4188	1.25×10^{-4}	1.26×10^{-4}
Poisson / PINN	SubTrack++	0.5623	3.94×10^{-4}	4.03×10^{-4}
Forced Burgers / PINN	SGD-M	0.8454	0.01289	0.01316
Forced Burgers / PINN	AdamW	2.568	0.001492	0.001489
Forced Burgers / PINN	Fixed	0.4933	0.006064	0.006414
Forced Burgers / PINN	Periodic SVD	0.4933	0.001853	0.001898
Forced Burgers / PINN	**GrassRoute**	0.4933	0.001953	0.002079
Forced Burgers / PINN	SubTrack++	0.8114	0.001515	0.001549
Darcy / DeepONet	SGD-M	0.7469	0.08382	0.08672
Darcy / DeepONet	AdamW	2.201	0.01937	0.01989
Darcy / DeepONet	Fixed	0.5206	0.02219	0.02226
Darcy / DeepONet	Periodic SVD	0.5206	0.03552	0.03698
Darcy / DeepONet	**GrassRoute**	0.4898	0.03734	0.03799
Darcy / DeepONet	SubTrack++	0.599	0.1036	0.1087
Burgers operator / FNO	SGD-M	0.9935	0.01471	0.01447
Burgers operator / FNO	AdamW	0.4612	0.00797	0.007823
Burgers operator / FNO	Fixed	0.7662	0.01922	0.01772
Burgers operator / FNO	Periodic SVD	0.7662	0.01526	0.01476
Burgers operator / FNO	**GrassRoute**	0.7062	0.02147	0.02013
Burgers operator / FNO	SubTrack++	0.7643	0.01422	0.01406

B.10. Accuracy–state evidence and independent follow-up

The practical claim is specific: retaining adaptive state only in selected directions can reduce persistent optimizer memory while preserving useful prediction quality, and continuous tracking can average noisy directional information. It does not require the same optimizer to win every final-error comparison. We expose both the favorable and unfavorable cases.

Measured state versus final prediction quality. On PubMed the original extension obtains 76.0% with GrassRoute versus 74.2% with AdamW, retaining 87,576 versus 1,031,216 state bytes: a 91.5% reduction and a 1.8 percentage-point accuracy gain in the single paired run. The periodic implementation is more accurate at 77.1% with the same state cost. On all-user MovieLens, GrassRoute retains 25.3% of AdamW's state and obtains 12.247% versus 12.363% Recall@20, a 0.116 percentage-point loss. This is a measured memory–quality trade-off, not a significance or noninferiority claim. On Poisson its relative solution error is 1.255×10^{-4} versus 2.987×10^{-4} for the original periodic implementation at the same persistent state cost, but AdamW is more accurate. Table 23 retains every completed paired task, including the unfavorable regression and operator-learning results.

Actual-implementation tracking test. We run the routed optimizer code on Gaussian gradient matrices with a known rotating population covariance, using rank 4 in dimension 64, signal/tail eigenvalues 1 and 0.03, and 16 or 128 gradient rows per update. The four rotation rates are $0, 0.001, 0.005, 0.02$ radians per step. All methods start from the same random frame and observe the same 2000 matrices. We compare GrassRoute at a fixed $\rho = 3$, its no-momentum ablation, a fixed router, and transport-corrected instantaneous SVD refreshes at periods 5, 25 and 100. The target projector is known analytically; no compressed-Gram proxy is used. Error is measured *before* updating on the current gradient, preventing a freshly computed SVD from being evaluated only on its fitting batch. With the fixed tracking rate $\rho = 3$, GrassRoute has lower steady tracking error than interval-25 instantaneous SVD in 8 of the 8 predeclared conditions. Table 26 includes the static and fast-drift controls, the no-momentum ablation, and interval-5 and interval-100 alternatives. These comparisons measure subspace

Table 19. **Exploratory drifting-regression controls.** A rank-four teacher in $\mathbb{R}^{16\times64}$ rotates at the stated radians per update; stationary is the negative control. Entries are mean independent test MSE over the last 20% of 1500 fresh steps after four equal-budget validation candidates per method. All conditions and outcomes are retained. Seed 42; lower is better. State reduction compares persistent GrassRoute optimizer state with AdamW. This synthetic stream is distinct from public-data generalization.

Drift	SGD-M	AdamW	Fixed	Periodic SVD	GrassRoute	State reduction
0	2.346×10^{-5}	9.139×10^{-4}	3.705×10^{-5}	3.425×10^{-5}	4.235×10^{-5}	68.8%
0.005	3.448×10^{-4}	4.767×10^{-3}	1.149×10^{-3}	6.947×10^{-4}	7.778×10^{-4}	68.8%
0.020	1.315×10^{-3}	1.002×10^{-2}	1.648×10^{-2}	1.694×10^{-2}	2.395×10^{-1}	68.8%

Table 20. **Focused Cora GCN architecture check.** Nine 100-epoch validation candidates per method and a fresh 1000-epoch formal run. Final public-test accuracy (%) on the Planetoid split, seed 42. All six methods and incomplete cells are shown. This reuses Cora and is an architecture replication, not a new dataset-family win.

Method	Test accuracy
SGD	81.10
AdamW	78.50
Fixed	21.40
Periodic SVD	80.50
GrassRoute	80.30
SubTrack++	78.60

identification on synthetic streams, not an end-to-end accuracy advantage.

Stronger smoothing controls and a faster-drift stress test. Instantaneous SVD does not average sample noise; that is a meaningful property of that baseline; full-covariance estimators can also average noise. A second, independently recorded protocol therefore adds normalized full-covariance EMA estimators with decays 0.9 and 0.99, each refreshed every 5 or 25 steps. It retains every original condition and adds rotation rate 0.1 at both row counts. The full 64×64 covariance is stored and counted. The same moment transport, starting frame, stream and pre-update metric are used. This control can exploit temporal smoothing without requiring Grassmann momentum, at an additional state cost. After adding four full-covariance EMA controls and the nonadiabatic rate 0.1, GrassRoute has the lowest error in 5 of 10 conditions and beats both instantaneous refresh baselines in 10 of 10. EMA SVD is stronger in the static or slow-drift conditions. At 128 rows and drift 0.02, GrassRoute has mean error 0.00541 versus 0.05440 for the best displayed EMA alternative, with 6,144 versus 22,528 persistent state bytes. This isolates a noise–lag–memory trade-off; the current implementation does not establish a CPU speed advantage.

Independent, larger validation budget on exact-solution PINNs. We return to the scientific-learning family motivated by the archived Burgers result and include *both* exact-solution PINNs, rather than searching only the favorable task. Each of AdamW, fixed routing, PERIODICDIAG, and GrassRoute receives nine 100-epoch validation screening runs. AdamW searches nine geometrically spaced learning rates from 10^{-4} to 0.03. Each routed method searches learning rates $\{0.001, 0.003, 0.01\}$, crossed with a method-specific parameter: residual-rate ratios $\{0.3, 1, 3\}$ for fixed routing; refresh periods $\{5, 25, 100\}$ for PERIODICDIAG; and tracking rates $\{0.3, 1, 3\}$ for GrassRoute. All candidates, including divergence records, are retained. Selection uses final validation relative solution error; the selected configuration is trained afresh for 2000 full epochs, and test error is evaluated once at the end. Parameter initialization and collocation streams are paired. The protocol and source hash were saved before the runs. This follow-up changes the search budget and periodic transport, so its endpoints remain separate from the 162-comparison proposal extension. The independent follow-up has completed 8 of 8 selected formal runs. The currently completed follow-up endpoints do not establish an all-baseline GrassRoute win. Its endpoints are not pooled with the original six-method extension. On Poisson, GrassRoute's selected relative solution error is 6.70×10^{-5} at 0.0379 MB of persistent optimizer state, versus 9.52×10^{-4} at 0.2673 MB for the independently selected AdamW configuration. This favorable comparison is selection-sensitive: the original, smaller search already found an AdamW endpoint of 9.28×10^{-5}, and PERIODICDIAG in the new search achieves 4.90×10^{-5}. On forced Burgers, AdamW remains better than GrassRoute. More screening candidates do not guarantee better long-horizon selection when screening lasts only 100 epochs; the complete endpoints, not only favorable pairs, are shown below.

Table 21. **Cora GraphSAGE fixed-hyperparameter seed replication.** Three fresh 1000-epoch runs use hyperparameters frozen from the original seed-42 validation selection. The public split and architecture are fixed; seeds 43–45 vary initialization and stochastic training. Accuracy or macro-F1 is in percent. Mean and sample standard deviation are descriptive. Persistent optimizer state is in decimal MB. All endpoints are shown; *fail* denotes a retained numerical failure.

Method	Seed 43	Seed 44	Seed 45	Mean ± SD	State MB
SGD	79.30	79.40	79.00	79.23 ± 0.21	1.475
AdamW	78.10	77.50	77.30	77.63 ± 0.42	2.950
Periodic SVD	78.10	*fail*	78.60	–	0.207
GrassRoute	78.30	75.70	76.70	76.90 ± 1.31	0.207

Table 22. **SMS text classification fixed-hyperparameter seed replication.** Three fresh 1000-epoch runs use hyperparameters frozen from the original seed-42 validation selection. The public split and architecture are fixed; seeds 43–45 vary initialization and stochastic training. Accuracy or macro-F1 is in percent. Mean and sample standard deviation are descriptive. Persistent optimizer state is in decimal MB. All endpoints are shown; *fail* denotes a retained numerical failure.

Method	Seed 43	Seed 44	Seed 45	Mean ± SD	State MB
SGD	96.30	94.63	94.60	95.18 ± 0.97	0.525
AdamW	95.76	95.20	95.48	95.48 ± 0.28	1.050
Periodic SVD	93.94	94.29	94.63	94.29 ± 0.35	0.071
GrassRoute	94.29	95.64	94.29	94.74 ± 0.78	0.137

Checking the graph result with equal, expanded search budgets. A separate follow-up tests Cora, CiteSeer and PubMed against all six methods, including corrected periodic transport and official SubTrack++. Each method receives nine 100-epoch validation candidates and 1000 fresh formal epochs. GrassRoute crosses three learning rates with tracking rates $\{0.3, 1, 3\}$; fixed routing searches residual-rate ratios, corrected periodic routing searches refresh intervals, and official SubTrack++ searches its initial geodesic step. AdamW and SGD each search nine learning rates. The protocol records every candidate before training, uses paired model/dropout streams and the public graph splits, and retains every selected endpoint. These transductive results test optimizer performance and do not validate the iid assumptions of the generalization theorem. Numerical failures, bounded process retries, and a device-only CPU adapter are recorded separately; they do not alter the nine unique hyperparameters or permit test-driven fallback (Appendix B.3). The independent Planetoid follow-up has completed 18 of 18 selected formal runs. Completed paired scores favor GrassRoute over both AdamW and corrected periodic routing on Cora / SAGE. Pending cells and full selection/state records are retained.

B.11. Complete new public-data suite

The six new tasks are retained as a complete evaluation set. The suite was motivated by the geometric hypothesis after examining the earlier archive; it is a follow-up study rather than an independent confirmation of universal optimizer superiority. A main-text subset is identified by the rule in Section 5.4; this appendix retains all six tasks and all methods regardless of their ranking. No unsuccessful run is deleted after observing its test result.

Text. 20 Newsgroups uses the official by-date train/test split through the scikit-learn loader, with headers, footers and quotes removed (scikit-learn developers, 2026). A stratified 20% of the official training set is reserved for validation. TF-IDF uses a vocabulary of at most 2048 terms, minimum document frequency 2, maximum document frequency 0.95, sublinear term frequency and row normalization; its vocabulary and inverse document frequencies fit only the training documents. Official test documents, including empty texts and exact-text overlaps in the released corpus, are retained and those counts are recorded; exact-text independence is not asserted. SMS Spam (Almeida & Hidalgo, 2011) removes exact stripped-message duplicates and conflicting duplicate labels before a stratified 70/15/15 split. It uses the same TF-IDF parameters. Both text models use a 64-unit ReLU hidden layer, dropout 0.1, and a linear classifier. Sparse inputs remain sparse in the first matrix product. Macro-F1, rather than majority-class accuracy, is the primary selection metric; predictions use the fixed argmax rule.

Structured data. Adult (Becker & Kohavi, 1996) retains its official test set and reserves 20% of its training rows for stratified validation. Continuous features are standardized with training statistics; categories are training-fitted one-hot

Table 23. All completed paired extension endpoints: state retained relative to AdamW, and final test metrics. Lower state is beneficial; metric direction is explicit. These are the original versioned periodic endpoints, not the transport-corrected follow-up. Wall time is not encoded as isolated speed.

Task	State / AdamW	AdamW	GrassRoute	Periodic
Matrix reg. / 128 ↓	62.5%	1.07×10^{-4}	0.484	1.07×10^{-4}
Matrix reg. / 256 ↓	56.2%	1.12×10^{-4}	0.341	1.12×10^{-4}
Matrix reg. / 512 ↓	53.1%	1.31×10^{-4}	0.366	1.31×10^{-4}
Low-rank reg. / 128 ↓	62.5%	1.04×10^{-4}	0.023	1.04×10^{-4}
Low-rank reg. / 256 ↓	56.2%	1.19×10^{-4}	0.00962	1.19×10^{-4}
Low-rank reg. / 512 ↓	53.1%	1.32×10^{-4}	0.0194	1.32×10^{-4}
MNIST / MLP ↑	6.5%	98.13	97.90	97.82
ResNet-18 / CIFAR-10 ↑	10.4%	94.50	89.99	92.85
Cora / SAGE ↑	7.0%	78.30	78.00	79.10
CiteSeer / SAGE ↑	6.6%	68.10	67.10	68.70
PubMed / SAGE ↑	8.5%	74.20	76.00	77.10
ogbn-arxiv / SAGE ↑	16.1%	55.05	51.86	53.16
Poisson / PINN ↓	14.2%	9.28×10^{-5}	1.26×10^{-4}	2.99×10^{-4}
Forced Burgers / PINN ↓	14.5%	0.00149	0.00208	0.0019
Darcy / DeepONet ↓	16.5%	0.0199	0.038	0.037
Burgers operator / FNO ↓	27.4%	0.00782	0.0201	0.0148
MovieLens-1M / MF ↑	25.3%	12.36	12.25	12.35
Amazon Books subset / MF ↑	25.1%	1.25	0.95	0.90
Yelp subset / MF ↑	25.1%	2.30	1.45	2.00

Table 24. Independent Planetoid follow-up: nine 100-epoch validation candidates per method, then 1000 fresh formal epochs, seed 42. All accuracies are final public-test percentages; – indicates unfinished training. The periodic method now transports both moments. SubTrack++ is the same pinned official implementation. These are transductive graph experiments.

Task	SGD-M	AdamW	Fixed	PeriodicDiag	GrassRoute	SubTrack++
Cora / SAGE	**79.70**	77.50	39.10	77.70	78.60	77.30
CiteSeer / SAGE	68.80	67.40	36.90	**69.10**	68.40	64.90
PubMed / SAGE	**76.90**	74.80	53.90	76.80	76.60	76.20

Table 25. **Tracking implementation resources.** Mean CPU optimizer-call time across all four drift rates at 128 gradient rows, including diagnostics; persistent state is decimal kB. Shared implementation timing is reported separately from task accuracy.

Method	CPU call / ms	State / kB
GrassRoute	0.1497	6.14
NoMomentum	0.1354	6.14
Fixed	0.1428	6.14
Periodic5	0.09495	6.14
Periodic25	0.0632	6.14
Periodic100	0.05768	6.14

Table 26. Complete predeclared tracking grid, 2000 steps, seed 42. Mean $r^{-1}\|\sin\Theta\|_F^2$ before each update on steps 1501–2000; lower is better. All methods see identical streams and start from the same random frame. Periodic variants use corrected diagonal transport. Bold marks the minimum among all displayed methods.

Rows	Drift	GrassRoute	No mom.	Fixed	Periodic5	Periodic25	Periodic100
16	0.000	**0.0307**	0.0410	0.9119	0.1455	0.1386	0.1403
16	0.001	**0.0307**	0.0410	0.9346	0.1455	0.1388	0.1429
16	0.005	**0.0308**	0.0411	0.9223	0.1456	0.1435	0.2070
16	0.020	**0.0322**	0.0427	0.9263	0.1489	0.2109	0.6477
128	0.000	**0.0038**	0.0050	0.9119	0.0156	0.0161	0.0169
128	0.001	**0.0038**	0.0050	0.9346	0.0157	0.0164	0.0205
128	0.005	**0.0039**	0.0051	0.9223	0.0159	0.0216	0.0973
128	0.020	**0.0054**	0.0067	0.9263	0.0201	0.0988	0.6060

features, unknown categories are ignored and missing categories are explicit. Spambase (Hopkins et al., 1999) uses a stratified 70/15/15 split and training standardization. Both models have 128 and 64 ReLU units, dropout 0.1 after the first hidden layer, and two output logits. The primary metric is AUROC; accuracy, macro-F1 and cross-entropy are retained as secondary diagnostics.

Inverse imaging. The Optdigits corpus (Alpaydin & Kaynak, 1998) preserves its official writer-disjoint test set, with 20% of the original training set reserved for stratified validation. The 64 pixel features are divided by 16. The denoising task adds independent Gaussian noise with standard deviation 0.3 and clips corrupted inputs to $[0, 1]$. The completion task independently masks half the pixels and concatenates the visibility mask to the input. Training uses fresh corruptions on every full epoch; validation and testing use fixed independent corruptions. The encoder–decoder widths are 128–32–128, with tanh hidden activations and sigmoid output. Denoising optimizes all-pixel MSE. Completion optimizes missing-pixel MSE; observed pixels are copied when computing full-image reconstruction MSE and PSNR. The two tasks do not constitute separate independent image datasets.

Search and state accounting. Ranks start at 8 and are capped below the input dimension and at half the output dimension. Small matrices and biases use full Adam fallback in routed methods. All weight decays are zero. AdamW searches learning

Table 27. Stronger tracking controls on all ten predeclared conditions. EMA SVD stores the full normalized covariance and uses decay .9 or .99; its refresh period is 5 or 25. Mean pre-update $r^{-1}\|\sin\Theta\|_F^2$ on steps 1501–2000. Bold identifies the best displayed error; no temporal observations are treated as independent replicates.

Rows	Drift	GrassRoute	SVD/5	SVD/25	EMA.9/5	EMA.9/25	EMA.99/5	EMA.99/25
16	0.000	0.03073	0.14550	0.13859	0.00636	0.00644	**0.00059**	0.00059
16	0.001	0.03074	0.14549	0.13882	**0.00649**	0.00698	0.01054	0.01269
16	0.005	0.03083	0.14565	0.14349	**0.00996**	0.01974	0.15492	0.19335
16	0.020	**0.03217**	0.14885	0.21093	0.06017	0.19458	0.42936	0.62078
16	0.100	**0.07407**	0.23144	0.65302	0.51743	0.60934	0.63732	0.54925
128	0.000	0.00381	0.01564	0.01613	0.00078	0.00077	0.00007	**0.00007**
128	0.001	0.00381	0.01566	0.01635	**0.00093**	0.00134	0.01021	0.01235
128	0.005	**0.00391**	0.01594	0.02159	0.00445	0.01420	0.15473	0.19328
128	0.020	**0.00541**	0.02008	0.09878	0.05440	0.18954	0.42786	0.62058
128	0.100	**0.04949**	0.11814	0.60853	0.50970	0.60812	0.70552	0.57523

Table 28. Independent expanded-search PINN follow-up with a transport-corrected periodic comparator. Nine validation-screen candidates per method (100 epochs each), then a fresh 2000-epoch run of the selected configuration. Final held-out solution relative L_2 error; lower is better. – denotes unfinished, not omitted, training.

Task	AdamW	Fixed	PeriodicDiag	GrassRoute
Poisson / PINN	9.52×10^{-4}	1.97×10^{-4}	$\mathbf{4.90\times10^{-5}}$	6.70×10^{-5}
Forced Burgers / PINN	**0.00239**	0.0128	0.00582	0.00427

Table 29. New-task data and primary metrics. Preprocessing fits the training split only. Counts are actual prepared examples, not nominal dataset sizes.

Task	Train	Validation	Test	Primary metric
20 Newsgroups	9051	2263	7532	Macro-F1 ↑
SMS Spam (deduplicated)	3612	774	774	Macro-F1 ↑
Adult	26048	6513	16281	AUROC ↑
Spambase	3220	690	691	AUROC ↑
Optdigits denoising	3058	765	1797	Pixel MSE ↓
Optdigits completion	3058	765	1797	Missing-pixel MSE ↓

rates $\{0.0003, 0.0005, 0.001, 0.002, 0.003, 0.005, 0.01, 0.02, 0.03\}$; SGD-M searches nine geometrically spaced rates from 0.001 to 3, with momentum 0.9. Routed methods cross rates $\{0.001, 0.003, 0.01\}$ with residual-rate ratios $\{1, 10, 100\}$. GrassRoute tracking rates $\{0.3, 1, 3\}$ and periodic intervals $\{5, 25, 100\}$ are assigned in a balanced cyclic design across those nine cells, not exhaustively crossed. Official SubTrack++ crosses the same three parameter rates with initial subspace steps $\{0.1, 1, 3\}$ and uses the device-only CPU adapter described above. Adaptive moments use $\beta_1 = 0.9, \beta_2 = 0.999$; GrassRoute uses routing momentum 0.9 and diagonal transport. Fixed and periodic comparators retain no unused routing momentum. Thin SVD uses a double-precision retry only if its single-precision decomposition raises a linear algebra error. Persistent state is measured from allocated optimizer storage, including the official projector state. Shared-host CPU wall times are not isolated speed measurements.

All screening runs use 200 complete epochs; final validation selects a configuration, which is retrained afresh for 1000 epochs before a single held-out test evaluation. Optimizer random initialization is isolated from dropout/corruption streams. Restartable checkpoints retain model, optimizer and random states. Every candidate, numerical failure, full trajectory, source hash and endpoint remains in the ledger. The frozen manifest is `revision/newtask_results/protocol.json`; the checked outcome ledger is `revision/newtask_status.json`. These new runs do not inflate the original 162-comparison completion count.

Table 30. Complete frozen new-task matrix, seed 42. Nine 200-epoch validation candidates per method, followed by 1000 fresh full epochs. Macro-F1 and AUROC are percentages; image tasks report MSE (missing pixels for inpainting). – means unfinished or numerically failed, with failures retained in the ledger. Best endpoints are bold only after all six methods complete.

Task	Metric	SGD-M	AdamW	Fixed	Thin SVD	GrassRoute	SubTrack++
Optdigits / denoising	MSE ↓	0.0236	0.0187	0.0325	0.0241	0.0249	**0.0178**
Optdigits / inpainting	MSE ↓	0.0412	**0.0296**	0.0507	0.0379	0.0429	0.0314
Spambase / MLP	AUC ↑	**97.87**	97.62	97.41	97.58	97.55	97.80
Adult / MLP	AUC ↑	89.29	88.91	90.75	**91.23**	90.27	89.14
SMS / TF-IDF MLP	F1 ↑	**95.48**	**95.48**	46.69	94.29	94.63	95.40
20 News / TF-IDF MLP	F1 ↑	**54.33**	50.95	52.96	53.49	52.31	50.38

Table 31. All classification secondary endpoints and state costs in the new suite. Text configurations were selected by macro-F1 and tabular configurations by AUROC, never by the secondary test metrics. Accuracy, F1 and AUC are percentages; cross-entropy is unscaled; persistent state is decimal MB. – means unfinished or inapplicable.

Task	Method	Accuracy	Macro-F1	AUROC	CE	State MB
Spambase / MLP	SGD-M	93.34	93.02	97.87	0.3269	0.063
Spambase / MLP	AdamW	94.07	93.76	97.62	0.5338	0.127
Spambase / MLP	Fixed	94.07	93.78	97.41	0.4692	0.021

Task	Method	Accuracy	Macro-F1	AUROC	CE	State MB
Spambase / MLP	Thin SVD	93.63	93.34	97.58	0.3465	0.021
Spambase / MLP	**GrassRoute**	92.19	91.81	97.55	0.3459	0.027
Spambase / MLP	SubTrack++	93.34	93.02	97.80	0.4853	0.056
Adult / MLP	SGD-M	84.15	77.23	89.29	0.4159	0.089
Adult / MLP	AdamW	84.14	76.98	88.91	0.3993	0.179
Adult / MLP	Fixed	85.25	78.59	90.75	0.3185	0.022
Adult / MLP	Thin SVD	85.58	79.16	91.23	0.3067	0.022
Adult / MLP	**GrassRoute**	84.86	78.31	90.27	0.3408	0.030
Adult / MLP	SubTrack++	83.71	76.94	89.14	0.4541	0.093
SMS / TF-IDF MLP	SGD-M	98.06	95.48	99.30	0.0799	0.525
SMS / TF-IDF MLP	AdamW	98.06	95.48	98.28	0.1752	1.050
SMS / TF-IDF MLP	Fixed	87.60	46.69	64.97	0.3601	0.071
SMS / TF-IDF MLP	Thin SVD	97.67	94.29	99.02	0.0952	0.071
SMS / TF-IDF MLP	**GrassRoute**	97.80	94.63	99.08	0.0931	0.137
SMS / TF-IDF MLP	SubTrack++	98.06	95.40	97.50	0.1415	0.153
20 News / TF-IDF MLP	SGD-M	55.12	54.33	–	2.1272	0.530
20 News / TF-IDF MLP	AdamW	51.66	50.95	–	2.4000	1.060
20 News / TF-IDF MLP	Fixed	54.25	52.96	–	1.4909	0.074
20 News / TF-IDF MLP	Thin SVD	54.30	53.49	–	1.6752	0.074
20 News / TF-IDF MLP	**GrassRoute**	53.21	52.31	–	1.7997	0.141
20 News / TF-IDF MLP	SubTrack++	50.94	50.38	–	5.9154	0.159

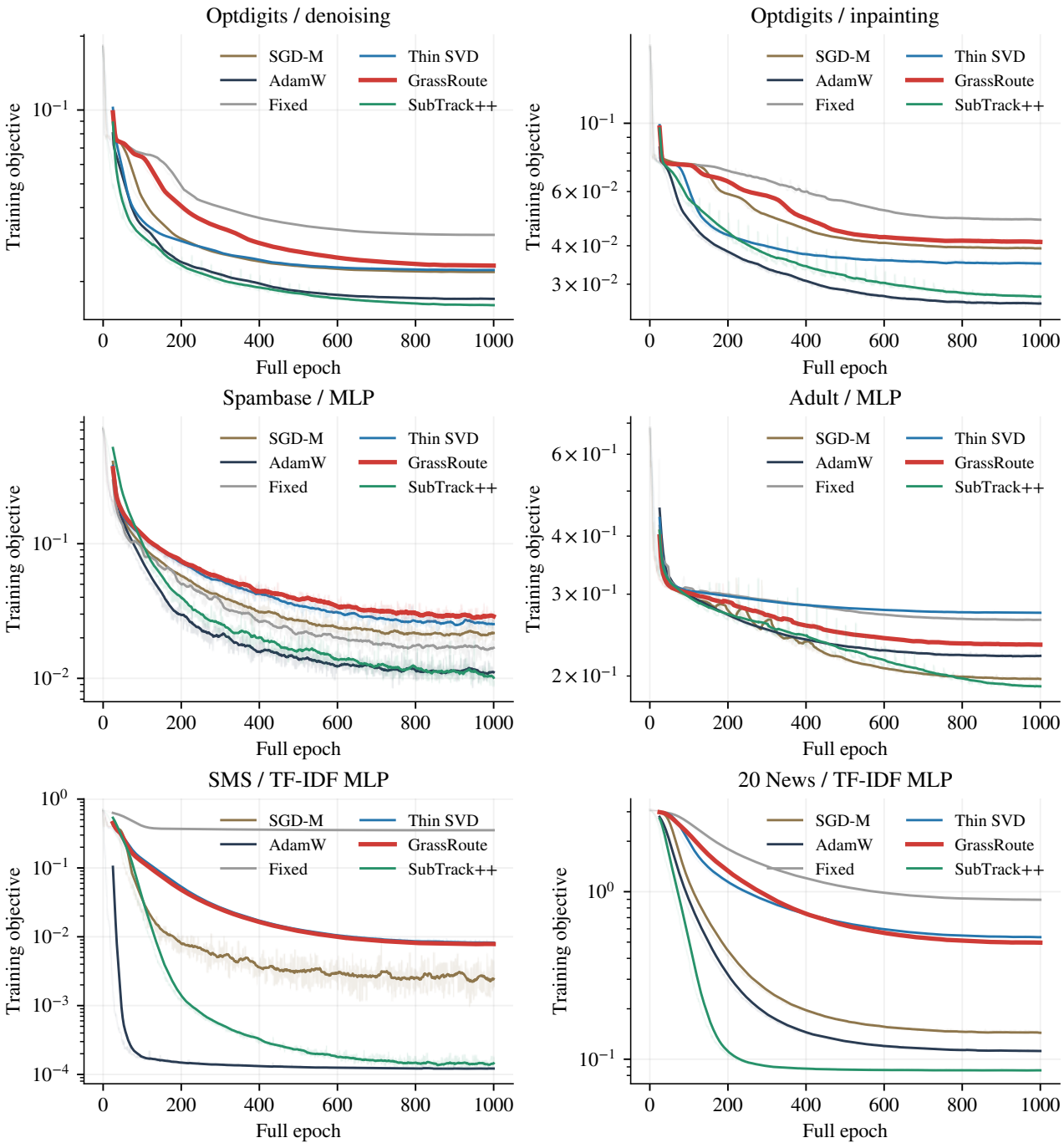


Figure 4. Complete new-task training-loss matrix, including every completed formal optimizer run. Missing curves indicate unfinished or failed formal training, never an omitted unfavorable endpoint. All full runs have 1000 epochs. Faint lines are raw objectives and solid lines are 25-epoch trailing means. Task selection for the main-text illustration does not remove any curve from this complete matrix.